Rational Number Project

Initial Fraction Ideas*

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ABOUT THE AUTHORS

Kathleen Cramer is a professor in the College of Education and Graduate Studies at the University of Wisconsin at River Falls. She teachers graduate and undergraduate mathematics techniques classes for students majoring in elementary education. She has taught mathematics in elementary school and in junior high school. Kathleen has a Ph.D. from the University of Minnesota in mathematics education. She has published articles and book chapters dealing with the teaching and learning of fractions and proportional reasoning. She has done numerous workshops for teachers dealing with fraction instruction.

Professor Cramer has been involved with the Rational Number Project (RNP) since 1980. She participated in the initial teaching experiments with fourth and fifth graders. She has taken the primary responsibility for revising the lessons developed from the research to form the two sets of RNP Fraction Lessons for the Middle Grades.

Merlyn Behr was for over 25 years a professor of mathematics education at Northern Illinois University in DeKalb, Illinois. He was also a faculty member at Florida State University where he received his Ph.D., and at Louisiana State University at Baton Rouge. Merlyn's primary interest was in children's learning of elementary- and middle-grades mathematical concepts. He contributed a great deal to our understanding of children's cognitive processes in these areas. He was very active in the research community and served on the editorial board of the Journal for Research in Mathematics Education (JRME) and as chair of the North American chapter of the research group of the Psychology of Mathematics Education. As a co-founder of the RNP, Merlyn was instrumental in charting its course and providing much valued intellectual leadership in many aspects of RNP activity.

Merlyn died in February 1995. His wit and professional contributions are sorely missed.

Thomas Post, former high school mathematics teacher in New York State, joined the faculty of the College of Education at the University of Minnesota in 1967 after receiving his Ed.D. from Indiana University. Professor Post's interest is closely allied with other RNP members, as he is especially interested in children's and teachers' perceptions of middle-school mathematics. He also has an interest in interdisciplinary approaches to curriculum. He was a co-founder of the RNP and has been active in the mathematics education research community. Along with Kathleen Cramer, Merlyn Behr and Richard Lesh, he has been one of the co-authors of some 70 papers, book chapters and technical reports produced by the RNP since it's inception in 1979. Tom has also served on the editorial board of the JRME and has been chair of the North American chapter of the research group Psychology of Mathematics Education.

Richard Lesh, former professor and dean at Northwestern University, received his Ph.D. from Indiana University. He spent 5 years overseeing computer software development in mathematics and science at WICAT systems in Provo, Utah. He then served as senior research scientist at ETS in Princeton, NJ where he developed innovative strategies and materials for assessing outcomes in mathematics classrooms. Professor Lesh has served as project manager of the program unit - Research on Teaching and Learning - at the National Science foundation. Currently, he is a professor of mathematics at the University of Massachusetts-Dartmouth helping to further advance our thinking about authentic assessment, principles and strategies. Dick is on of the original co-founders of the RNP and has worked on each of its six grants since 1979. He currently leads the Massachusetts site of the RNP's Middle-Grades Teacher Enhancement Project, which is the latest of the projects funded by NSF.

Merlyn Behr was a founding member of the Rational Number Project. He developed the first draft of the fraction materials that was used in the initial research with children. His insight into children's learning, and his thoughtful analyses if students' thinking are an integral part of this curriculum.

These Lessons are dedicated to Merlyn Behr. After a long and illustrious career in mathematics education, he passed away in February, 1995.

The authors would like to acknowledge the important contributions made by the fourth- and fifth-grade teachers in the Rosemount-Apple Valley, MN school district who participated in the pilot study of these materials. Their suggestions for improving the lesson have been incorporated into the materials. The quality of these lessons is due in part from the input these teachers provided.

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Preface

The **Rational Number Project** (RNP) is a cooperative research and development project funded by the National Science Foundation. Project personnel have been investigating children's learning of fractions, ratios, decimals and proportionality since 1979. This **book of fraction lessons** is the product of several years of working with children in classrooms as we tried to understand how to organize instruction so students develop a deep, conceptual understanding of fractions.

The lessons were originally published in 1997 under the title: RNP: *Fraction Lessons for the Middle Grades Level 1*. They have been revised (August, 2009) and renamed to better reflect its content. *Initial fraction ideas* include developing meaning for fractions using a part-whole model, constructing informal ordering strategies based on mental representations for fractions, creating meaning for equivalence concretely, and adding and subtracting fractions using concrete models. Initial fraction ideas do not include formal algorithms, and instruction with formal algorithms was not part of this original RNP curriculum module.

A companion module has been developed with NSF support. This module, *Fraction Operations and Initial Decimal Ideas*, extends students' fraction ideas to develop fraction operations of addition, subtraction, multiplication and division with symbols. The new module also introduces students to decimal ideas – naming decimals, order, equivalence, addition and subtraction. This module can be found on the RNP website at this address:

http://www.cehd.umn.edu/rationalnumberproject/rnp2.html

- **These lessons** provide teachers with an alternative to the textbook scope and sequence for fraction instruction and are appropriate for students in grades 4 – 6 but have also been shown to be effective in remedial settings with older students.

- **These lessons** help students develop number sense for fractions because they invest time in the development of concepts, order and equivalence ideas.

- **These lessons** provide students with daily "hands-on" experiences. Fraction circles, chips and paper folding are the manipulative models used in these lessons to develop initial fraction ideas.

- **These lessons** provide teachers with daily activities that involve children in large group and small group settings. All the lessons involve students using manipulative materials. Our work with children has shown that students need extended periods of time with manipulatives to develop meaning for these numbers.

- **These lessons** offer teachers insight into student thinking as captured from the RNP research with children. The "Notes to the Teacher" section shares examples of students' misunderstandings, provides anecdotes of student thinking, and contains information on using manipulative materials.

- **These lessons** will help teachers and students attain the goals set for fractions by the National Council of Teachers of Mathematics in their *Principals and Standards for School Mathematics (2000).*

- Understand fractions as parts of unit wholes, as parts of a collection, as locations on number lines, and as divisions of whole numbers;
- Use models, benchmarks, and equivalent forms to judge the size of fractions;
- Recognize and generate equivalent forms of commonly used fractions;
- Develop and use strategies to estimate computations involving fractions;
- Use visual models, benchmarks, and equivalent forms to add and subtract commonly used fractions.

- **These lessons** delay development of operations with fractions until students have developed meaning for fractions. The lessons develop understanding of the operations with fractions by using story problems and fraction circles. Estimation is emphasized throughout.

- **These lessons** have been used with over 1600 students. Teachers' response to them has been uniformly enthusiastic. Results of pilot testing show significant differences in students' performance between students who used these lessons as compared to students who used the textbook for fraction instruction. Students using the RNP lessons outperformed students using the district's commercial textbook. Marked differences in thinking were noted in the pilot study. Students using the RNP lessons thought about fractions in a conceptual manner, while students using the textbook thought about fractions in a procedural manner.

- **These lessons are for teachers who know from their own experience with children that there must be a better way to teach fractions!!! Students who use these lessons will develop concepts and will be able to operate on fractions meaningfully.**

Teacher's Guide
RNP: Initial Fraction Ideas

Introduction

These lessons reflect research on children's fraction learning conducted by the National Science Foundation-sponsored Rational Number Project (RNP). Since 1978, Merlyn Behr (Northern Illinois University), Kathleen Cramer (University of Minnesota), Thomas Post (University of Minnesota) and Richard Lesh (Indiana University) have studied how elementary-aged children learn to extend their understanding of numbers to include fraction ideas.

The RNP staff conducted teaching experiments with fourth- and fifth-grade children. In a teaching experiment, researchers entered the classroom as teachers and worked with a curriculum created from a well thought-out theoretical framework. During the teaching and learning process, researchers study children's learning as they progress through the lessons. This is done through classroom observations, student interviews and written assessments.

The curriculum created for the teaching experiments that evolved into these lessons reflected the following beliefs: (a) Children learn best through active involvement with multiple concrete models, (b) physical aids are just one component in the acquisition of concepts: verbal, pictorial, symbolic and real-world representations also are important, (c) children should have opportunities to talk together and with their teacher about mathematical ideas, and (d) curriculum must focus on the development of conceptual knowledge prior to formal work with symbols and algorithms.

The teaching experiments were conducted in two parts. The first phase of the project was ten weeks long with small groups of fourth and fifth graders at two different sites (Minnesota and Illinois). The second phase was conducted with a classroom of 30 students. A class of fourth graders participated from January of their fourth grade, through January of their fifth grade. Instruction was four days per week and covered the following topics: part-whole model for fractions, ratio and quotient models for fractions, decimals, and number lines.

Throughout both teaching experiments a subset of children were interviewed every two weeks. The interviews provided information on children's thinking about fraction ideas. We were interested in what role manipulative materials played in their thinking as well as what understandings and misunderstandings children have with fractions.

Our work with children helped explain why children have so much difficulty with fractions. It also informed us as to the type of experiences children need to develop a deep, conceptual understanding of fractions. Consider a few of the insights garnered from these teaching experiments:

1. Children have difficulty internalizing that the symbol for a fraction represents a single entity. When asked if $\frac{2}{3}$ was one or two numbers, many children would say that the symbol represented two numbers. When students consider $\frac{2}{3}$ as two numbers then it makes sense to treat them like whole numbers. For example, when students add two fractions by adding the numerators and then denominators, they are interpreting the symbols as four numbers, not two. Many errors with fractions can be traced to students' lack of mental images for the quantity the symbol represents.

2. Ordering fractions is more complex than ordering whole numbers. Comparing $\frac{1}{4}$ and $\frac{1}{6}$ conflicts with children's whole number ideas. Six is greater than four, but $\frac{1}{4}$ is greater than $\frac{1}{6}$. With fractions, **more** can mean **less**. The **more** equal parts you partition a unit into, the **smaller** each part becomes. In contrast, $\frac{3}{5}$ is greater than $\frac{2}{5}$ because 3 of the same-size parts are greater than 2 of the same-size parts. In this case, **more** implies **more**. Being able to order plays an important part in estimating fraction addition and subtraction. Ideally when a student adds, for example, $\frac{1}{4}+\frac{1}{3}$, she should be able to reason from her mental images of the symbols that (a) the answer is greater than $\frac{1}{2}$, but less than one and (b) $\frac{2}{7}$ is an unreasonable answer because it is less than $\frac{1}{2}$.

3. Understanding fraction equivalence is not as simple as it may seem. Some children have difficulty noting equivalence from pictures. Imagine a circle partitioned into fourths with one of those fourths partitioned into three equal parts. Some children we worked with were unable to agree that $\frac{3}{12}$ equals $\frac{1}{4}$ even thought they agreed that physically the two sections were the same size. Children said that once the lines were drawn in, you could not remove them. [Therefore $\frac{3}{12} \neq \frac{1}{4}$]. In reality, that is just what must be done to understand fraction equivalence from a picture.

4. Difficulties children have with fraction addition and subtraction come from asking them to operate on fractions before they have a strong conceptual understanding for these new numbers. They have difficulty understanding why common denominators are needed so they revert to whole number thinking and add numerators and denominators.

The RNP Curriculum

The RNP curriculum offers an alternative scope and sequence to one suggested in fourth- or fifth-grade textbooks. The RNP philosophy is that extended periods of time invested with manipulative materials developing concepts, order, and equivalence ideas are needed before students can operate on fractions in a meaningful way. We call these skills, initial fraction ideas. These goals are consistent with the instructional goals set forth in the National Council of Teachers of Mathematics in their *Principles and Standards for School Mathematics*. The RNP curriculum provides teachers with carefully researched lessons to meet these goals.

The RNP Level 1 materials develop the following topics: (a) part-whole model for fractions, (b) concept of unit, (c) concepts of order and equivalence and (d) addition and subtraction of fractions at the concrete level. The concrete models used are fraction circles, paper folding and chips. It de-emphasizes written procedures for ordering fractions, finding fraction equivalences, and

symbolic procedures for operating on fractions. Instead it emphasizes the development of a quantitative sense of fraction.

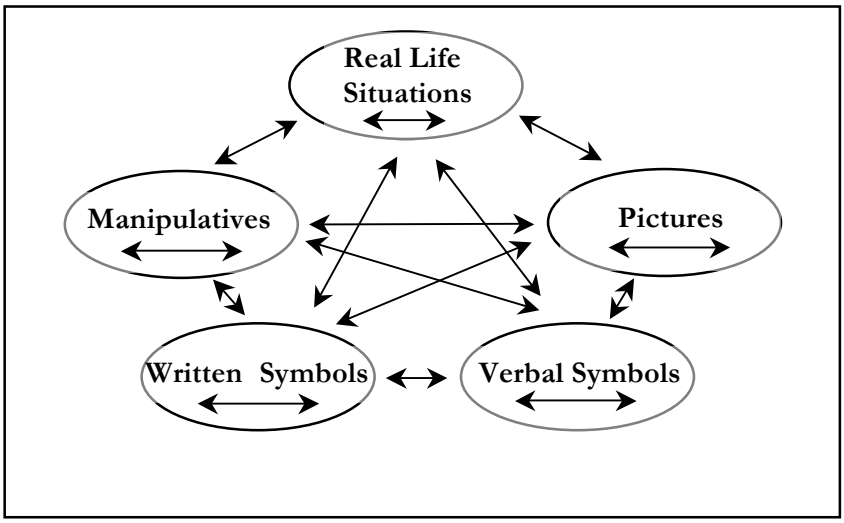
To think quantitatively about fractions, students should know something about the relative size of fractions and be able to estimate a reasonable answer when fractions are operated on. Below, find an example of a fourth-grad student's thought process for estimating a fraction addition problem. This student used the RNP curriculum; her thinking reflects a quantitative sense of fraction. Students using the RNP lessons develop this type of understanding for fractions.

> **Problem: John calculated the problem as follows: 2/3 + 1/4 = 3/7. Do you agree?**
>
> *Student*: I don't agree. He did it weird. You don't add the top numbers and bottom numbers.
>
> *Teacher*: What would be an estimate?
>
> *Student*: It would be…greater than 1/2 because 2/3 is greater than 1/2.
>
> *Teacher*: Would it be greater or less than one?
>
> *Student*: Less than one. You'd need 1/3 and 1/4 is less than 1/3.
>
> *Teacher*: What about 3/7?
>
> *Student*: 3/7 is less than 1/2.
>
> *Teacher*: How do you know?
>
> *Student*: Because 3/7 isn't 1/2. I just know.

Theoretical Framework

Children using these lessons will be using several manipulative models and will consider how these models are alike and different. They will work in small groups talking about fraction ideas as well as interacting with the teacher in large group settings. They will be drawing pictures to record their actions with fraction models. They will be solving story problems using manipulatives to model actions in the stories.

This model for teaching and learning reflects the theoretical framework suggested by Jean Piaget, Jerome Bruner, and Zoltan Dienes. Richard Lesh, a long time RNP member, suggested an instructional model that clearly shows how to organize instruction so children are actively involved in their learning. Consider this picture.



Lesh suggests that mathematical ideas can be represented in the five ways shown here. Children learn by having opportunities to explore ideas in these different ways and by making connections between different representations. This model guided the development of the RNP curriculum.

Lesson Format

The lessons reflect a classroom organization that values the important role a teacher plays in student learning as well as the need for students to work cooperatively, talking about ideas, and using manipulative models to represent rational number concepts. Each lesson includes an overview of the mathematical idea developed. Materials needed by teachers and students are noted. The lesson begins with a class **Warm Up**. Warm Ups are used to review ideas developed in previous lessons and should take only 5-10 minutes of class time. There is a **Large Group Introduction** section in each lesson. The teacher's lesson plans provide problems and questions to generate discussion and target the exploration. **Small Group/Partner Work** is included in each lesson where

students together continue the exploration of ideas introduced in the large group. The class ends with a **Wrap Up**. A final activity is presented to bring closure to the lesson. At times this will be a presentation by students of select problems from the group work. We found that students like to share their thinking. At other times the Wrap Up will be another problem to solve as a group. The amount of time needed for each lesson will vary from classroom to classroom. A single lesson does not necessarily reflect one day's work, though teachers often will find that one day is sufficient to cover the material.

An important part of each lesson is the "Comments" section. Here insights into student thinking captured from the initial RNP teaching experiments are communicated to teachers. These notes clarify a wide variety of issues, such as why mastery at the symbolic level is not the primary objective for many of the earlier lessons. The notes also share examples of students' misunderstandings for teacher's reflection and anecdotes of student thinking from earlier RNP projects. These notes to the teachers also clarify methods for using manipulative materials to model fraction ideas.

Manipulative Materials

Fraction circles, two-sided colored counters and paper folding are the manipulative models used. Our research has shown that the fraction circles are the most important manipulative model for developing mental images of fraction symbols.

Fraction Circles

The master for the fraction circles are in the appendix with a page showing the different partitions and colors used for the fraction circles. The circles should be duplicated on index using colors noted on each master. Teachers who have used the fraction circles have relied on their students, parents or teacher-aids to cut out the circles and to organize them in two-pocket folders. If you choose to send home the fraction circles to be cut out with the parent's

help, you will find in the appendix a parent and child activity sheet for them to do together once the circles are cut out.

Counters

Two-sided colored counters are available from most publishers of mathematics manipulative materials. A less expensive way is to purchase from a tile store, one square inch tiles (white on one side, tan on other). These cost less than 1.5 cents per tile. Thirty per student should be enough.

Paper Folding

Use 8.5″ by 11″ sheets of paper cut into strips 1″ by 8.5″. Have lots on hand for students to use for lessons 7 and 10.

Pilot Testing

The RNP Level 1 lessons have been piloted in the Rosemount-Apple Valley, MN school district. Some 66 teachers participated in this study. Thirty-three fourth- and fifth-grade teachers used the RNP lessons while thirty-three other fourth- and fifth-grade teachers used the textbook. RNP students outperformed textbook students in all areas assessed. Particular differences were noted in students' thinking. RNP students thought about fractions in a conceptual manner, while textbook students generally thought about fractions procedurally. RNP students were much more able to verbalize their thinking about fractions than textbook students.

Teachers from this pilot who used the RNP curriculum gave us detailed feedback on lessons. The lessons have been revised to reflect the input from these teachers.

Assessment

Four quizzes are provided in the appendix to assess students' fraction understanding as they work through all 23 lessons. A written test is provided for use at the end of the lessons. Items reflect information directly taught in the lessons as well as extensions. Another important source of information on

children's thinking can come from talking to students. Interviews are included to be used at certain intervals in the lessons. You may want to select three students representing a range of understanding and interview them as they progress through the lessons. The interview items come from the interviews we used with children and reflect questions we feel gave us the most information on children's thinking.

Interview questions assess understanding on content addressed in the lessons as well as questions on content to be addressed in upcoming lessons. It is interesting to see how students transfer information to new situations. Of particular interest is interview #3 which, if used, should be given before lesson 19. This interview asks students to add and subtract fractions before the lessons on addition and subtraction. We found it interesting to note how many students were able to construct their own reasonable process for adding and subtracting fractions. A written test and four interviews can be found in the appendix.

Special Notes on Students' Thinking

From our interviews with children we noted that they constructed what we now refer to as informal strategies for ordering fractions. These strategies reflect students' use of mental images of fractions to judge the fraction's relative size. These informal strategies do not rely on procedures usually taught: least common denominators and cross-products. We have named the four strategies noted in students' thinking as: same numerator, same denominator, transitive and residual strategies.

When comparing $\frac{2}{3}$ and $\frac{2}{6}$ (fractions with the same numerator) students can conclude that $\frac{2}{3}$ is the larger fraction because thirds are larger than sixths and two of the larger pieces must be more than two of the smaller pieces. This strategy involves understanding that an inverse relationship exists between the number of parts a unit is partitioned into and the size of the parts.

The same denominator strategy refers to fractions like $\frac{3}{8}$ and $\frac{2}{8}$. In this case, the same denominator implies that one is comparing parts of the unit that are the same size. Three of the same-size parts are greater than two of the same-size parts.

The student strategy that has been termed the *transitive* strategy can be modeled by comparing $\frac{3}{7}$and $\frac{5}{9}$. When making this comparison, a student can conclude that $\frac{3}{7}$ is less than $\frac{5}{9}$ because $\frac{3}{7}$ is less than $\frac{1}{2}$, while $\frac{5}{9}$ is greater than $\frac{1}{2}$. This is the transitive strategy because students use a single outside value to compare both fractions.

When comparing $\frac{3}{4}$ and $\frac{5}{6}$, a student can reflect that both fractions are one "piece" away from the whole unit. Because $\frac{1}{6}$ is less than $\frac{1}{4}$, $\frac{5}{6}$ must be closer to the whole and is therefore the bigger fraction. This thinking strategy has been called a *residual* strategy because students focus on the part "leftover" in judging the relative size of the fractions.

These four strategies closely parallel students' actions with manipulatives. They are in contrast to the paper and pencil procedures, which require changing both fractions to common denominators or calculating cross-products. RNP lessons developed only these student-constructed strategies. The order questions on the interviews will assess whether students construct these strategies. Students who have constructed these strategies have developed are on the way to developing number sense for fractions.

Final Comments

You will find at the end of each lesson a form for you to record your adaptations for each lesson. Any curriculum will need to be "personalized" by the teacher who uses it, so it best meets the needs of his/her students. This form will act as a reminder about changes you feel are important to make the next time you teach the lesson.

The RNP Lessons
Initial Fraction Ideas

RATIONAL NUMBER PROJECT
Initial Fraction Ideas
Scope and Sequence

LESSON	MANIPULATIVE	TOPIC
1	Fraction Circles	Exploration with the circles.
2	Fraction Circles	Model and verbally name: 1-half, 1-third, 1-fourth.
3	Fraction Circles	Model and verbally name unit fractions with denominators greater than 4.
4	Paper Folding	Compare paper folding to fraction circles. Model and name (verbally and with written words) unit and non-unit fractions.
5	Fraction Circles	Model fractions and record with symbols a/b.
6	Fraction Circles	Model the concept that the greater the number of parts a unit is divided into, the smaller each part is.
7	Paper Folding	Reinforce the concept that the greater the number of parts a unit is divided into, the smaller each part is.
8	Fraction Circles	Fraction Equivalence
9	Fraction Circles Pictures	Fraction Equivalence
10	Paper Folding	Fraction Equivalence.
11	Fraction Circles	Order fractions by comparing to 1-half.
12	Chips	Introduce new model for fractions less than one by comparing to a familiar model.

LESSON	MANIPULATIVE	TOPIC
13	Chips	Model fractions using several units for the same fraction.
14	Chips	Model fractions using chips; determine fractions that can be shown given a set of chips.
15	Chips	Fraction Equivalence.
16	Fraction Circles	Reconstruct the unit given the fraction part.
17	Fraction Circles	Model fractions greater than one sing mixed and improper fraction notation.
18	Fraction Circles	Fraction equivalence for 1-half based on a number pattern.
19	Fraction Circles	Estimate sum of two fractions within story contexts.
20	Fraction Circles	Find the sum of two fractions using fraction circles.
21	Fraction Circles	Estimate and solve concretely fraction subtraction using "take-away" and "difference" contexts.
22	Fraction Circles	Estimate and solve fraction subtraction using "difference" and "how many more" contexts.
23		Summary activities to tie together students' number sense and addition and subtraction.

Rational Number Project

Initial Fraction Ideas Lesson 1: Overview	Materials
Lesson provides guided exploration with fractions circles. Students start to become familiar with colors and relationships like 3 browns cover 1 black and 1 brown is bigger than 1 red.	∞ Fraction Circles for students and teacher ∞ Student Page A ∞ Transparency 1

Teaching Actions

Large Group Introduction

1. Start the Lesson by asking children to sort through their fraction circles to answer these questions:

   (a) How many blues cover the black circle?

   (b) Which is bigger, 1 brown or 1 gray?

   (c) How many pinks cover 1 yellow?

   (d) How many browns cover the black?

   (e) Which is bigger, 1 brown or 2 reds?

   (f) How many purples cover 1 yellow?

   (g) How many dark blues are there? Light blues?

Small Group/Partner Work

2. Explain to the students that they are to continue their exploration by using the circles to complete Student Page A.

Wrap Up

3. End the lesson by working through Transparency 1. The figure on the left represents the circle part you want to cover. To the right are the circle parts. Students are to determine which combination of parts will cover the shape on the left.

Comments

Students need to play with the fraction circles before developing a formal language for describing relationships among the pieces.

There are two different blues: a set of 4 dark blue pieces; a set of 7 light blue pieces. In the lessons the color "blue" refers to the set of 4 dark blue pieces. "Light blue" will refer to the set of 7 blues.

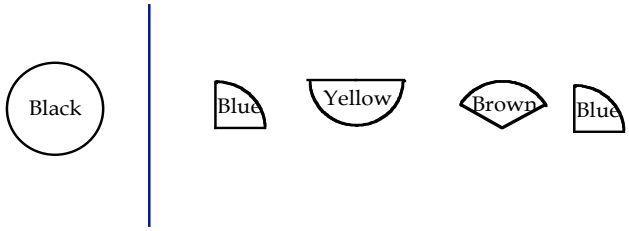
Different ways to approach Student Page A:

Students do page individually and then compare with a partner.

Students do page with a partner.

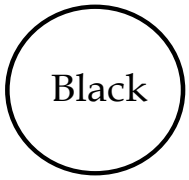
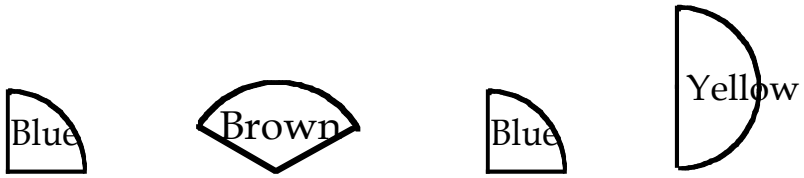
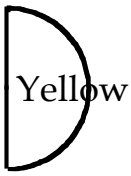
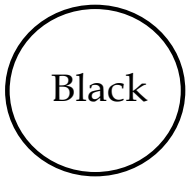
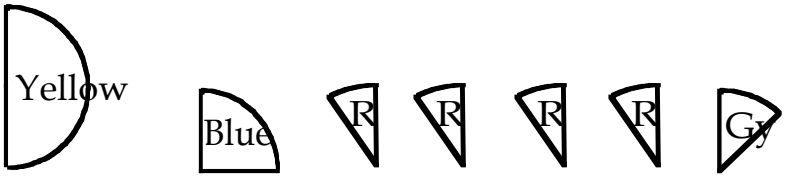
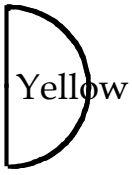
Do a few problems together and then students finish on their own.

If some students finish Student Page A ahead of others, ask them to create their own problems and record them on the back of the page or put them on the board for others to solve.

Teaching Actions	Comments
All pieces selected do not have to be of the same color. Example  	You may want to duplicate Transparency 1 for students.
4. Encourage students to guess first and then use their fraction circles to find the exact combination. In the above example, 2 blues and 1 yellow would cover the circle.	To encourage students to guess you might want to emphasize making "hypotheses". Write the word hypothesis on the board. Record students' guesses, test them out and reach a group consensus.

Translations

- ∞ Verbal to manipulative
- ∞ Picture to manipulative to verbal
- ∞ Manipulative to written symbols

 Black	 Blue Brown Blue Yellow
 Yellow	 Blue Blue Pink Pink Gy
 Blue	 R R R Gy
 Black	 Yellow Blue R R R R Gy
 Brown	 Blue Pink R Pink
 Yellow	 Brown Brown Pink Pink R

Exploring with the Fraction Circles

1. _____ browns equal 1 whole circle.
2. 1 whole circle equals _____ pinks.
3. _____ reds equal 1 whole circle.
4. _____ pinks equal 1 brown.
5. 1 brown equals _____ reds.
6. 1 brown is (less than, equal to, greater than) 1 pink.
7. 1 red is (less than, equal to, greater than) 1 brown.
8. 1 yellow is (less than, equal to, greater than) 1 brown.
9. 1 yellow and 1 brown and 1 _____ equals 1 whole circle.
10. 1 yellow equals 1 brown and 2 _____ .
11. 3 pinks and 1 _____ equal 1 whole circle.
12. _____ grays and 1 blue and 1 yellow equals 1 whole circle.
13. 2 grays and _____ blue equals 1 yellow.
14. 1 pink equals _____ reds.
15. 4 _____ equal 1 yellow.

Rational Number Project

Initial Fraction Ideas **Lesson 2: Overview**	Materials
Students explore relationships among circle pieces, modeling and orally naming fraction amounts for: 1-half, 1-third, and 1-fourth.	∞ Fraction Circles for students and teacher ∞ Student Page A, B

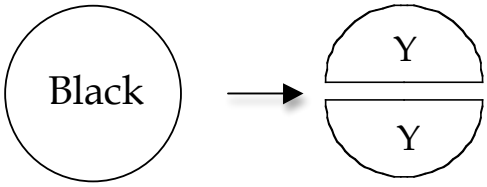
Teaching Actions

Warm Up

Find three different ways to cover 1 yellow piece.
Find three different ways to cover 1 brown piece.

Large Group Introduction

1. Ask students to take out a black circle. Model how to divide the black circle into 2 equal parts by showing that 2 yellow parts cover the whole circle.

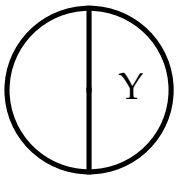


[2 yellows cover 1 black circle]

2. Note that 1 black equals 2 yellows or 2 yellows equal 1 black. Ask: Are the 2 parts covering the whole equal?

3. Conclude by stating that when 2 equal parts equal one whole, each part (pick up 1 yellow) is called one-half. This yellow piece is one-half of the black circle.

4. Show one-half by placing 1 yellow on the black circle. *"1 yellow covers half of the black circle."*



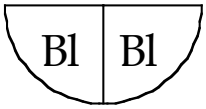
Comments

Most lessons will have a "Warm Up" problem that reviews ideas from previous lessons. These warm ups are not meant to take more than 5 minutes of class time.

Flexibility of unit is stressed right from the beginning by having students find multiple representations for 1/2, 1/3, and 1/4.

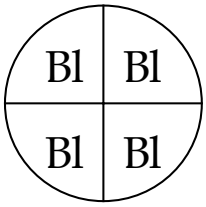
The critical variable with fractions is that a unit is divided into equal parts. A single part can be given a fraction name that depends on what it is being compared to.

Example:



2 blues equal 1 yellow so 1 blue is one-half of the yellow. [Here 1 yellow is the unit]

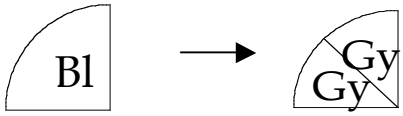
Example:



4 blues equal 1 black. 1 blue is one-fourth of the black circle. [Here black is the unit]

Teaching Actions

5. Continue by looking for other examples for one-half. Show one blue piece and ask: If one blue is my unit, how can we divide this piece into 2 equal parts? What color pieces will do this?

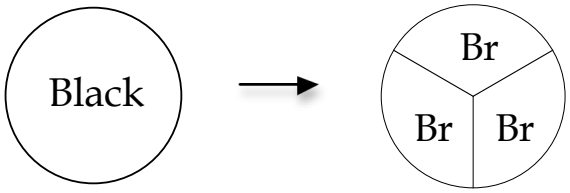


6. Use these questions: Are the 2 parts equal? 1 gray is 1 of 2 equal parts; what fraction of the blue piece is 1 gray? [1-half]

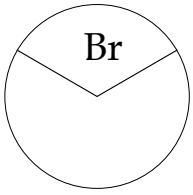


7. Show 1 yellow and ask students to consider the yellow as the unit, divide it into 2 equal parts and orally name each part.

8. Model, using the black circle as the unit, representations for thirds.



9. 3 browns cover 1 black; 1 brown is 1 of 3 equal parts; 1 brown is one-third of the black. Show as:



10. Find other examples for 1-third using 1 yellow and 1 brown and then 1 blue as the unit.

11. Model fourths using 1 black, 1 yellow, and 1 brown as units.

Comments

Students are naming fractions in the verbal mode only. In the next lesson students will record as: 1-fourth.

You may want to show how other units can be partitioned into halves:

1 pink → 2 reds
1 brow → 2 pinks
1 orange → 2 purples

Teaching Actions	Comments
12. End the development part of the lesson with a non-example. Show how 2 blues and 1 yellow cover the black circle. Pick up 1 blue and say that this piece is 1 of 3 parts of the circle so it is one-third of the circle. Ask: Is this true? If I wanted to know what part of the black circle 1 blue is, what must I do? 13. [Repeat showing 2 browns and 2 pinks covering the black circle. 1 pink does not equal 1-fourth]. **Small Group/Partner Work** 14. Student pages A & B present problems similar to ones presented in large group as well as problems within realistic contexts. Assign to students in pairs, as they are to answer the questions orally. **Wrap Up** 15. To assess the "big idea" in this lesson present the following scenario: Lianna said that 1 red piece is one-third; Rodrigo said 1 red is one-fourth. Who is correct? [Note that 1 red is one-third of the blue; 1 red is also one-fourth of the brown. Both Lianna and Rodrigo are correct once you know what unit they are comparing the red to].	Another way to assess the lesson's big idea is to put one of each of these colors: yellow, blue, pink, and red on the overhead and ask: "You have called all of these 1-half, yet they are different sizes. How is that possible?"

Translations

- ∞ Manipulative to verbal
- ∞ Pictures to verbal

Find three different ways to cover 1 yellow piece. Find three different ways to cover 1 brown piece.

Name_____ Lesson 2 / Student Page A

The class will work together in groups or in pairs on these problems. Answers are to be given orally or by drawing a picture. On some of the problems children may want to use the fraction circles to help solve the problem.

1. The yellow piece is the unit.

   How many blues cover the yellow piece? _____

   1 blue is _____ of the yellow.

   (Say the word)

2. The blue piece is the unit.

   How many reds cover the blue piece? _____

   1 red is _____ of the blue.

   (Say the word)

3. The brown piece is the unit.

   How many reds cover the brown piece? _____

   1 red is _____ of the brown.

   (Say the word)

4. What color is 1-half of the blue? _____

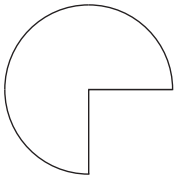
5. What color is 1-third of the yellow? _____

6. Draw a picture of a pizza. Show on your drawing the pizza cut into 2 fair shares.

   Each fair share is _____ of the whole pizza.

   (Say the word)

7. Here is a picture of a pizza with one piece removed.



The piece is _____ of the whole pizza.
(Say the word)

8. Here is a picture of a candy bar that someone has started to cut into pieces.

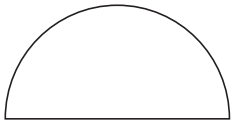


The small piece is _____ of the whole candy bar.
(Say the word)

Draw lines to finish cutting the candy bar into equal parts.

9. Mary's patio is a whole circle. Draw a picture of Mary's patio. Show on your drawing that the patio is in 3 equal-size parts. Each part is _____ of Mary's patio. (Say the word)

10. John has a patio that looks like this:



Draw on John's patio to show it divided into 3 equal-size parts. Each part is _____ of John's patio.
(Say the word

Mary said. "John's patio is really one-half (not a whole)." What would you say to Mary?

Rational Number Project

Initial Fraction Ideas Lesson 3: Overview	Materials
Students model and name (orally and in written words) unit fractions with denominators greater than 4.	∞ Fraction Circles for students and teacher ∞ Student Page A

Teaching Actions

Warm Up

Find the piece that is 1-half of each of these colors: yellow, blue, brown, orange.

Large Group Introduction

1. Show one yellow piece. Ask students to divide it into six equal parts.

2. Explain that since 6 reds cover 1 yellow, 1 red is one-sixth of the yellow.

3. Ask students to divide a black circle into 6 equal parts. What fraction piece is one-sixth of the black?

4. Make this chart to show the relationship between the number of equal parts a unit is divided into and the word name for that number of divisions.

Number of Equal Parts Unit is divided into	Word Name	
2	half	halves
3	third	thirds
4	fourth	fourths
5	fifth	fifths
6	sixth	sixths
7	seventh	sevenths
[Continue to include 8, 9, 10, 12, 15]		

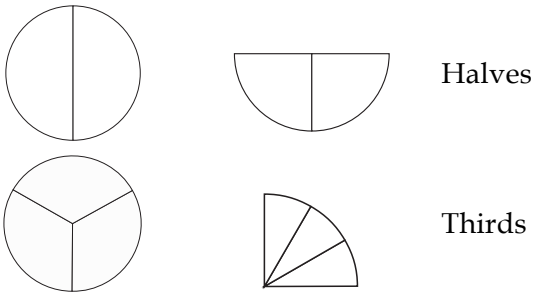
Comments

Make a large classroom chart for students to use as a reference for the rest of the fraction unit. You might include a third column showing a picture of a unit (not always a whole circle) divided into the appropriate number of equal parts.

You may want to have students make their own personal chart.

Teaching Actions

For each item in the chart show at least 2 physical models. For example:



Halves

Thirds

Students should model each example with their fraction circles.

5. Students should help you find these different representations.
   - ∞ You may suggest the unit and ask them to divide it into a certain number of equal parts.
   - ∞ You might ask students to suggest the unit. For example, say: *"The next value in the chart is to model sixths. What unit can we use?"*

6. Once the chart is completed, work through these problems:
   - ∞ Using the black circle as the unit, ask students to find the color that divides the unit into 4 equal parts. Hold up 1 of 4 parts, call it "one-fourth", and record the written name as 1-fourth.
   - ∞ Using the yellow circle as the unit, ask students to find the color that divides the unit into 4 equal parts. Hold up all for parts; call it "one-fourth"; record 1-fourth.
   - ∞ Ask: "How are the two models for 1-fourth alike? Different?

7. Repeat for sixths and twelfths using two different units.

Comments

Students initially record fractions in words like: *1-fourth*; *1-sixth*. Research suggests that students make fewer reversals with the symbols (for example, writing 3/2 for 2/3) when they first write fractions in words.

You may want to do more examples.

Teaching Actions	Comments
8. To prepare students for Student Page A ask the following questions. Have students record answers using word names. o The blue piece is the unit. What fraction name can you give 1 gray piece? 1 red piece? o The brown piece is the unit. What fraction name can you give 1 pink? 1 white? 1 gray?	You might consider assigning students Page A without this introduction. This will make the activity more of a problem solving activity.
Small Group/ Partner Work 9. Assign Student Page A.	
Wrap Up 10. End the class with this game: Teacher says: "Two of the colors I am thinking of equal one yellow. What color is it? What fractional name can I give each piece?	Extra challenges: If the yellow piece is the unit, what value does the black circle have? If the blue piece is the unit, what value does the yellow piece have? The black one? These questions may lead to a nice discussion. Students may question how to express the answer. If the yellow piece is the unit (or one) then the black circle is 2 units, 2 wholes or just 2.

Translations

∞ Manipulative to verbal to written symbols

Find the piece that is 1-half of each of these pieces:

- 1- yellow
- 1- blue
- 1- brown
- 1-orange

Naming Fraction Amounts Using Circles

Use fraction circles to find the names of the different fraction pieces.

I. The black circle is the unit. What fraction name can you give these pieces?

1 yellow 1-half 1 brown _____

1 blue _____ 1 gray _____

1 white _____ 1 green _____

1 red _____ 1 pink _____

II. Now make 1 yellow unit. What fraction name can you give these pieces?

1 blue _____ 1 gray _____

1 pink _____ 1 red _____

III. Change the unit to 1 blue. What fraction name can you give these pieces?

1 gray _____ 1 red _____

IV. Change the unit to 1 orange. What fraction name can you give these pieces?

1 purple _____ 1 green _____

Rational Number Project

Initial Fraction Ideas Lesson 4: Overview	Materials
Students use paper folding to model and name unit and non-unit fractions. Students compare the paper-folding model to fraction circles. Students record fractions in words: one-fourth, two-thirds.	∞ Paper strips for folding for students ∞ Fraction Circles for teacher ∞ Student Pages A-L

Teaching Actions

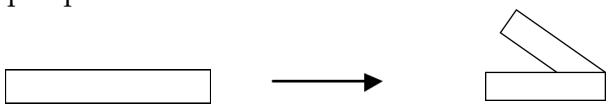
Warm Up

Name the red piece in three different ways by changing the unit. What different units did you use?

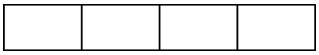
Large Group Introduction

1. Prior to using paper strips to model fractions it is necessary to practice folding strips into 2, 3, 4, 6, 8, and 12 equal parts.

   Ask students to follow along with you as you model how to fold paper strips. Fold paper strip into two equal parts:



2. Keep it folded. Now fold it again into two equal parts. Ask: how many equal parts do you think we have? Unfold:



3. Ask students to verbalize how to fold paper strips to form four equal parts.

4. Model folding into three equal parts. Form the letter "S" with a paper strip to get close to 3 equal parts. Press down on paper.

Comments

This lesson may take two class periods. Students are still recording fractional amounts using word names; symbols are introduced in lesson 5.

Cut paper strips from 8.5" by 11" sheets of paper about 1 inch wide and 8.5" long.

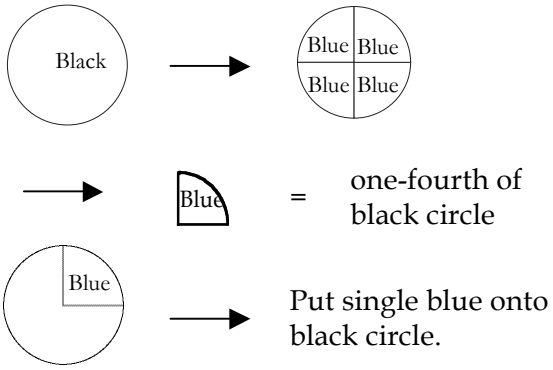
Teaching Actions

5. Model sixths. Fold paper strip into thirds and then fold into two equal parts. Have students do this and guess, before unfolding, the number of equal parts they expect.

6. Ask students if they could have obtained sixths by folding first in halves and then in thirds? Try it.

7. Ask students to think of strategies for folding 8ths and 12ths. Encourage trial and error strategy. Have them verbalize successful ways. For 12ths reinforce multiple ways.

8. Students can shade equal parts of paper strips to show fractions. Using fraction circles, show one-fourth using a black circle as the unit.



Say: To show one-fourth of a black circle I divided it into four equal parts. Pick up one of the parts to show one-fourth.

9. Ask: How can you show me one-fourth with a paper strip? Have students fold into 4 equal parts and shade in one of the 4 equal parts. Record fraction name as 1-fourth.

10. Discuss how the two displays for one-fourth are alike and different.

11. Repeat for 1-third, 1-eighth, 1-twelfth.

Comments

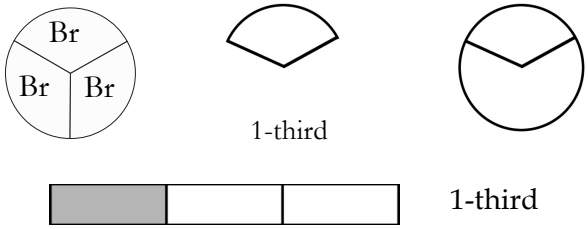
Students often will expect 5 equal parts (3+2). They are more apt to think additively than multiplicatively.

<u>To get 12ths</u>
Halves → halves → thirds
Thirds → halves → halves
Halves → thirds → halves

The similarity between the two displays is what's important. A unit is divided into <u>equal</u> parts and one or more equal parts are highlighted in some way. This is a *manipulative to manipulative* translation.

Teaching Actions

12. Look at two displays for one-third:

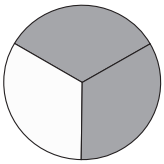


1-third

13. Shade in another third on the paper strip.



Ask: how many thirds are shown now? How can I show two-thirds with circles? (Pick up two browns and say these are two-thirds of black.) State that 2-thirds is 1-third and 1-third more:



14. Now draw a picture of a square. Divide it into 4 equal parts and shade 3 of 4 parts. Ask students to fold paper to show the same fraction that you drew. Record fractions as 3-fourths: 1-fourth + 1-fourth + 1-fourth.

15. Return to fraction circles. Model problems as in lesson 3, this time with non-unit fractions.

Examples:

- The black circle = 1. What is the value of 1 blue; 3 blues; 1 brown; 2 browns; 3 reds.
- The yellow piece = 1. What is the value of 1 blue; 2 reds; 3 grays; 2 pinks.

Comments

Non-unit fractions are introduced as sums of unit fractions: 2-fourths is 1-fourth and 1-fourth.

Students now have seen two models for fractions. Practice pages that follow this lesson give students a chance to apply their new learning to pictures of units in different shapes.

Teaching Actions

Small Group/Partner Work

16. There are several student pages in this lesson. **Select the most appropriate ones for your students.** Students may need some assistance to do some of the pages. See Comments for clarification.

Wrap Up

17. Go over problems 6 and 7 from Student Page B. Have students share their solutions. Pick and choose other problems for students to share.

Comments

Teacher Notes for Student Pages:

B: Problems 6 and 7 provide some problem solving. Students reconstruct the unit given one part. For example if  equals 1-half, then the whole must be two of those parts:



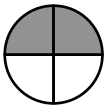
If  equals 1-fourth, then the whole must be four of those parts:



G: Clarify with students that a picture may need to be modified to determine if 2-fourths are shaded in. For example:



Is 2-fourths shaded?



2-fourths can easily be seen once the picture is completed by drawing in the needed lines.

Translations

- Manipulative to verbal
- Manipulative to manipulative to verbal
- Manipulative to verbal to written symbols (word names)

Name the red piece in three different ways by changing the unit. What different units did you use?

1. Here is a picture of a candy bar.



Draw to show the candy bar divided into 5 equal-sized pieces.

2. Here is a picture of a pan of brownies.

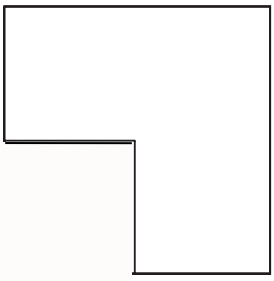


The pan of brownies is cut into _____ equal-sized parts.

Each piece is _____ of the whole pan.

3. O-So-Good candy bars come in the shape of a square. After Janis ate one pieces of an O-So-Good candy bar, it looked like the shape below.

The piece that Janis ate is _____ of the whole candy bar.



4. Hamdi's garden is a rectangle. Draw a picture of Hamdi's garden. Show on your drawing that the garden is in 9 equal-sized parts.

Each part is _____ of Hamdi's garden.

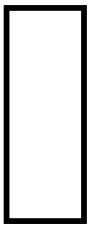
5. Devan's garden is in the shape of square. Draw a picture of Devan's garden.

   Draw on Devan's garden to show it divided into 3 equal-sized parts.

   Each part is _____ of Devan's patio.

6. One-half of a coffee cake was left after a party was over.

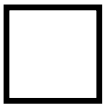
   The half looked like this:



   Draw a picture of the whole cake.

   Explain to your classmates how you solved the problem.

7. Willis, Vang, Ellen, and Marta shared part of a candy bar equally. Marta's share looked like this:

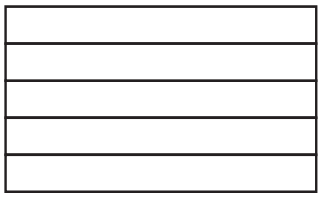


   Draw a picture to show the whole candy bar.

   Explain to your classmates how you solved the problem.

For each exercise, look at the figure and then answer the question and write the word name for each fraction.

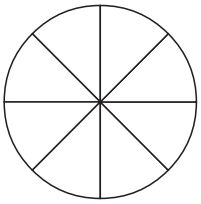
1.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

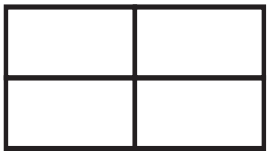
2.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

3.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

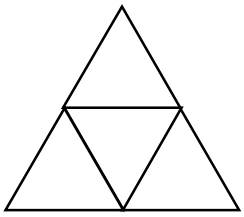
4.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

5.

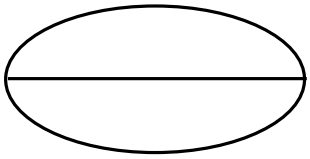


_____ equal-sized parts.

Each part is _____ _____ of the whole.

Name_____ Lesson 4 / Student Page D

6.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

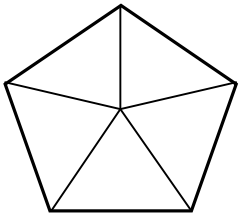
7.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

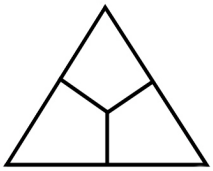
8.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

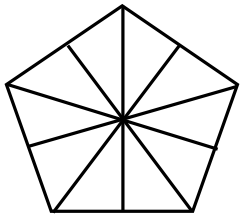
9.



_____ equal-sized parts.

Each part is _____ _____ of the whole.

10.



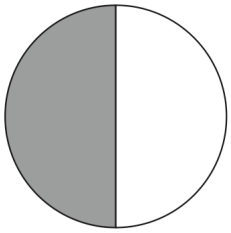
_____ equal-sized parts.

Each part is _____ _____ of the whole.

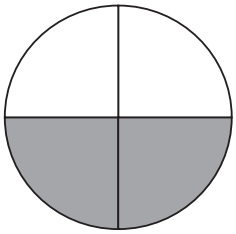
Directions:

You'll need paper strips for folding. For any four of the figures shown below, fold paper strips to model the fraction that the figure models. After you have folded and shaded your paper, write on it the fraction you have shown (use words, not symbols).

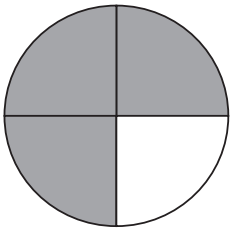
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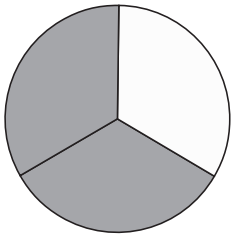
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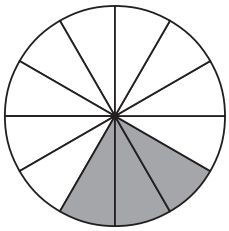
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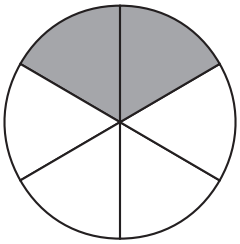
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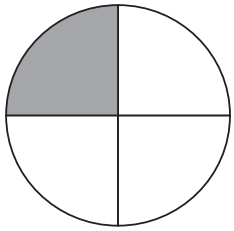
5.



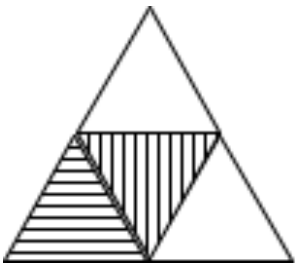
6.



7.



8.



Directions:

You'll need paper strips for folding. For any four of the figures shown below, fold paper strips to model the fraction that the figure models. After you have folded and shaded your paper, write on it the fraction you have shown (use words, not symbols).

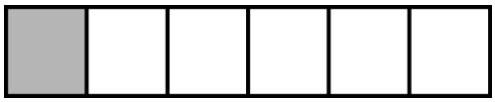
1.



2.



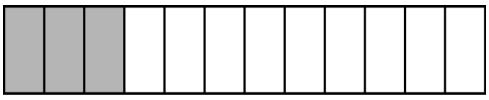
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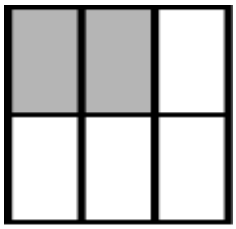
4.



5.



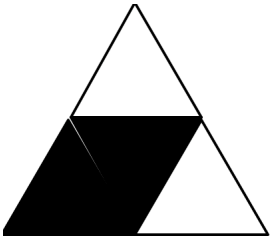
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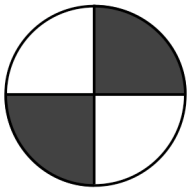
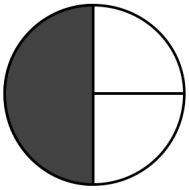
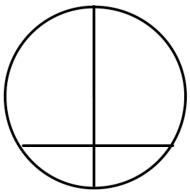
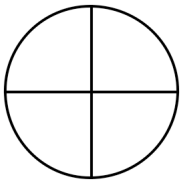
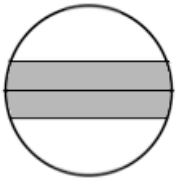
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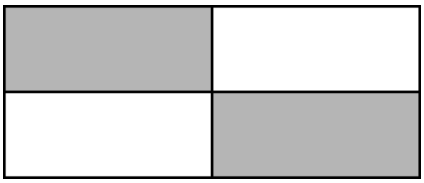
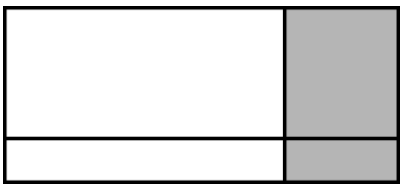
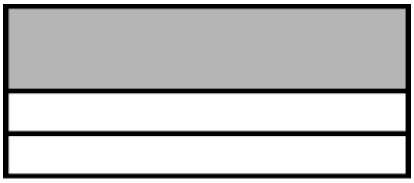


8.



Look at each picture carefully. Place an "X" beside each picture that shows 2-fourths shaded in. You may need to draw in lines to determine if 2-fourths are shaded.




For each diagram, fill in the blanks to tell about the diagram.

a.



Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is _____ -sixth

b.



Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is 1- _____

c.

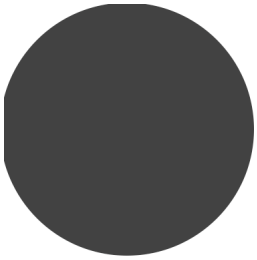


Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is _____

d.

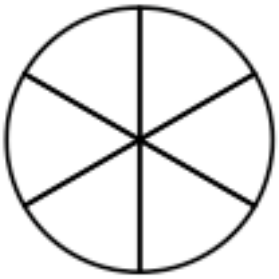


Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is _____

e.

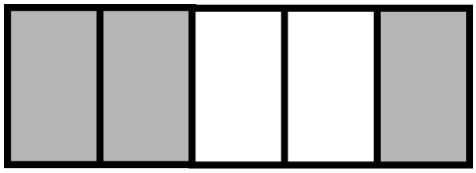


Number of equal parts _____

Number of equal parts shaded _____

The fraction shaded is _____

f.

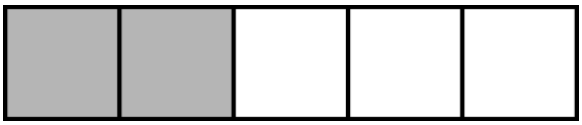


Number of equal parts _____

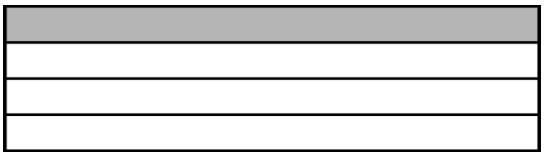
Number of equal parts shaded _____

The fraction shaded is _____

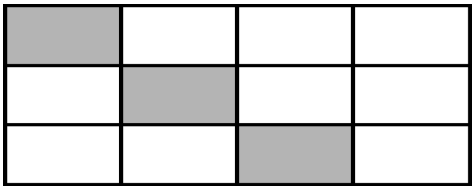
Write words like 2-fourths, 3-fifths, and so on for the fraction shaded by each diagram.



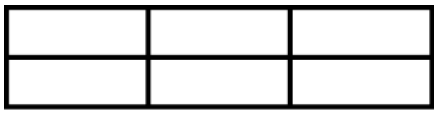
Write _____



Write _____



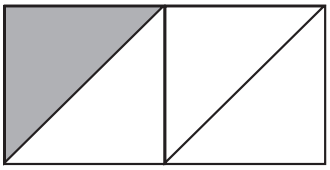
Write _____



Write _____



Write _____



Write _____

Write the fraction that is shown in words:

a.



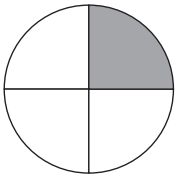
_____

b.



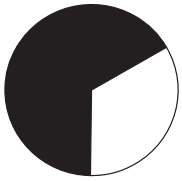
_____

c.



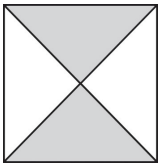
_____

d.



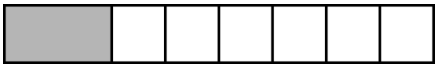
_____

e.

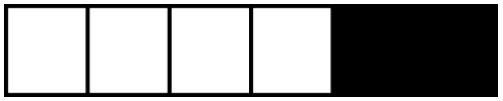


_____

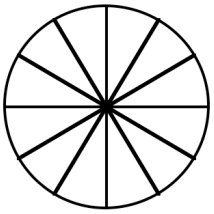
f.



_____

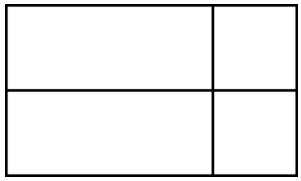
g. 

_____

h. 

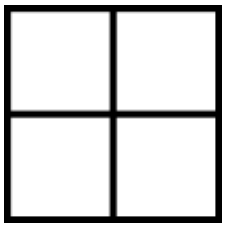
_____

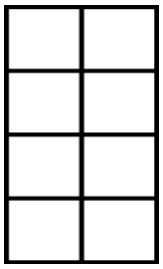
Circle the figures that have equal-sized parts.

1. 

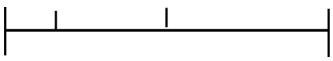
2. 

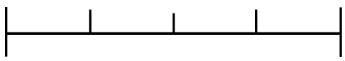
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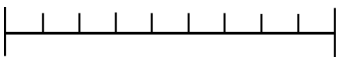
4. 

5. 

6. 

7. 

8. 

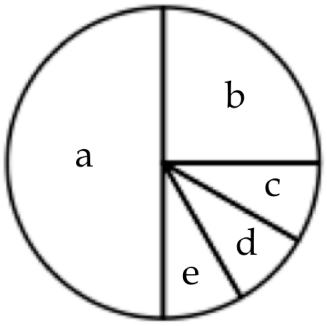
9. 

Name_____ Lesson 4 / Student Page L

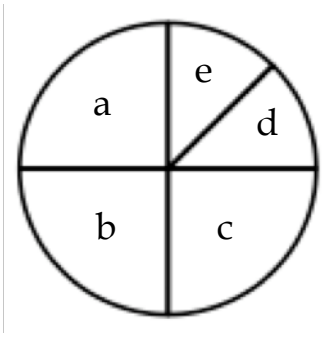
Problem Solving

Directions:

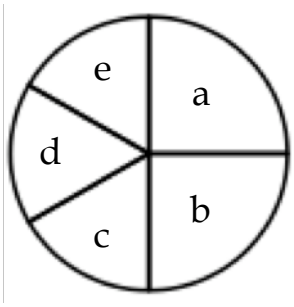
For each of the drawings write the color corresponding to the part marked a, b, c, and so on. Then write the word name for the fraction that the color represents. You can use fraction circles if you need them. Your teacher will help you with exercise 1.



Color	Fractions in Words
a. yellow	1-half
b. blue	1-fourth
c.	
d.	
e.	



Color	Fractions in Words
a.	
b.	
c.	
d.	
e.	



Color	Fractions in Words
a.	
b.	
c.	
d.	
e.	

Rational Number Project

Initial Fraction Ideas **Lesson 5: Overview**	Materials
Students are introduced to fraction symbols by translating from manipulatives to verbal to symbols.	∞ Fraction Circles for students and teacher ∞ Student Pages A – E

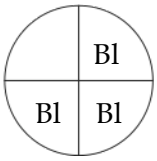
Teaching Actions

Warm Up

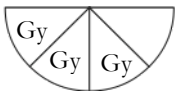
Use paper strips to show these fractions. Which is the largest? $\frac{1}{3}$ $\frac{1}{12}$ $\frac{1}{4}$

Large Group Introduction

1. Ask students to use fraction circles to show 3-fourths. They are to show two models. For example:



3 blues are 3-fourths of 1 black.



3 grays are 3-fourths of 1 yellow.

2. Ask how the two models are alike.

3. Record in words fraction name: 3-fourths. Explain that there is also a symbol name for 3-fourths and it is $\frac{3}{4}$.

4. Discuss the meaning of $\frac{3}{4}$. Ask how many equal parts each unit is divided into? Point to the bottom of the fraction symbol and explain that this4 tells us that. The 3 tells us that we are interested in 3 of

Comments

It's not important for students to memorize the words: numerator and denominator.

It's very important to help children verbalize the meaning of fraction symbols.

Have them talk through what they are doing with the fraction circles.

The action on the manipulative reinforces the meaning of the symbol.

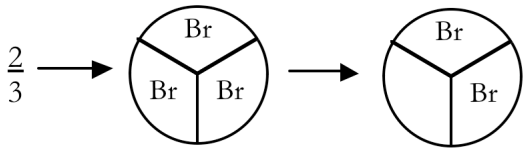
You can also return to previous student pages and have students record answers in symbol form.

Teaching Actions

these 4 equal parts. The fraction means $\frac{1}{4}$ and $\frac{1}{4}$ and $\frac{1}{4}$.

5. Write $\frac{2}{3}$ on the board and ask students to show that fraction with the fraction circles. Have them verbalize why their model does indeed represent $\frac{2}{3}$.

   First divide the whole circle into 3 equal parts …
   then explain



"I divided the circle into 3 equal parts to find what color is thirds. Then I only want two of them so



shows 2 of 3 equal parts. It is $\frac{1}{3}$ and $\frac{1}{3}$ more."

6. Repeat for $\frac{3}{5}, \frac{2}{6}, \frac{4}{8}, \frac{3}{3}$.

   Embed examples in context:

   A spinner for a game was divided into 5 equal parts. 3/5 of the spinner was blue. Show that amount with the fraction circles.

   A pizza was cut into 6 equal parts. You ate 2/6 of the pizza. Show that amount with the fraction circles.

Small Group/Partner Work

7. Student pages that follow reinforce the meaning of the symbol. Select the most appropriate (and amount of) practice that your students need.

Comments

Teaching Actions	Comments
Wrap Up 8. Ask students to describe 2-3 instances that fractions are used in everyday life or in science class. 9. Record situations from these examples that lead to recording a fraction with symbols. For example, to make chocolate chip cookies, you need to use $\frac{3}{4}$ of a cup of brown sugar. Draw a picture of a measuring cup, partition it into 4 equal parts and show $\frac{3}{4}$.	

Translations

- ∞ Manipulative to verbal to written symbols
- ∞ Written symbols to manipulative to verbal
- ∞ Real life to manipulative to written symbols
- ∞ Written symbols to written symbols
- ∞ Written symbols to pictures
- ∞ Pictures to written symbols

Use paper strips to show these fractions. Which is the largest?

$$\frac{1}{3} \qquad \frac{1}{12} \qquad \frac{1}{4}$$

1. Write each fraction in words.

a. $\frac{2}{4}$ _____ e. $\frac{7}{10}$ _____

b. $\frac{3}{7}$ _____ f. $\frac{7}{15}$ _____

c. $\frac{6}{8}$ _____ g. $\frac{3}{12}$ _____

d. $\frac{3}{11}$ _____ h. $\frac{7}{9}$ _____

2. Write the word name and the symbol name for each fraction described.

a. 3 of 5 equal-size parts are shaded. _____ _____

b. 5 of 7 equal-size parts are shaded. _____ _____

c. 3 of 13 equal-size parts are shaded. _____ _____

d. 12 of 17 equal-size parts are shaded. _____ _____

e. 0 of 3 equal-size parts are shaded. _____ _____

3. Write the fraction symbol for each fraction world.

a. 9-tenths _____ e. 13-twenty-firsts _____

b. 7-eights _____ f. 17-eighteenths _____

c. 6-sixths _____ g. 0-fourths _____

d. 15-nineteenths _____

4. Imagine a circle divided into 4 equal parts.

   Three $\frac{1}{4}$ parts are shaded!

   What fraction tells how much is shaded in all? _____

   Draw a picture.

5. Imagine a rectangle divided into 5 equal parts.

   Four $\frac{1}{5}$ parts are shaded!

   What fraction tells how much is shaded in all? _____

   Draw a picture.

6. Write the word name and the symbol name each fraction describes.

   a. A rectangle is folded into 7 equal-size parts.
      5 parts are shaded.

   b. A circle is folded into 8 equal-size parts.
      4 parts are shaded.

Directions:
Match each picture with its symbol or word name by writing the letter of the picture next to its symbol. The first one is done for you.

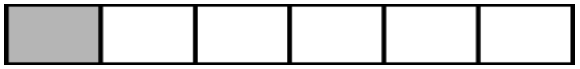
A.  $\frac{1}{6}$ F

B.  2-halves _____

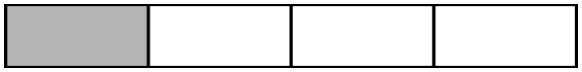
C.  $\frac{3}{4}$ _____

D.  2-thirds _____

E.  $\frac{3}{3}$ _____

F.  1-fourth _____

G.  $\frac{6}{6}$ _____

H.  $\frac{1}{3}$ _____

I.  3-sixths _____

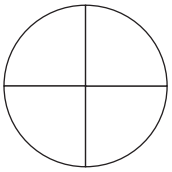
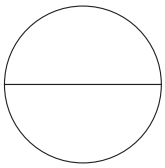
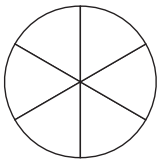
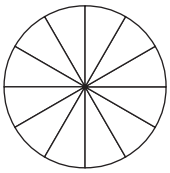
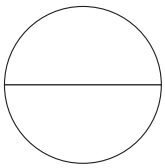
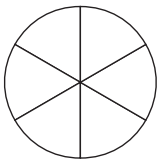
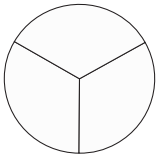
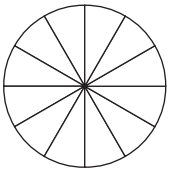
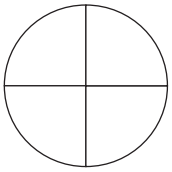
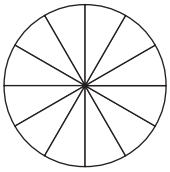
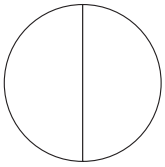
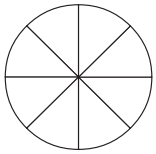
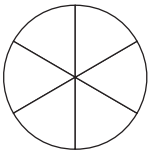
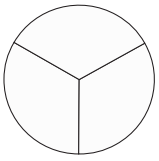
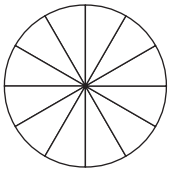
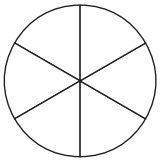
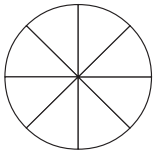
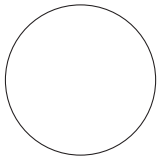
J.  $\frac{4}{6}$ _____

K.  2-fourths _____

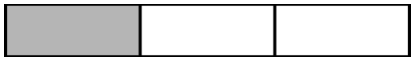
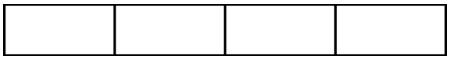
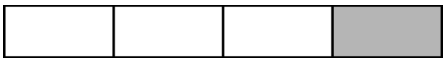
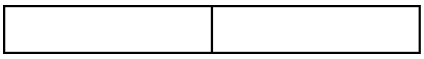
L.  $\frac{1}{2}$ _____

Name_____ Lesson 5 / Student Page D

Shade each circle to show the fractional amount.

$\frac{1}{4}$		$\frac{2}{2}$		$\frac{1}{6}$	
$\frac{5}{12}$		$\frac{0}{2}$		$\frac{5}{6}$	
$\frac{1}{3}$		$\frac{11}{12}$		$\frac{4}{4}$	
$\frac{2}{12}$		$\frac{1}{2}$		$\frac{6}{8}$	
$\frac{1}{6}$		$\frac{0}{3}$		$\frac{6}{12}$	
$\frac{6}{6}$		$\frac{8}{8}$		*You Decide*	

Write the name for the shaded part of each rectangle in words and then in symbols.

1. 1 – half $\frac{1}{2}$ 	9. 
2. 	10. 
3. 	11. 
4. 	12. 
5. 	13. 
6. 	14. 
7. 	15. 
8. 	16. 

Rational Number Project

Initial Fraction Ideas Lesson 6: Overview	Materials
Students observe with circles that as the unit is divided into more and more equal parts, the unit parts become smaller.	∞ Fraction Circles for students and teacher ∞ Student Pages A, B, C

Teaching Actions

Warm Up

Show these fractions with your fraction circles using two different units. Then draw pictures for each display: $\frac{3}{4}$ $\frac{5}{6}$ $\frac{2}{3}$

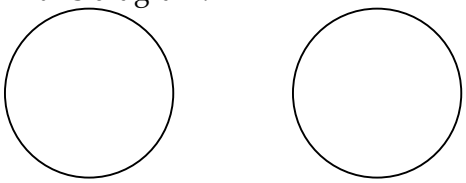
Large Group Introduction

1. Start the lesson by reviewing ordering of whole numbers. For example, ask a student to select the larger of these 2 numbers, 720 or 702, and to explain his/her strategy for doing so.

2. Give another example using a context. José earns $42,175 a year. Mara earns $51,275 a year. Who earns more?

3. Introduce the idea of ordering fractions with this example. Kara entered the Pizza Factory. She saw 2 friends in 1 booth and 3 friends in another booth. Both groups have just been served a large pizza. Which group should she sit with so that she gets the most to eat?

4. Draw this diagram:



5. Ask students to show Kara's share in booth 1 (with 2 friends) and in booth 2 (with 3 friends).

Comments

To think quantitatively about fractions, students should know something about the relative size of fractions. Lesson 6 is the first of several lessons to help students construct informal strategies for ordering fractions. At Level 1, we want to provide the concrete experiences that students need if they are ever to reason intuitively about fraction symbols.

Activities in this lesson will lead students to reason, for example, that 1/4>1/8 because if you divide a circle into 8 equal parts, the parts will be smaller than if you divide the same unit into 4 equal parts.

This notion of more and greater can lead to misunderstandings. Some students may want to say that 1/8 > 1/4 because 8 > 4 or because with eighths, you have <u>more</u> pieces than you do with fourths. Whole-number reasoning has a strong influence on how children think about fractions.

Children need to be reminded that to compare fractions, we look at the "size of piece," not the "number of pieces".

Teaching Actions

6. Which group has the most people? In which group does a person have the smallest share of pizza?

7. Conclude that 1/3 of the pizza is more than 1/4 of the pizza. [Repeat with 6 people at a table; 5 people at a table.]

8. Develop this idea of more implying less, by using Student Page A.

9. Ask students to use their fraction circles as you work together; name the black circle as the unit.

10. Ask: How many brown pieces cover the whole circle? How many orange? Which color takes more pieces to cover the whole unit? Which color has the smaller pieces?

11. Record that information in a chart.

Color	How many cover 1 circle	Which color takes more…	Which color has smaller…
1. Brown	3		
Orange	5	√	√
2.			

12. When completed, ask students if they see any patterns between the number of pieces to fill the whole unit and the size of the pieces.

13. As a group, write a rule similar to either of these:
    - ∞ As the number of pieces needed to fill the whole decreases, the size of each piece gets larger.
    - ∞ As the number of pieces needed to fill the whole increases, the size of each piece gets smaller.

14. Once the rule is generated use it in examples without the circular pieces.

    Examples:
    20 purples = 1 whole
    80 greens = 1 whole
    Which is larger, 1 purple or 1 green?

Comments

Teaching Actions

18 goos = 1 whole
12 boos = 1 whole
Which is smaller, 3 boos or 3 goos?

15. Conclude by asking: Does more always mean less with fractions? Give this example: Imagine that it takes 10 maroon pieces to cover the whole circle. Which is smaller, 2 maroon pieces or 3 maroon pieces? How do you know?

16. Ask: How is this example different from all the rest we've talked about today?

Small Group/Partner Work

17. Assign Practice Pages B and C to reinforce the day's lesson.

Wrap Up

18. Ask students to explain their reasoning for the 2 pairs of fractions on Student Page B they were asked to order without using manipulatives. Ask several students to explain their thinking. Ask students to describe the picture they have in their mind that helps them order these two fraction pairs.

Comments

Students tend to over generalize. This lesson leads children to order fractions with the same numerator, but different denominators (1/3 vs. 1/2, 2/5 vs. 2/10, 40/100 vs. 40/90). The same reasoning will not work for comparing fractions with the same denominator but different numerators (3/4 vs. 2/4).

If a circle is divided into the same size pieces (4ths) the ordering decision is made by looking at the numerator –"2 of same-sized pieces is greater than 2 of same-sized pieces." Now more does mean more. No wonder students have trouble with fractions!

Translations

- ∞ Real life to picture to verbal
- ∞ Manipulative to written symbols to verbal
- ∞ Written symbols to manipulative
- ∞ Real life to manipulative to pictures

Show these fractions with your fraction circles using two different units. Then draw pictures for each display:

$\frac{3}{4}$ $\frac{5}{6}$ $\frac{2}{3}$

Directions: **Use fraction circles to fill in the table.**

Color	How many cover 1 whole circle?	Which color takes MORE pieces to cover 1 whole?	Which color has SMALLER pieces?
1. Brown	**3**		
Orange	**5**	√	√
2. Orange			
White			
3. Purple			
White			
4. Gray			
Green			
5. White			
Green			
6. Orange			
Purple			
7. Gray			
Brown			
8. Brown			
Green			

Name_____ Lesson 6 / Student Page B

Directions:
Use fraction circles to compare the two fractions. Circle the larger fraction.

$\frac{1}{3}$	$\frac{2}{3}$	$\frac{1}{4}$	$\frac{3}{4}$
$\frac{4}{5}$	$\frac{3}{5}$	$\frac{2}{7}$	$\frac{2}{6}$
$\frac{6}{7}$	$\frac{2}{7}$	$\frac{4}{12}$	$\frac{4}{15}$
$\frac{8}{12}$	$\frac{11}{12}$	$\frac{6}{7}$	$\frac{3}{7}$
$\frac{2}{7}$	$\frac{2}{9}$	$\frac{9}{10}$	$\frac{3}{10}$
$\frac{4}{8}$	$\frac{4}{6}$	Try these without manipulatives.	
$\frac{2}{3}$	$\frac{2}{8}$	$\frac{13}{100}$	$\frac{27}{100}$
$\frac{7}{10}$	$\frac{7}{9}$	$\frac{9}{10}$	$\frac{9}{100}$

Use fraction circles to solve problems.

1. Mr. Hickman made a large apple pie. His daughter ate $\frac{1}{2}$ of the pie. His son ate $\frac{1}{3}$ of the pie. Who ate less? Draw a picture to show your thinking.

2. Spinner A was divided into 6 equal parts shaded green. Spinner B was divided into 10 equal parts with 4 parts shaded green. Which spinner had the larger amount of green? Explain "in your own words" your reasoning.

3. Jessica and Kim shared a large pizza. Jessica ate $\frac{2}{6}$ of a pizza. Kim ate $\frac{3}{6}$ of the pizza. Who ate more? Draw a picture to show your thinking?

4. Mathew and Cassandra shared a bag of candy. Mathew ate $\frac{2}{3}$. Cassandra ate $\frac{2}{5}$. Who ate more? Explain your thinking.

5. Andrew spent $\frac{1}{2}$ of his allowance on candy. Ellen spent $\frac{1}{3}$ of her allowance on a movie. Is it possible that Ellen spent more than Andrew? Explain. [Use the back of the page].

Rational Number Project

Initial Fraction Ideas Lesson 7: Overview	Materials
This lesson reinforces the idea that as the number of parts the unit is divided into increases, the size of the parts decreases.	∞ 8.5" X 1" strips of paper for each student and teacher ∞ Student Pages A, B, C

Teaching Actions

Warm Up

Order these fraction pairs. Write an explanation for each pair (use pictures in your explanation).

$\frac{3}{4}\ \frac{3}{10}$ $\frac{5}{7}\ \frac{3}{7}$ $\frac{1}{9}\ \frac{1}{4}$

Large Group Introduction

1. Ask children to fold a strip of paper into 4 equal parts. Using the same strip of paper ask them how they can increase the number of equal parts to 8. Have them do so, but before they open up the strip of paper to show eighths ask: Before you open up the strip, can you tell me if the size of the equal parts will be larger or smaller than fourths? Why?

2. Repeat for:
   - ∞ 3rds changed to 6ths
   - ∞ Then to 12ths
   - ∞ 4ths changed to 12ths

3. Now ask students to fold, shade, and label these fractions with paper folding:

$\frac{1}{3}$ $\frac{1}{4}$
$\frac{2}{3}$ $\frac{2}{6}$
$\frac{3}{4}$ $\frac{1}{4}$

[Do more if needed]

Comments

Children need opportunities to use new ideas in order to ensure they internalize them.

Many experiences with physical models are needed to overcome the influence of children's whole number thinking.

In this lesson students use paper folding to reexamine the relationship between size of piece and number of pieces the whole is divided into.

Encourage children to explain their ordering. Don't let them refer to only one part of the fraction, as for example: 1/3 vs. 1/4 "thirds are bigger". Thirds may be bigger, but that information is enough to order 2 fractions only if the numerators are the same. "Thirds are bigger so 1 of a larger piece is greater than 1 of a smaller pieces." By talking like this children are coordinating numerator and denominator to approximate the size of the fraction. You want to build the notion of a fraction as a single entity!

Students may over generalize and think bigger is always more. Check for this.

Teaching Actions	Comments
4. You may want to refer back to the pizza problem from lesson 6. Model with paper folding or pictures the answer to the question in that story.	Some children may be able to compare without manipulatives $\frac{1}{3}$ vs. $\frac{1}{5}$; $\frac{2}{10}$ vs. $\frac{2}{20}$
Small Group/Partner Work	but there is no need to push abstraction at this level.
5. Put students in pairs and assign Student Page A. Student 1 will make fraction 1 with paper folding; student 2 will make fraction 2. They will then compare and circle the larger fraction.	Some students may try to compare fractions without the manipulatives and make errors. Encourage them to use paper folding at least to verify their guesses.
6. Student Pages B and C offer extra practice.	
Wrap Up	
7. Conclude the lesson by first asking children to create their own context for comparing 2 fractions. Examples: Mary had 2/4 of large pizza; Joan had 2/4 of large pizza. Who ate more? Lianna ate 4/8 parts of a candy bar Rodrigo ate 4/7 of same-sized candy bar. Who ate more?	Challenge Student Page B: The problem here is that the two units are not the same. 1/5 < /2 only if the two units are the same. Comparing fractions assumes equal units.
8. Now ask students this question: Jose and Mara both ate ½ of a pizza. Jose said he ate more than Mara. Mara said they ate the same amount. Could Jose be correct?	This problem is similar to a NAEP item given to 4th graders. Only 24% were able to explain that if Jose's pizza was larger than Mara's then his ½ would be more.

Translations

- ∞ Written symbols to manipulative to verbal

Order these fraction pairs. Write an explanation for each pair (use pictures in your explanation).

$\frac{3}{4}$ $\frac{3}{10}$

$\frac{5}{7}$ $\frac{3}{7}$

$\frac{1}{9}$ $\frac{1}{4}$

Name_____ Lesson 7/ Student Page A

Directions:
Circle the larger fraction. Use your paper strips to determine the answers.

$\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{6}$	$\frac{2}{12}$
$\frac{2}{4}$	$\frac{2}{6}$	$\frac{1}{3}$	$\frac{2}{3}$
$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{9}$	$\frac{3}{3}$
$\frac{3}{8}$	$\frac{3}{4}$	$\frac{5}{6}$	$\frac{4}{6}$
$\frac{1}{3}$	$\frac{1}{12}$	$\frac{5}{12}$	$\frac{5}{8}$
$\frac{1}{2}$	$\frac{1}{12}$		

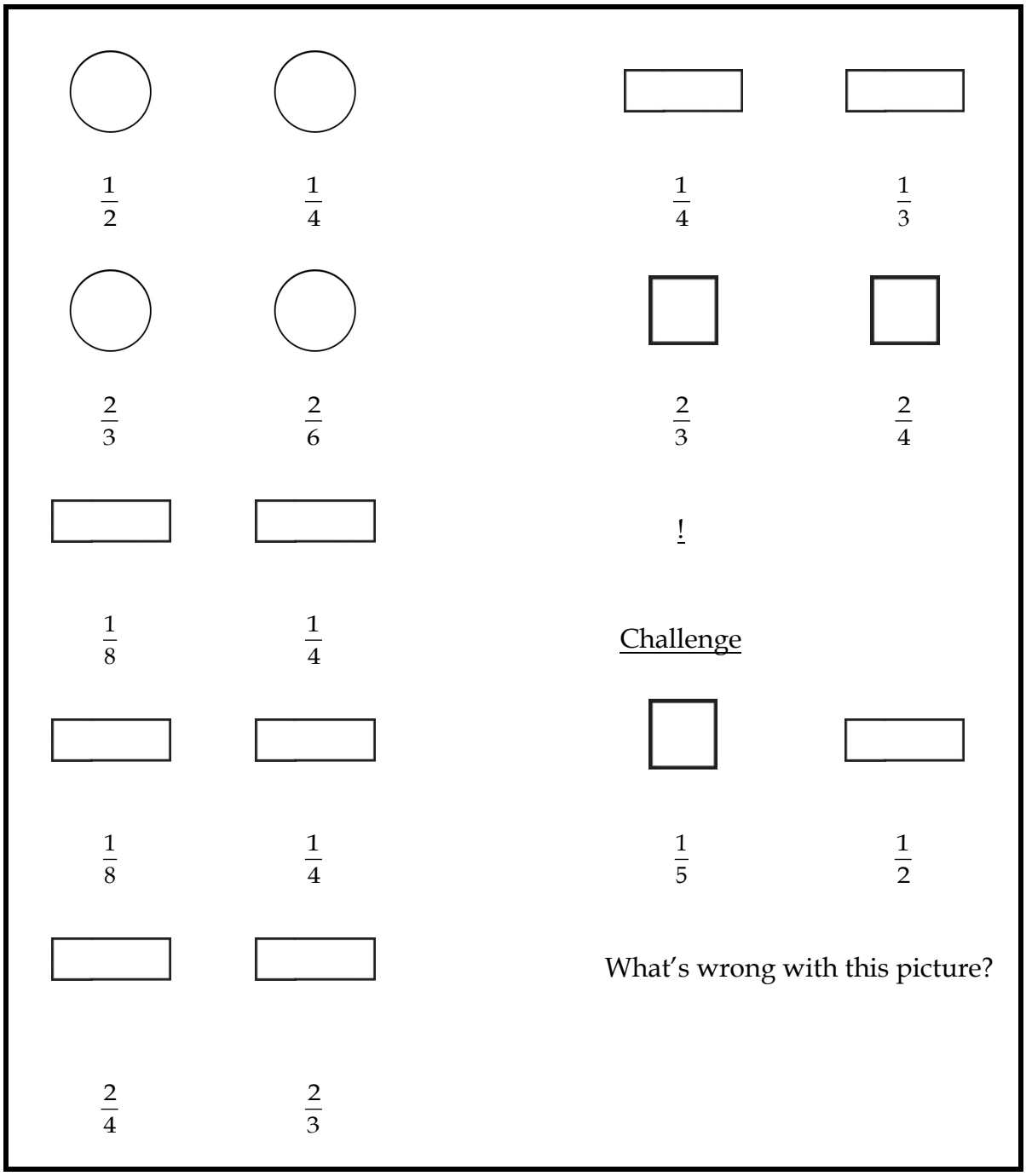
Without your paper strips, circle the larger fractions.

$\frac{1}{100}$	$\frac{1}{99}$
$\frac{3}{40}$	$\frac{3}{50}$
$\frac{2}{10}$	$\frac{4}{10}$

$\frac{5}{12}$	$\frac{8}{12}$

Name_____ Lesson 7/ Student Page B

Directions:
Shade each picture to show the fraction. Circle the SMALLER fraction.



Directions

A friend has been out of school for two days and missed the math lessons dealing with comparing fractions. Write your friend a letter explaining how to compare fractions like the ones you have been working with. [You may want to draw pictures.]

Rational Number Project

Initial Fraction Ideas
Lesson 8: Overview

Students explore fraction equivalence by naming fractions equal to 1/2 with fraction circles and by finding other fraction equivalences with fraction circles.

Materials

- Fraction Circles for students and teacher
- Large sheet of chart paper for teacher; Equivalence Chart for students
- Fraction Fill board for students and numeral cards for teacher
- Circle page divided into twelfths

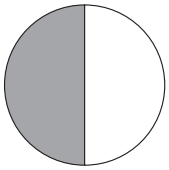
Teaching Actions

Warm Up

Use paper folding strips to show these two fractions: $\frac{3}{4}$ and $\frac{11}{12}$. Compare the strips – which fraction is larger? Why?

Large Group Introduction

1. At the overhead, cover the whole circle with 1 yellow and ask students to find different ways to cover the remaining half of the circle. Record answers by color.

Ex: 

1 blue and 2 grays
2 blues
•
•
•
•
•

2. Repeat this activity, but this time, specify that they have to use one color to cover half the circle. Record results by color and fraction name.

1 yellow	=	2 blues
1 yellow	=	3 pinks

3. Ask what each display has in common. [They all cover

Comments

The idea of equivalence is a prerequisite for fraction operations. To add $\frac{1}{2}+\frac{3}{4}$, you will explain that $\frac{1}{2}$ can be exchanged for $\frac{2}{4}$ because $\frac{1}{2}=\frac{2}{4}$.

Equality should first be developed from concrete models before explaining a rule that generates equal fractions $\left(\frac{1}{2}\times\frac{2}{2}=\frac{2}{4}\right)$

You are defining equivalent fractions by showing that fractions are equivalent if they cover the same amount of the circle. Partitioning is different so the digits in the fraction symbols are different; but 1 of 2 equal parts covers the same amount as 2 of 4 equal parts.

Teaching Actions

1-half of the black circle therefore they are all equivalent].

4. Continue to model fraction equivalences by completing the equivalence chart. Make a large classroom chart of this picture or use a transparency of student's chart.

Red	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{3}{12}$	$\frac{4}{12}$	$\frac{5}{12}$	$\frac{6}{12}$	$\frac{7}{12}$	$\frac{8}{12}$	$\frac{9}{12}$	$\frac{10}{12}$	$\frac{11}{12}$	$\frac{12}{12}$
White												
Gray												
Pink												
Blue												
Brown												
Yellow												

5. Start by asking students to cover 2/12 of the whole circle. Now ask them if they can cover the same amount with whites (without cutting the pieces). Check other colors. Because 1 pink equals 2 reds, record on the chart, 1/6 under the 2/12 column across from "pink".

Red	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{3}{12}$	$\frac{4}{12}$	$\frac{5}{12}$	$\frac{6}{12}$	$\frac{7}{12}$	$\frac{8}{12}$	$\frac{9}{12}$	$\frac{10}{12}$	$\frac{11}{12}$	$\frac{12}{12}$
White												
Gray			$\frac{2}{8}$									
Pink		$\frac{1}{6}$										
Blue			$\frac{1}{4}$									
Brown												
Yellow												

6. Repeat for 3/12, recording in fraction symbols 2/8 and 1/4 under the 3/12 column [see above].

Small Group/Partner Work

7. Continue, completing the rest of the chart. Students can do this in groups or with a partner.

Comments

Give students a copy of the master for the circle divided into 12ths for this activity. Students try to cover $\frac{2}{12}$ of the circle by placing fraction pieces on top of this circle page.

Teachers have found that filling in the equivalence chart to be easy for some and difficult for others. Consider completing the chart in pairs.

Teaching Actions

Red	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{3}{12}$	$\frac{4}{12}$	$\frac{5}{12}$	$\frac{6}{12}$	$\frac{7}{12}$	$\frac{8}{12}$	$\frac{9}{12}$	$\frac{10}{12}$	$\frac{11}{12}$	$\frac{12}{12}$
White				$\frac{3}{9}$				$\frac{6}{9}$				$\frac{9}{9}$
Gray			$\frac{2}{8}$			$\frac{4}{8}$			$\frac{6}{8}$			$\frac{8}{8}$
Pink		$\frac{1}{6}$		$\frac{2}{6}$		$\frac{3}{6}$		$\frac{4}{6}$		$\frac{5}{6}$		$\frac{6}{6}$
Blue			$\frac{1}{4}$			$\frac{2}{4}$			$\frac{3}{4}$			$\frac{4}{4}$
Brown						$\frac{1}{2}$						$\frac{2}{2}$
Yellow												

8. Use this chart as a reference for fraction equivalencies. Students can use it as they play the following game. If you are making a classroom chart then after students complete their own charts take time to share their work to fill out the classroom chart. Discuss how the chart shows fraction equivalences.

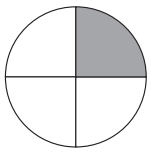
Wrap Up

9. Play the Fraction Fill game. Students use their equivalence chart to help them find equivalences.

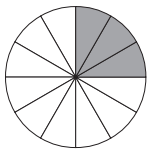
MATERIALS**:** Fraction numeral cards**
Fraction Fill Board

DIRECTIONS**:** Teacher randomly selects a numeral card and shows it to students. Students choose to shade that amount on one of the circles. They can only shade 1 representation for that fraction amount.

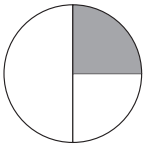
Ex: 1/4 Student can shade



or



but not both



Student cannot shade the circle divided into two equal parts by adding lines to the circle to divide it into fourths.

Comments

One-half equivalences are the most important ones for children to learn. Students with good "fraction number sense" use $\frac{1}{2}$ as a reference point for estimating the size of other fractions.

A student will use his/her concept of $\frac{1}{2}$ to estimate, for example, $\frac{2}{4}+\frac{2}{3}$: "$\frac{2}{4}$ equals $\frac{1}{2}$; $\frac{2}{3}$ is greater than $\frac{1}{2}$; so the sum is greater than 1."

This reasoning is an excellent example of thinking quantitatively. Thinking quantitatively is not rule-bound, but relies on the mental images children have for fractions. Images for $\frac{1}{2}$ are critical.

Just as students learn the basic +, -, x, ÷ facts, you want them to learn basic fraction equivalents for $\frac{1}{2},\frac{1}{3},\frac{2}{3},\frac{1}{4},\frac{2}{4},\frac{3}{4}$. On this level, they don't need rules, but many physical/visual examples with these fractions and their equivalents.

Fraction Fill is a game that can be played throughout the rest of the lessons.

Teaching Actions	Comments
Continue showing numeral cards. Students refer to equivalence chart to make selections. The first to shade two complete circles says "Fraction Fill." ** Make a set of cards for $\frac{1}{2},\frac{1}{3},\frac{2}{3},\frac{1}{4},\frac{2}{4},\frac{3}{4},\frac{1}{6},\frac{2}{6},\frac{3}{6},\frac{4}{6},\frac{5}{6},$ $\frac{1}{12},\frac{2}{12},\frac{3}{12},\frac{4}{12},\frac{5}{12},\frac{6}{12},\frac{7}{12},\frac{8}{12},\frac{9}{12},\frac{10}{12},\frac{11}{12}$	

Translations

- ∞ Written symbols to manipulative to written symbols
- ∞ Written symbols to pictures

$\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$
$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	
$\frac{1}{12}$	$\frac{2}{12}$	$\frac{3}{12}$	$\frac{4}{12}$	$\frac{5}{12}$	$\frac{1}{12}$
$\frac{7}{12}$	$\frac{8}{12}$	$\frac{9}{12}$	$\frac{10}{12}$	$\frac{11}{12}$	

Use paper folding strips to show these two fractions:

$$\frac{3}{4} \text{ and } \frac{11}{12}$$

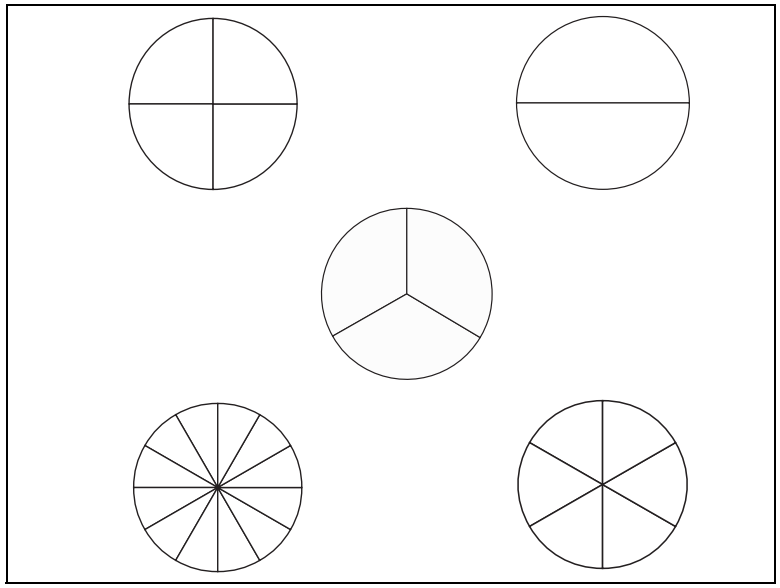
Compare the strips – which fraction is larger? Why?

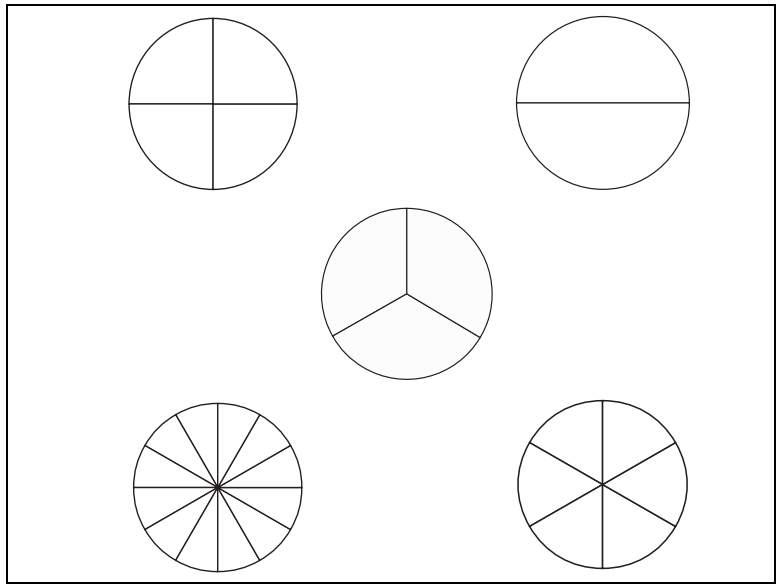
Name_____ Lesson 8 / Student Page A

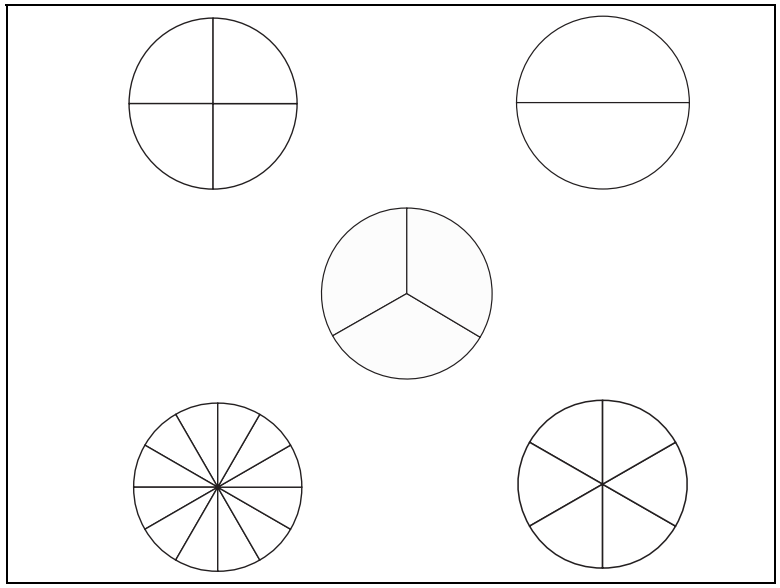
Red	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{3}{12}$	$\frac{4}{12}$	$\frac{5}{12}$	$\frac{6}{12}$	$\frac{7}{12}$	$\frac{8}{12}$	$\frac{9}{12}$	$\frac{10}{12}$	$\frac{11}{12}$	$\frac{12}{12}$
White												
Gray												
Pink												
Blue												
Brown												
Yellow												

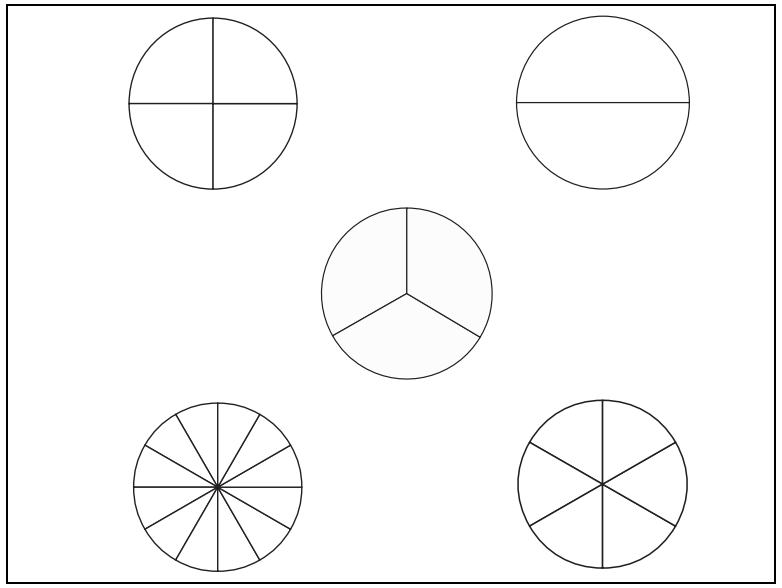
Name_____

Fraction Fill

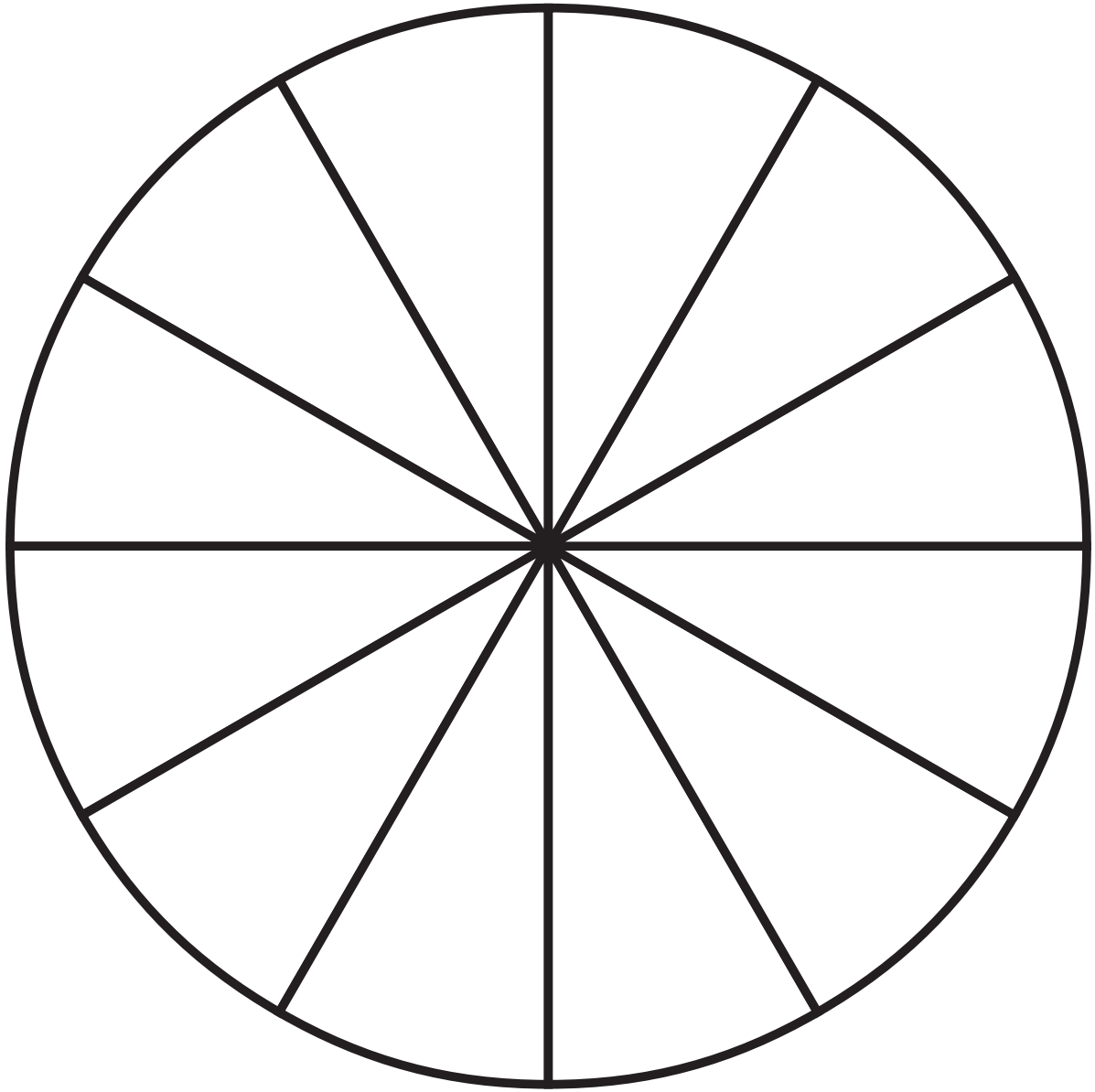








Name_____



Rational Number Project

Initial Fraction Ideas Lesson 9: Overview	Materials
Students continue to explore equivalence with pictures and fraction circles.	∞ Transparencies 1 & 2 ∞ Student Pages A, B, C ∞ Fraction Circles for students

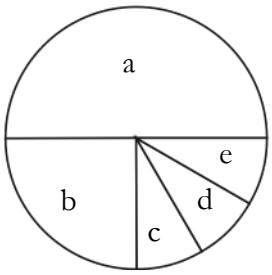
Teaching Actions

Warm Up

Joey and Ty each had a Hershey's candy bar. Joey ate $\frac{6}{8}$ of his candy bar while Ty ate $\frac{3}{4}$. Who ate more?

Large Group Introduction

1. Show transparency 1 to the class.



2. Ask students to name section **a**; section **b**; section **c**. [Also ask what color fraction-circle piece matches each part]. Have them explain their reasoning.

3. Ask students if fractional parts can have more than one name. Ask students to name section **a** in two different ways. Record on the transparency what they say with words and/or symbols:

   Examples:

   1 yellow = $\frac{1}{2}$;

   1 blue = $\frac{1}{4}$

   1 yellow = 2 blues: $\frac{1}{2} = \frac{2}{4}$

Comments

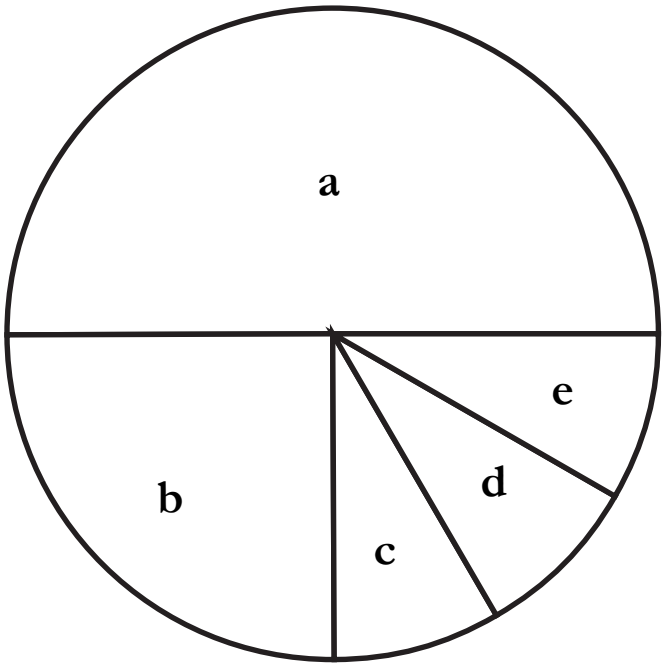
Seeing equivalence from pictures is not the same as seeing it with manipulatives. Some children are better at adding and taking out lines drawn in a diagram. Don't be surprised to see differences in how children respond to these pictures.

Teaching Actions	Comments
4. Point to the section (c + d + e). Ask: How are **b** and (**c + d + e**) alike? *[Cover the same amount]*	
5. As a group write sentences using colors and symbols that describe equivalences in the picture. Examples 1 blue = 3 reds; 1/4 = 3/12 1 blue and 3 reds = 1 yellow; ¼ + 3/12 = 1/2 6 reds = 1 yellow; 6/12 = 1/2	
6. Show transparency 2 to the class and talk through the naming of each part: a, b, c, (b + c), d, (d + e), (d + e+ f + g) in several ways. Record symbolic sentences. Examples: a = $\frac{1}{6}$; b = $\frac{2}{6}$; (b+c) = $\frac{1}{2}$; c = (d+e); $\frac{1}{6} = \frac{4}{12}$	
7. Repeat for the second rectangle at the bottom of the page.	
Small Group/Partner Work	Note: Problem 1 is already completed; this was the same as the problem on Transparency 1.
8. Assign in pairs Student Pages A, B, C. For problems 1, 2 and 3, children refer to their fraction circles; for the last 3 problems, children rely on diagrams. They may need to draw on the pictures. Encourage them to do so.	
Wrap Up	
9. Ask students to come to the board and share their strategies for solving problems on Student Page C.	

Translations

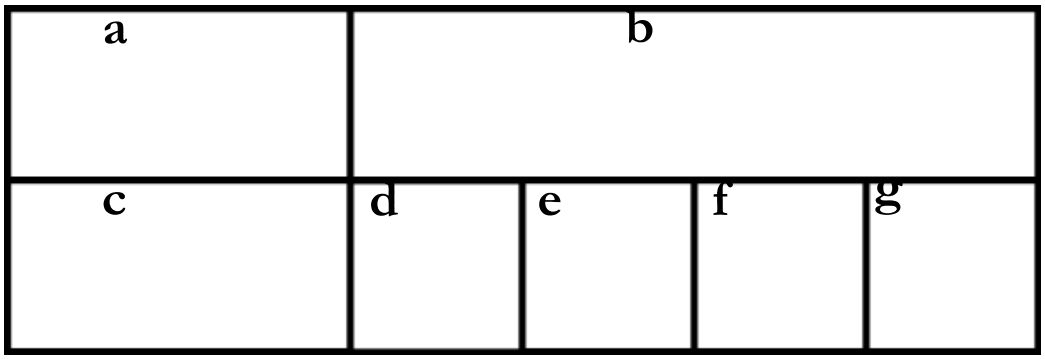
- ∞ Pictures to verbal to written symbols
- ∞ Pictures to manipulative to written symbols

Transparency 1

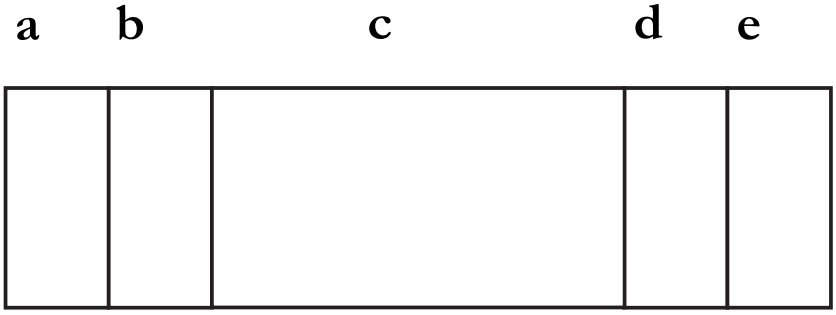


Sentences I can write about the parts:

Transparency 2



Sentences I can write about the parts:



Sentences I can write about the parts:

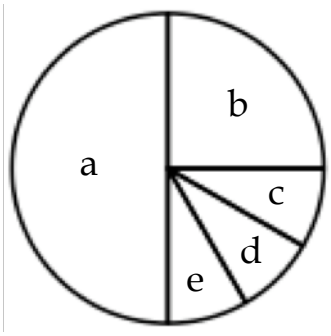
Joey and Ty each had a Hershey's candy bar. Joey ate $\frac{6}{8}$ of his candy bar while Ty ate $\frac{3}{4}$. Who ate more? Explain your thinking.

Problem Solving

Directions:

For each of the drawings write the color corresponding to the part marked a, b, c, and so on. Then write a sentence that is true about all of the color-coded parts altogether. User your fraction circles to help you, if you need them.

1.



Color	Fractional Part of Whole Circle
a. yellow	$\frac{1}{2}$
b. blue	$\frac{1}{4}$
c. red	$\frac{1}{12}$
d. red	$\frac{1}{12}$
e. red	$\frac{1}{12}$

Sentences I can write about the parts:

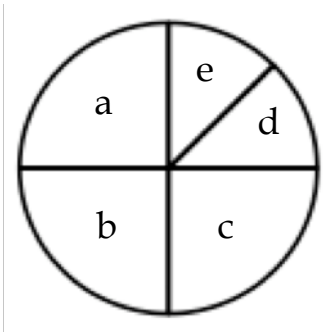
a) 1 yellow and 1 blue and 3 reds equal 1 whole circle.
$\frac{1}{2}$ and $\frac{1}{4}$ and $\frac{3}{12}$ = 1 whole.

b) 1 blue and 3 reds equal 1 yellow. $\frac{1}{4}$ and $\frac{3}{12} = \frac{1}{2}$.

c) 3 reds equal 1 blue. $\frac{3}{12} = \frac{1}{4}$.

d) 6 reds equal 1 yellow. $\frac{6}{12} = \frac{1}{2}$.

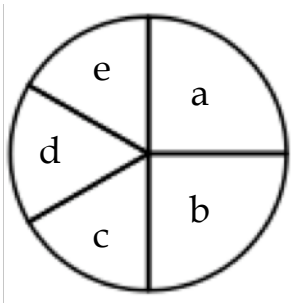
2.



Color	Fractional Part of Whole Circle
a.	
b.	
c.	
d.	
e.	

Sentences I can write about the parts:

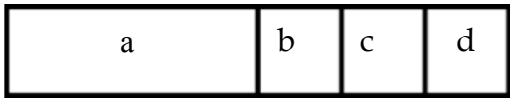
3.



Color	Fractional Part of Whole Circle
a.	
b.	
c.	
d.	
e.	

Sentences I can write about the parts:

4.



Fractional Part of Rectangle

a. _____

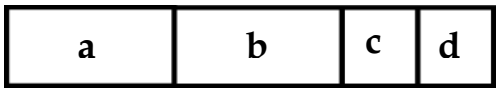
b. _____

c. _____

d. _____

Sentences I can write about the parts:

5.

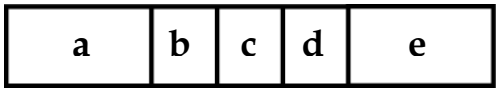


Fractional Part of Rectangle

a. _____
b. _____
c. _____
d. _____

Sentences I can write about the parts:

6.



Fractional Part of Rectangle

a. _____
b. _____
c. _____
d. _____
e. _____

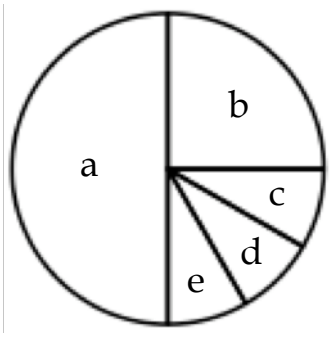
Sentences I can write about the parts:

Problem Solving

Directions:

For each of the drawings write the color corresponding to the part marked a, b, c, and so on. Then write a sentence that is true about all of the color-coded parts altogether. User your fraction circles to help you, if you need them.

1.



Color	Fractional Part of Whole Circle
a. yellow	$\frac{1}{2}$
b. blue	$\frac{1}{4}$
c. red	$\frac{1}{12}$
d. red	$\frac{1}{12}$
e. red	$\frac{1}{12}$

Sentences I can write about the parts:

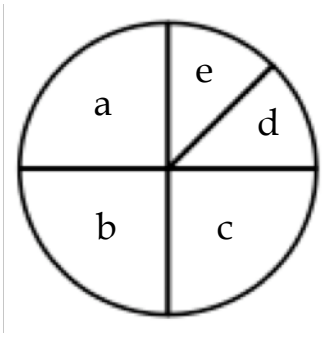
a) 1 yellow and 1 blue and 3 reds equal 1 whole circle.
$\frac{1}{2}$ and $\frac{1}{4}$ and $\frac{3}{12}$ = 1 whole.

b) 1 blue and 3 reds equal 1 yellow. $\frac{1}{4}$ and $\frac{3}{12}=\frac{1}{2}$.

c) 3 reds equal 1 blue. $\frac{3}{12}=\frac{1}{4}$.

d) 6 reds equal 1 yellow. $\frac{6}{12}=\frac{1}{2}$.

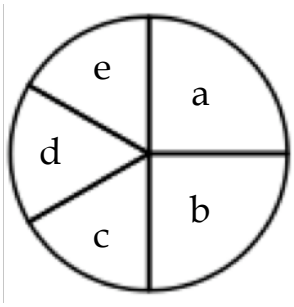
2.



Color	Fractional Part of Whole Circle
a.	
b.	
c.	
d.	
e.	

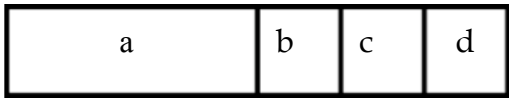
Sentences I can write about the parts:

3.



Color	Fractional Part of Whole Circle
a.	
b.	
c.	
d.	
e.	

Sentences I can write about the parts:



Fractional Part of Rectangle

a. _____

b. _____

c. _____

d. _____

Sentences I can write about the parts:

5.

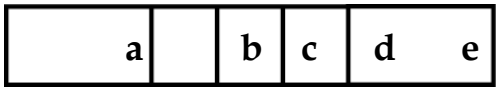


Fractional Part of Rectangle

a. _____
b. _____
c. _____
d. _____

Sentences I can write about the parts:

6.



Fractional Part of Rectangle

a. _____
b. _____
c. _____
d. _____
e. _____

Sentences I can write about the parts:

Rational Number Project

Initial Fraction Ideas Lesson 10: Overview	Materials
Students explore equivalence ideas with paper folding.	∞ Scissors and glue ∞ Strips of paper (8.5″ x 1″) for folding for each student. ∞ Student Pages A-F

Teaching Actions

Warm Up

Using your fraction circles find three fractions equal to $\frac{1}{4}$. Draw a picture for each equivalent fraction. Record the equivalent fraction under each picture.

Large Group Introduction

1. Throughout this activity, teacher and students do the examples. Teacher may choose to use larger strips of paper for demonstration purposes.

2. Ask students to fold strips of paper into thirds and shade $\frac{2}{3}$ of the paper. Write the symbol for amount shaded on that strip.

3. Now have students fold the same strip to show 6 equal parts. Before they actually open up the folded paper, ask them to guess the number of shaded parts.

4. Open up the amount and record on the paper strip.

5. Ask: Do you have more than 1 fraction written on your paper? Explain why.

6. Ask students to refold and create 12 equal parts. Unfold and note how many total equal parts; how many parts are shaded? Record that amount.

Comments

Students will benefit from seeing equivalent fractions with more than one manipulative.

Remember all this work with manipulatives is an investment that will pay off later as children learn to operate with fractions. The manipulative experiences will give them the mental images they need to operate (+, -, x, √) on fractions in a meaningful way.

Notes regarding Student Pages:

B and C: After cutting out rectangles and sorting into equivalent groups you may want students to paste groups onto paper and record fraction names on that paper, omitting Student Page C.

D and E: Students may want to just draw lines on the paper strips instead of folding.

Teaching Actions	Comments
7. Ask: In what way are $\frac{2}{3}$, $\frac{4}{6}$, and $\frac{8}{12}$ alike? Different?	
8. Replicate the above starting with $\frac{1}{2}$; go to $\frac{2}{4}$, $\frac{4}{8}$	
Small Group/Partner Work	
9. Practice is provided in Student Pages A-F	
Wrap Up	
10. Record on the board this set of equivalences for 1-half: $\frac{1}{2}=\frac{2}{4}$; $\frac{1}{2}=\frac{3}{6}$; $\frac{1}{2}=\frac{5}{10}$. Ask students if they see any interesting number patterns in fraction pairs equal to ½.	Students may notice that for 1-half, the denominator is twice the size of the numerator; they may notice that that if the numerator is doubled the denominator is doubled or if the numerator is tripled then the denominator is tripled. You are not formalizing the symbolic rule for equivalence, but just helping students start to notice the multiplicative nature of fractions starting with ½.

Translations

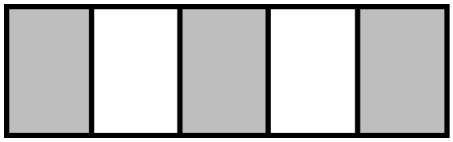
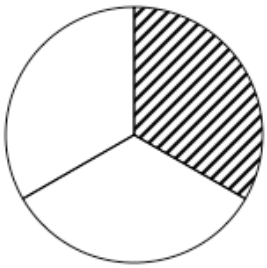
- ∞ Written symbols to manipulative to written symbols
- ∞ Pictures to written symbols

Using your fraction circles find three fractions equal to $\frac{1}{4}$.

Draw a picture for each equivalent fraction.

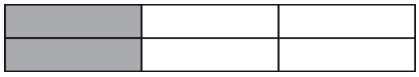
Record the equivalent fraction under each picture.

Name_____ Lesson 10 / Student Page A

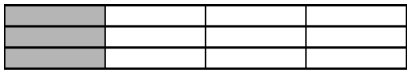
1. 	A. Fraction shaded: _____ B. Make into six equal-sized parts drawing in lines on picture. C. Fraction shaded now: _____
2. 	A. Fraction shaded: _____ B. Make into 15 equal-sized parts drawing in lines on picture. C. Fraction shaded now: _____
3. 	A. Fraction shaded: _____ B. Make into 12 equal-sized parts drawing in lines on picture. C. Fraction shaded now: _____
4. 	A. Fraction shaded: _____ B. Make into six equal-sized parts drawing in lines on picture. C. Fraction shaded now: _____

Directions:
Cut out the rectangles. Put the rectangles together into groups so that each rectangle in the group has the same amount shaded. After you have grouped them, fill in the table by writing the fraction name for each rectangle in the group. Fractions in the same group are called equivalent fractions.

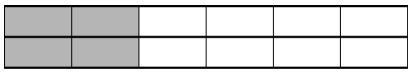
a) 

b) 

c) 

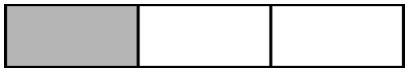
d) 

e) 

f) 

g) 

h) 

i) 

j) 

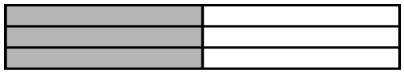
k) 

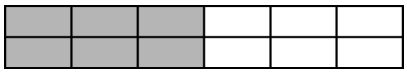
l) 

m) 

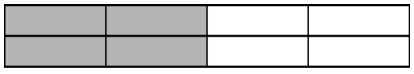
n) 

o) 

p) 

q) 

r) 

s) 

t) 

Name_____ Lesson 10 / Student Page C

Record Sheet for Student Page B

Write equivalent fractions for each group you made.

$\frac{1}{2}$				

$\frac{1}{3}$			

$\frac{2}{3}$		

$\frac{3}{4}$		

$\frac{5}{6}$		

PAPER FOLDING AND EQUIVALENT FRACTIONS

Cut the strips on Student page E so you can fold them to solve these problems.

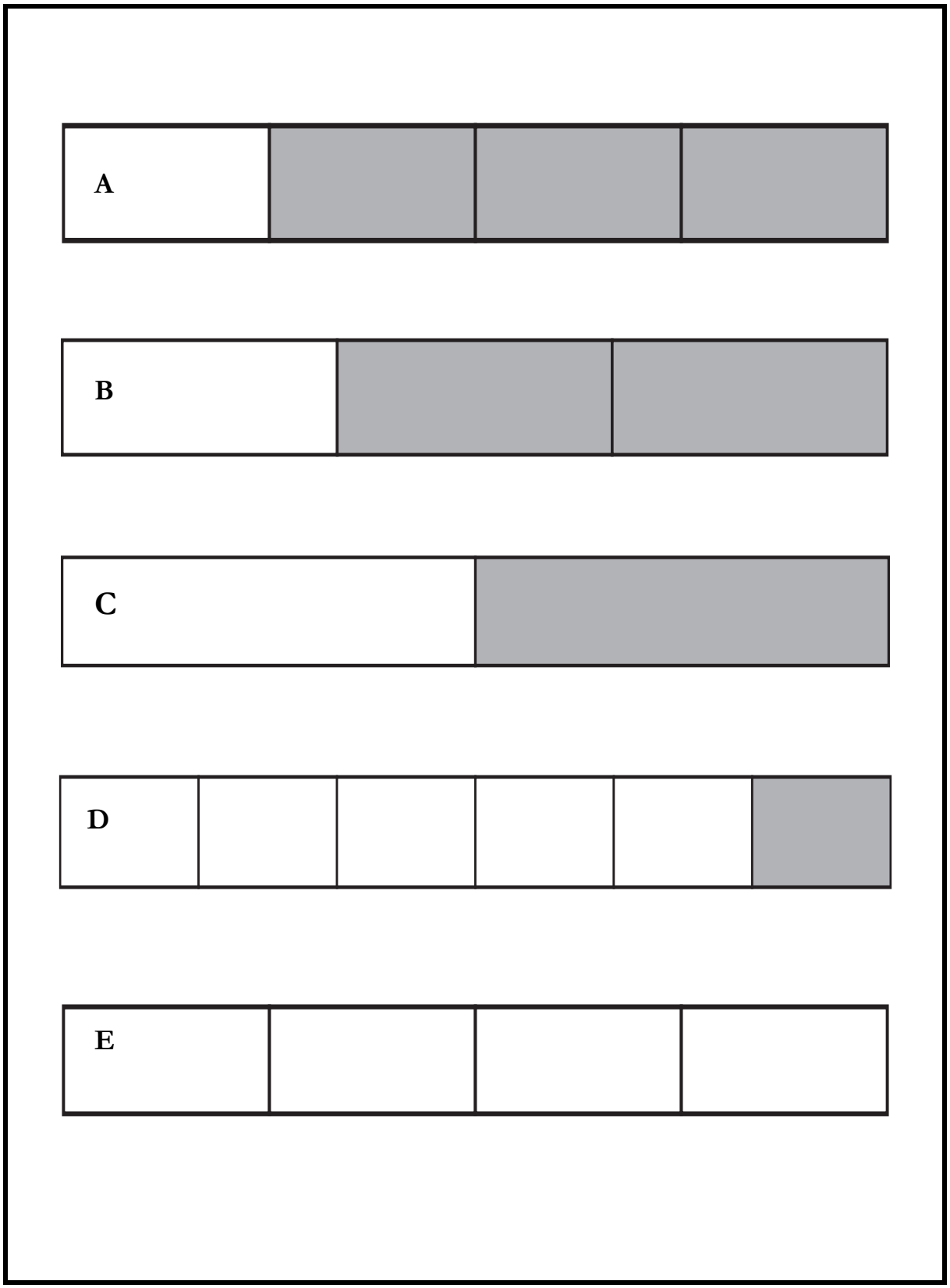
1. Write the symbol for the fraction shaded on strip A:
   _____
   Fold to make **8 equal sized parts.**
   Write the symbol for the fraction which is shown: _____

2. Write the symbol for the fraction shaded on strip B:
   _____
   Fold to make **12 equal sized parts.**
   Write the symbol for the fraction which is shown: _____

3. Write the symbol for the fraction shaded on strip C:
   _____
   Fold to make **6 equal sized parts.**
   Write the symbol for the fraction which is shown: _____

4. Write the symbol for the fraction shaded on strip D:
   _____
   Fold to make **12 equal sized parts.**
   Write the symbol for the fraction which is shown: _____

5. Be careful! Think! Take folding strip E.
   Fold to make 3 equal parts.
   Shade 2 of the 3 equal parts.
   Write the fraction for the amount shaded in strip E:
   _____


A
B
C
D
E

Paper Folding and Equivalent Fractions

1) Show 3 fractions equal to 1/2. (Hint: you will need to start with 3 sheets of paper folded into 2 equal parts with one part shaded. Draw a picture to show your answers.

2) Use paper folding to find out which of these are true statements.

$\frac{1}{3}=\frac{2}{6}$ $\frac{1}{4}=\frac{2}{6}$

$\frac{3}{4}=\frac{9}{12}$ $\frac{4}{6}=\frac{2}{3}$

3) Use paper folding to find these equivalences..

$\frac{1}{2}=\frac{}{8}$ $\frac{2}{6}=\frac{}{3}$

$\frac{1}{4}=\frac{}{12}$ $\frac{3}{6}=\frac{}{2}$

Rational Number Project

Initial Fraction Ideas Lesson 11: Overview	Materials
Students use fraction circles to order 2 fractions by comparing them to one-half.	∞ Fraction Circles for students and teacher ∞ Student Pages A, B

Teaching Actions

Warm Up

Draw a picture of paper folding strips to show the fraction $\frac{2}{3}$. Now partition your picture to show how many ninths equal $\frac{2}{3}$.

Large Group Introduction

1. Ask students to take out the black circle and to cover one-half of the circle with 1 yellow.

2. Show on the overhead that 3 blues, which is 3/4 of the black, is greater than 1 yellow ($\frac{1}{2}$ of the black).

   Record: 3 blues > 1 yellow so
   $$\frac{3}{4} > \frac{1}{2}$$

3. Ask students to find 4 other fractions greater than $\frac{1}{2}$. Model and record their responses on the overhead.

4. Now ask them to imagine fraction pieces greater than 1 yellow or $\frac{1}{2}$ of the circle. Have them write down at least 3 estimates for amounts greater than $\frac{1}{2}$. Encourage students to share their estimates and explain what they thought of or pictured.

   Ex: A child may say, "I can see that 3 pinks are the same as 1 yellow, so 5 pinks must be greater than $\frac{1}{2}$."

Comments

Students need many experiences with concrete materials to develop mental images of fractions so they can develop a quantitative notion of fraction.

Comparing to $\frac{1}{2}$ is a powerful strategy for judging the relative size of fractions and is a characteristic of having a quantitative notion of fraction.

Looking at specific numerical relationships between numerator and denominator to determine if fractions are greater or less than $\frac{1}{2}$ is not the goal for all students. Some students may show that see number patterns for $\frac{1}{2}$.

We encourage students to rely on their mental images related to the fraction circles or paper folding to guide their ordering strategies.

Teaching Actions

5. Have students verify each guess with their circles and record results with fraction notation.

Small Group /Partner Work

6. Student Page A provides independent practice with circles comparing fractions to $\frac{1}{2}$.

7. Student Page B provides more practice with ordering and equivalence ideas developed so far.

Wrap Up

8. End class by presenting these problems for discussion. Emphasize student verbalization of their thinking as they order these fractions. They may or may not use the circles.

   Which is bigger or are they equal?

   Examples:

$\frac{1}{3}$	$\frac{3}{4}$
$\frac{1}{4}$	$\frac{1}{3}$
$\frac{6}{7}$	$\frac{3}{7}$
$\frac{4}{100}$	$\frac{4}{70}$
$\frac{6}{8}$	$\frac{4}{6}$
$\frac{4}{12}$	$\frac{2}{4}$
$\frac{4}{6}$	$\frac{2}{3}$

Comments

Ordering fractions using common denominator rule is not part of these lessons. Many students should be able to order these fraction pairs using mental images for fractions.

You can use the problems in the wrap up to evaluate which students can order fractions using their mental images of fraction circles. Keep returning to order tasks like these to informally assess students' number sense.

A common error students make is to look only at the denominator to make an order decision. For example, when comparing $\frac{1}{3}$ vs. $\frac{3}{4}$, a student may say $\frac{1}{3}$ is the bigger fraction since thirds are larger than fourths.

Encourage students to reflect on the numerator and denominator to determine the fraction's relative size.

Translations

- ∞ Written symbols to manipulative
- ∞ Manipulative to verbal to written symbols
- ∞ Written symbols to verbal

Draw a picture of paper folding strips to show the fraction $\frac{2}{3}$.

Now partition your picture to show how many ninths equal $\frac{2}{3}$.

Name_____ Lesson 11 / Student Page A

Exploring $\frac{1}{2}$ With Fraction Circles

Use the whole circle as your unit. Make the fraction $\frac{2}{5}$ with the fraction circles.

Decide if $\frac{2}{5}$ is greater or less than $\frac{1}{2}$.

Record your response in the box:

$\frac{2}{5}$ | *is less than* | $\frac{1}{2}$

Complete the problems below. Use these choices:

is less than
is greater than or
is equal to

$\frac{2}{3}$		$\frac{1}{2}$	$\frac{1}{4}$		$\frac{1}{2}$
$\frac{5}{12}$		$\frac{1}{2}$	$\frac{8}{10}$		$\frac{1}{2}$
$\frac{3}{4}$		$\frac{1}{2}$	$\frac{1}{5}$		$\frac{1}{2}$
$\frac{2}{8}$		$\frac{1}{2}$	$\frac{4}{6}$		$\frac{1}{2}$
$\frac{7}{12}$		$\frac{1}{2}$	$\frac{9}{15}$		$\frac{1}{2}$

Using Fraction Circles to Order Fractions

Use fraction circles to show each fraction. Compare the fractions. Circle the largest fraction. If the fractions are equivalent, circle both.

(1) $\frac{3}{4}$ $\frac{1}{2}$

(2) $\frac{1}{3}$ $\frac{8}{12}$

(3) $\frac{4}{6}$ $\frac{2}{3}$

(4) $\frac{4}{12}$ $\frac{2}{4}$

(5) $\frac{9}{12}$ $\frac{2}{6}$

(6) $\frac{1}{3}$ $\frac{1}{4}$

(7) $\frac{4}{8}$ $\frac{1}{2}$

(8) $\frac{3}{4}$ $\frac{6}{8}$

(9) $\frac{1}{8}$ $\frac{1}{12}$

(10) $\frac{4}{8}$ $\frac{5}{8}$

(11) $\frac{2}{4}$ $\frac{3}{6}$

(12) $\frac{2}{6}$ $\frac{2}{12}$

(13) $\frac{3}{6}$ $\frac{5}{6}$

(14) $\frac{6}{8}$ $\frac{8}{8}$

(15) $\frac{4}{8}$ $\frac{6}{12}$

Rational Number Project

Initial Fraction Ideas Lesson 12: Overview	Materials
Students are introduced to chips as a fraction model. They learn to represent a given fraction using different sets of chips as a unit.	∞ Chips or tiles with a different color on each side ∞ Paper strips for folding ∞ Display chips or overhead chips for teacher ∞ Student Page A

Teaching Actions

Warm Up

Which fraction is larger? Share your answer and rationale to a partner. Did you both agreed on the larger fraction? Did you think about it in the same way or differently?

$\frac{3}{4}$ $\frac{11}{12}$

Large Group Introduction

1. You use chips while students use paper folding to show the same fraction.

2. Model the fraction $\frac{2}{3}$ with the display chips you made [see description at end of lesson].

   Say: I have 6 chips (show 6 chips with the white side up). I'm going to partition them into 3 equal groups. Can you partition your paper strip into 3 equal parts?

   Ask: How are the displays alike and different?



3. Turn over 2 of the 3 equal groups of chips to show the tan side of the chips. Ask children to model this action on their paper strips by shading 2 of 3 equal parts.

Comments

This will be a challenging lesson for students. You may want to use two class periods to cover this material.

To reinforce important fraction concepts students are introduced to a new model for fractions (chips) by relating this model to a previous one (paper folding). This is a translation from one physical model to another. Seeing similarities between models helps students abstract important concepts.

Common error: Students model $\frac{2}{3}$ by making **groups of three** instead of **three equal groups**.

Teaching Actions

Say: I made 2 of 3 equal-sized groups tan.

Ask: What fraction of the chips is tan?

Ask: How are the chips and paper folding models alike? Different?



4. Summarize what you did by writing on the board.

   I started with a unit of _____ white chips.

   I divided the unit into _____ equal groups.

   I made _____ groups tan.

   _____ of _____ equal-sized groups are tan.

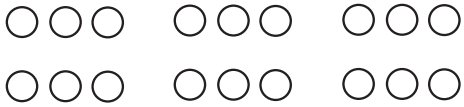
   What fraction of the groups is tan? ($\frac{2}{3}$)

5. Ask students to verbalize what they did with paper in a similar way. Conclude that there are many different models to show fractions.

6. Model other fractions using chips. Students should have their own chips. For the sake of consistency, use the white side to show the unit and use the tan side to show "amount shaded."

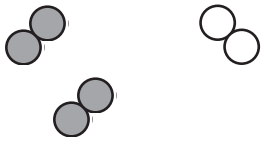
7. SAY**:** I have 18 chips. I want to show $\frac{4}{6}$ using these chips as my unit.

   ASK**:** How many equal-sized groups will I need? (six). Now divide the 18 chips into 6 equal groups.



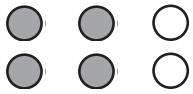
Comments

Students often need to exaggerate the grouping of the chips to highlight the separate groups.



Ex: $\frac{2}{3}$

They will spread out the sets and arrange the chips to touch to show a group. At this point don't use the array-model.

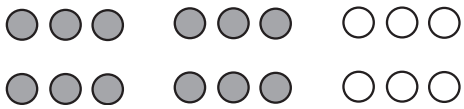


Some students say that 4/6 is covered. Informal discussion of equivalence is encouraged, but don't rush the idea.

Student verbalization is important. You might ask them to write a description for showing a fraction with chips.

Teaching Actions

ASK: To show $\frac{4}{6}$, how many equal groups must I make tan? (four)



8. Repeat for several more fractions.

Fraction	Unit
$\frac{2}{3}$	15
$\frac{2}{3}$	21
$\frac{6}{7}$	7
$\frac{3}{4}$	20

Small Group/Partner Work

9. Student Page A provides independent practice.

Wrap Up

10. Present these two stories to students. Ask them to decide which model, paper folding or chips, would be best to show the fraction the story.

    Story 1: Devan had 15 m & m's. She shared them equally between herself and 2 friends. What fraction of candy did she get?

    Story 2: Lianna has a Nestle Crunch bar. She plans to share it between herself and 2 friends. What fraction of the candy bar will each get?

Comments

Students will struggle a bit. As you move from group to group you may want to model the fraction using paper folding, going through each step as the students use the chips to copy your steps.

Students confuse the number of groups with the number of chips in each group so emphasize that $\frac{3}{4}$ means 3 out of 4 equal-sized groups. It does not mean to put 4 chips into a group.

Translations:

- ∞ Manipulative to manipulative
- ∞ Verbal (written words) to manipulative to pictures

Which fraction is larger? Share your answer and rationale to a partner.

Did you both agreed on the larger fraction?

Did you think about it in the same way or differently?

$$\frac{3}{4} \quad \frac{11}{12}$$

Modeling Fractions with Chips

1. Show 3-fourths with chips. Use 20 chips in all. Draw a picture of your display.

2. Show 3-fourths with chips. Use 8 chips in all. Draw a picture of your display.

3. Show 2-sixths with chips. Use 12 chips in all. Draw a picture of your display.

4. Show 2-sixths with chips. Use 6 chips in all. Draw a picture of your display.

5. Show 4-fifths with chips. Use 20 chips in all. Draw a picture of your display.

6. On the back of this page, describe steps you would take to show $\frac{3}{7}$ using 21 chips.

Rational Number Project

Initial Fraction Ideas Lesson 13: Overview	Materials
Students continue practicing showing fractions with chips. They determine several units that can be used to model a fraction and what units can't be used to model fractions.	∞ Chips for students and teacher ∞ Student Page A

Teaching Actions

Warm Up

Order these fractions from smallest to largest. Be ready to explain your thinking.

$\frac{2}{3}$ $\frac{2}{5}$ $\frac{5}{6}$ $\frac{2}{7}$ $\frac{5}{10}$

Large Group Introduction

1. Present this picture:



Say: I want to model this fraction using chips as my unit instead of paper. What fraction is shown? If I use 12 chips as my unit, tell me the steps to show $\frac{3}{4}$.

2. Vary the unit by asking students what they'd do if you used 4 chips as a unit and then 20 chips as the unit. Ask how these chip models are alike and how they are different.

3. Summarize by showing that to show $\frac{3}{4}$ you used 4, 12, and 20 chips. Ask if you could have used other sets of chips as your unit.

4. Ask students to show the fraction $\frac{2}{3}$ with chips. Allow them to choose the unit. Ask students to tell you what units they used.

Comments

Flexibility of unit is emphasized with chips, as was done with the fraction circles. Students should know that to show $\frac{2}{3}$, a number of sets can be used - 3 chips, 6 chips, 9 chips…

Regardless of the number of chips, the same action to model the fraction is used. (Partition into 3 equal groups and show 2 of the 3 groups tan.

Teaching Actions

5. Present this chart to students. Ask them to list 3 possible units that they could use as the unit for each fraction.

Fraction	Units you could use
$\frac{4}{5}$	
$\frac{2}{7}$	
$\frac{3}{4}$	
$\frac{1}{2}$	

6. Conclude the introduction by asking students to show the fraction $\frac{3}{4}$ with 14 chips. Discuss why this cannot be done. Ask for units that cannot be used to show the fractions in the above chart. [Make another column in the table].

Small Group/Partner Work

7. Assign Student Page A as a way to practice showing fractions with chips.

Wrap Up

8. End this lesson with some problem solving using chips. Present these two problems and ask students to solve them using chips. Then have a few students share how they solved the problems.

   Joe ate 4 jellybeans. This was $\frac{1}{5}$ of all the jellybeans in the bag. How many jellybeans were in the bag?

   Marta ate 18 jellybeans. This was $\frac{3}{5}$ of all the jellybeans in the bag. How many jellybeans in the bag?

Comments

The possible units are multiples of the denominator.

$\frac{4}{5}$ 5, 10, 15, 20… are all possible units.

Look closely at Student Page A. This page provides some problem solving for the students.

This page will need some initial guidance. Do the first 4 or 5 rows together. Students can finish the rest individually, in pairs or small groups.

To solve the challenges students have to reconstruct the unit. If 4 jellybeans equals 1-fifth, the there must be 20 jellybeans in the bag as the whole unit is made up of 5-fifths.

If 18 equals 3-fifths, then 1-fifth is 6 jellybeans. Therefore, 5-fifths would be 30 jellybeans.

Translations:

- Picture to manipulative to verbal
- Written symbols to manipulative

Order these fractions from smallest to largest. Be ready to explain your thinking.

$$\frac{2}{3} \qquad \frac{2}{5} \qquad \frac{5}{6} \qquad \frac{2}{7} \qquad \frac{5}{10}$$

Directions:
Use chips and complete the chart. The first one is done for you.

Number of chips in units	Number of equal-size parts	Number of chips in each equal-size part	Number of parts tan	Fraction of parts tan
15	3	5	2	$\frac{2}{3}$
15				$\frac{2}{5}$
9				$\frac{1}{3}$
21	7		5	
21		7		$\frac{2}{3}$
		2		$\frac{4}{5}$
10	2		1	
18	6		4	
18				$\frac{2}{3}$
18	9		4	
				$\frac{2}{4}$
12	4		2	
12	2		1	

Rational Number Project

Initial Fraction Ideas Lesson 14: Overview	Materials
Students continue to model fractions with chips. They determine possible fractions that can be shown with different sets of chips.	∞ Chips for students ∞ Student Pages A, B

Teaching Actions

Warm Up

Show $\frac{3}{4}$ using chips. You decide on the unit.

Draw a picture of your display.
Repeat using another unit.

Large Group Introduction

1. Put this chart on the board and ask children to tell you what fractions can be shown with each unit and why.

Unit in chips	Fractions you can show
6 chips	sixths, halves, thirds; not fourths
8 chips	
12 chips	
15 chips	
20 chips	

2. Ask students to name fractions that cannot be shown with the above sets.

3. Present these problems to students. Students model with chips. You draw pictures of models as students verbally describe their models to you.

   Show $\frac{1}{4}$ with chips.

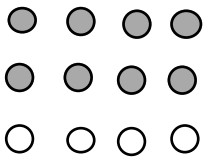
   Use 8 chips as a unit.

   Show $\frac{1}{4}$ with chips.

   Use a unit other than 8 chips.

Comments

To encourage communication, have students write out steps showing a fraction with chips.

Teaching Actions	Comments
Show $\frac{1}{6}$ with chips in 2 different ways. Use 6 chips and 12 chips as your units. Show $\frac{3}{5}$ with chips in two different ways. Use any two units. **Small Group/Partner Work** 4. Assign Student Pages A, B to reinforce chip model for fractions. **Wrap Up** 5. The next lesson is on equivalence with chips. Introduce this idea with this problem to end lesson 14: Ari showed the fraction $\frac{2}{3}$ using 12 chips. Erin said that his display really showed $\frac{8}{12}$ while Hamdi said it was $\frac{4}{6}$. Who is correct? Ari's display: 	

Translations:

- ∞ Written symbols to manipulative to verbal
- ∞ Written symbols to manipulative to pictures
- ∞ Real life to pictures

Show $\frac{3}{4}$ using chips. You decide on the unit.

Draw a picture of your display.

Repeat using another unit.

1. Show each of these fractions with chips in two ways.
   You decide on the unit.
   Draw pictures of your model.

	model 1	model 2
$\frac{2}{3}$		
$\frac{1}{5}$		
$\frac{1}{4}$		
$\frac{3}{6}$		
$\frac{4}{5}$		
$\frac{1}{2}$		

1. Show each of these fractions with chips in two ways.
   You decide on the unit.
   Draw pictures of your model.

	model 1	model 2
$\frac{2}{3}$		
$\frac{1}{5}$		
$\frac{2}{3}$		
$\frac{3}{6}$		
$\frac{4}{5}$		
$\frac{1}{2}$		

Draw pictures to model each story.

2. $\frac{2}{3}$ of Mr. Vega's math class are girls. There are 21 students in the class.

3. You can buy a box of 16 gum drops for 35 cents. If you share the box with three others, what fraction will each receive? How many gum drops will each receive?

4. William and his friend shared a small pizza evenly. How much did each eat?

5. Jessica and Jennifer shared their bag of m & m's with LeAnna. If each received a fair share, how much of the bag did Jessica and Jennifer get together?

Rational Number Project

Initial Fraction Ideas Lesson 15: Overview	Materials
Students explore fraction equivalence using chips.	∞ Chips for students ∞ Display chips for teacher ∞ Student Pages A-E

Teaching Actions

Warm Up

Jess ate $\frac{2}{3}$ of the peanuts in the bag. There are 7 peanuts left. How many did Jess eat? How many were there in the bag originally?

Large Group Introduction

Tell the students this story:

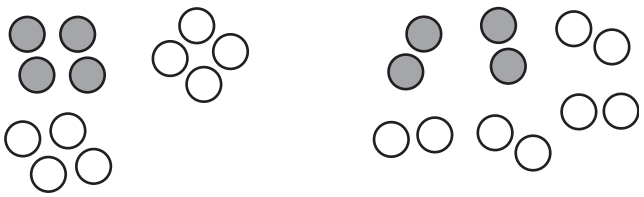
1. There are 12 pieces of hard candy in a bag. William ate 1/3 of the candy. Sonya ate 2/6 of the same-size bag of hard candy. Who ate more?

2. Have students model with chips William's share of 12 pieces of candy.

   Ask: What fraction of the bad did William eat? How many equal-sized groups will I divide 12 into? How can I show 1 of 3 equal-sized groups? (Turn chips to tan side).

3. Repeat for Sonya's share.

4. Have students look at each model and ask how they are alike and different. Ask again: who ate more?



$\frac{1}{3}$ $\frac{2}{6}$

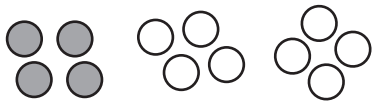
Comments

This will be a challenging lesson for students. You may want to use two class periods to cover this material.

Teaching Actions

5. Explain that you can name a fraction in more than one way. These two examples show that $\frac{1}{3}$ and $\frac{2}{6}$ are equal because the same unit is used to show both fractions, and the same number of chips are tan

6. Show 12 chips grouped into thirds:



$\frac{1}{3}$

Say: This shows 1 of 3 equal-sized groups. If I reorganize the chips into different groups, then I can see a different name.

7. Show:



1 of 3 groups

[You may want chips in the same group to touch].



2 of 6 groups

The second picture shows 2 of 6 equal-sized groups are tan or 2/6. Since the same number of chips is tan as in the previous model, $\frac{1}{3}$ and $\frac{2}{6}$ are equal.

8. Show



Say: I separated 12 chips into 12 groups with one chip in each group. What fraction is tan? The amount of tan chips is the same for $\frac{1}{3}$, $\frac{2}{6}$, and $\frac{4}{12}$. What is true about these fractions?

9. Let's show $\frac{3}{5}$ with 10 chips. Can you give me another name for the fraction? How can you tell?

Comments

Comparing fractions with chips depends on using the same number of chips in each unit. You cannot compare $\frac{1}{3}$ and $\frac{2}{6}$ if the units are different. For example:



$\frac{1}{3}$



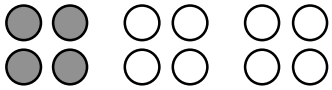
$\frac{2}{6}$

Here it seems that $\frac{2}{6} > \frac{1}{3}$.

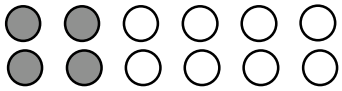
Reorganize groups to resemble arrays. You may still need to spread out groups to help children see different groups.

Ex:

$\frac{1}{3}$

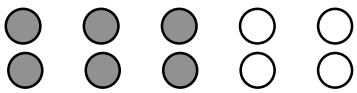


$\frac{2}{6}$

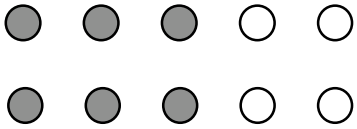


Emphasize the physical reorganization of the chips as a strategy for seeing different ways of grouping and naming fractional amounts.

$\frac{3}{5}$



$\frac{6}{10}$



Teaching Actions	Comments
10. Repeat for $\frac{2}{4}$, using 8 chips as the unit; $\frac{4}{6}$ using 24 chips as the unit; $\frac{1}{2}$ using 12 chips as the unit. **Small Group/Partner Work** 11. Assign Student Pages A-E. (You may want to do pages D and E in a large group setting). **Wrap Up** 12. End the lesson by asking for their thoughts on this story: Mark receives \$8 a month for an allowance. Janna receives \$12 a month for an allowance. Mark spent $\frac{1}{4}$ of his allowance. Janna spent $\frac{1}{6}$ of her allowance. Since $\frac{1}{4} > \frac{1}{6}$, Janna spent more. Do you agree?	

Translations:

- ∞ Real life to manipulative
- ∞ Manipulative to manipulative to verbal
- ∞ Written symbols to manipulative to written symbols to pictures

Jess ate $\frac{2}{3}$ of the peanuts in the bag.

There are 7 peanuts left.

How many did Jess eat?

How many were there in the bag originally?

Use your chips to do these problems. Name each fraction amount in more than one way if possible.

1. 8 chips is the unit. What is the value of each of these sets of counters?

   1 chip 4 chips

   2 chips 6 chips

2. 4 chips is the unit. What is the value of each of these sets of counters?

   1 chip 3 chips

   2 chips 5 chips

3. 6 chips is the unit. What is the value of each of these sets of counters?

   2 chip 3 chips

   5 chips 4 chips

Counters and Fractions

1. Show $\frac{1}{4}$ with chips. Use 8 chips as your unit. Draw a picture of your model. Give another name for $\frac{1}{4}$.

2. Show $\frac{2}{3}$ with your chips. Use 15 chips as your unit. Draw a picture of your model. Give another name for $\frac{2}{3}$.

3. Show $\frac{1}{6}$ with chips in two different ways. Use 6 chips and 12 chips as your units. Draw a picture of each model.

4. Show $\frac{1}{2}$ with 12 chips as the unit. Then show $\frac{1}{2}$ with 3 other units. Draw pictures of your models and name each one in more than one way if possible.

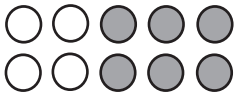
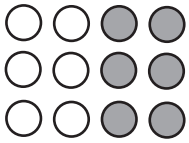
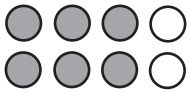
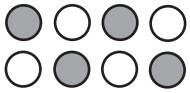
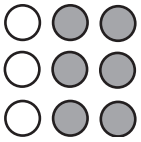
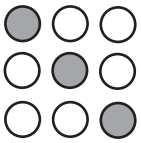
5. Show three fractions greater than $\frac{1}{2}$ with your chips. Show three fractions less than $\frac{1}{2}$ with your chips. Draw pictures of your models and name each one.

Fractions greater than $\frac{1}{2}$

Fractions less than $\frac{1}{2}$

Directions:

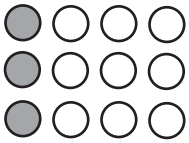
In the pictures below, ○ is the light side of the chip and ● is the dark side.
Give two fractions in symbols which tell the fraction of the chips which are light. For each exercise, complete the number sentence.

	Fraction 1	Fraction 2	Number Sentence
1. 	____	____	____ = ____
2. 	____	____	____ = ____
3. 	____	____	____ = ____
4. 	____	____	____ = ____
5. 	____	____	____ = ____
6. 	____	____	____ = ____

Name_____ Lesson 15 / Student Page E

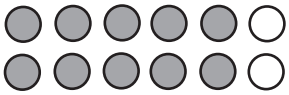
For exercises 7-11, write the equivalent fractions which are shown in the diagram.

7.



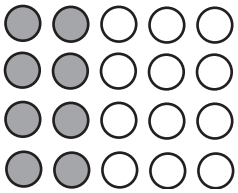
$\frac{\quad}{\quad} = \frac{\quad}{\quad}$

8.



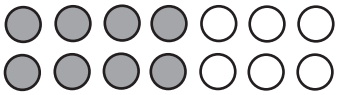
$\frac{\quad}{\quad} = \frac{\quad}{\quad}$

9.



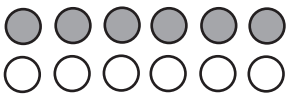
$\frac{2}{\quad} = \frac{4}{\quad} = \frac{8}{\quad}$

10.



$\frac{\quad}{\quad} = \frac{\quad}{\quad}$

11.



$\frac{1}{\quad} = \frac{\quad}{4} = \frac{\quad\quad}{12} = \frac{\quad}{6}$

Rational Number Project

Initial Fraction Ideas Lesson 16: Overview	Materials
Students extend their fraction concepts by reconstructing the unit when given the fraction part.	∞ Fraction Circles for students and teacher ∞ Student Page A

Teaching Actions

Warm Up

Order fractions from smallest to largest. Explain your thinking.

$\frac{6}{7}$ $\frac{2}{3}$ $\frac{99}{100}$ $\frac{9}{10}$ $\frac{3}{4}$

Large Group Introduction

1. Lead a discussion around the concept of unit. Possible questions include:

   ∞ To show $\frac{1}{3}$, what possible units could I use if I used fraction circles?
   ∞ If I used chips, what units could I use?
   ∞ If I used paper folding, how would I show $\frac{1}{3}$?
   [Use paper as a unit; partition into equal-sized parts; highlight a certain number of parts]

2. Explain that so far we have done a lot of problems in which we started with a unit and divided it into equal sized parts. Now we will reverse the process. You will know one or more of equal-sized parts and have to find the unit.

3. Model the idea of reconstruction the unit. Show 1 pink piece and say that this is 1 of 3 equal parts – it is $\frac{1}{3}$ of some amount, some unit.

4. Show 

and ask: because this is 1 of 3 equal sized parts, how many more parts do I need to build a whole unit? What size parts do I need? (All must

Comments

The activities in this lesson and the next reinforce the idea that, for example, 2 halves equal 1 whole, 3 thirds equal 1 whole, and so on. It also reinforces the notion that non-unit fractions are iterations of unit fractions ($\frac{3}{5}=\frac{1}{5}+\frac{1}{5}+\frac{1}{5}$).

Time spent on these activities continues to help children solidify mental images for fraction symbols.

Solutions of these problems will focus on the unit fraction and constructing the unit from the unit fraction.

Teaching Actions	Comments
be pink). 5. Place and count  1 part 2 parts 3 parts The whole unit is 3 pinks or 1 yellow. We know that 1 pink is $\frac{1}{3}$ of 1 yellow. We found the unit starting with $\frac{1}{3}$ of it. 6. Ask students to take out 1 blue piece. State that this blue is $\frac{1}{4}$ of some whole unit. 7. Ask: Will the unit be bigger or smaller? How many fourths make a whole unit? Use your circles to find the unit. 8. Repeat for these pieces and values: gray = $\frac{1}{4}$ [unit is yellow] red = $\frac{1}{3}$ [unit is blue] pink = $\frac{1}{2}$ [unit is brown] red = $\frac{1}{4}$ [unit is brown] gray = $\frac{1}{2}$ [unit is blue] 9. Present this example and explain that it is tricky: 1 gray = $\frac{1}{3}$. Find the unit. Ask the student to explain how to construct the unit. Then ask how we can describe the unit. Is there 1 piece to cover this amount? Since there isn't, students can name the unit as 3 grays. You can trace the 3 grays and say this amount is the unit.	Have students do these independently at their desks and then have them verbalize the process in a large group. In each of the examples so far, the answer could be expressed as a single piece. Ex: 1 gray = $\frac{1}{4}$, so the unit equals 4 grays or 1 yellow. This won't always be the case. If 1 blue = $\frac{1}{3}$, then the unit equals 3 blues.

Teaching Actions

10. Repeat for this example:

    1 blue = $\frac{1}{3}$. Unit = ?

Small Group/Partner Work

11. Student Page A provides practice.

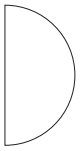
Wrap Up

12. End the class with this problem:

    I know that this blue piece is half of something.
    How can I find the value of these pieces?

    1 red 2 pinks
    1 gray 2 yellows

13. Accept suggestions. Then model a solution by reconstructing the unit (using their new skill). Show  . Explain that if  is $\frac{1}{2}$, then  and  make the unit. The unit is 1 yellow:

    
    unit

    Show 6 reds covering 1 yellow so 1 red is $\frac{1}{6}$.
    Now ask students to do the other 3.

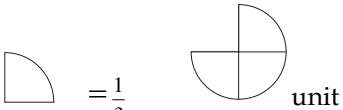
    [1 gray – $\frac{1}{4}$; 2 pinks = $\frac{2}{3}$; 2 yellows = 2]

14. Repeat for this problem: blue = $\frac{1}{3}$. Find the value of these pieces.

    1 red 1 yellow
    1 gray 9 reds

    [1 red = $\frac{1}{9}$; 1 yellow = $\frac{2}{3}$; 1 gray = $\frac{1}{6}$; 9 reds = 1]

Comments



No single piece covers 3 blues. Some children may express the unit as 3 blues; others as 1 yellow and 1 blue. Others will think their reasoning is wrong because there isn't one piece to cover 3 blues.

To name the other pieces, student should find the unit and then compare the other pieces to that unit. Don't try to solve this abstractly:

blue = $\frac{1}{2}$

red is $\frac{1}{3}$ of blue, so $\frac{1}{2}$ of $\frac{1}{3} = \frac{1}{6}$.

This problem is challenging. Step back and let students problem solve. Ask them to explain their strategies and reasoning.

Translations

- ∞ Written symbols to manipulative to verbal
- ∞ Written symbols to manipulative to written symbols

Order fractions from smallest to largest. Explain your thinking.

$\frac{6}{7}$ $\frac{2}{3}$ $\frac{99}{100}$ $\frac{9}{10}$ $\frac{3}{4}$

Problem Solving and Fraction Circles

I. Find the unit given the following information. Explain how you solved the problem. [You may want to draw pictures]

a) The red piece is $\frac{1}{4}$ of some amount. Find that amount. _____

b) The gray piece is $\frac{1}{6}$ of some amount. Find that amount. _____

c) The green piece is $\frac{1}{5}$ of some amount. Find that amount. _____

II. If the pink piece is $\frac{1}{4}$ what value do these have? Explain your reasoning.

a) 1 brown b) 1 red c) 1 white

Challenge: If the yellow piece is $\frac{2}{3}$ what value does one gray piece have?

Rational Number Project

Initial Fraction Ideas Lesson 17: Overview	Materials
Students name fractions greater than 1 with fraction circles. Students name fractions using both mixed numbers and improper fractions.	∞ Fraction Circles for students and teacher ∞ Transparency 1 ∞ Student Pages A, B, C

Teaching Actions

Warm Up

Imagine a tower made of 1- inch cubes. You can't see my tower but I will tell you that 12 cubes would be $\frac{2}{3}$ the height of my tower. How many cubes in my tower?

Large Group Introduction

1. Ask students to use their fraction circles with the black circles as the unit to show $\frac{2}{2}$, $\frac{4}{4}$, $\frac{5}{5}$ and $\frac{12}{12}$. In each example, ask for another name for the amount shown (1 whole or just 1).

2. Have students show $\frac{6}{8}$ using the whole circle as the unit. Ask if $\frac{6}{8}$ is greater or less than 1 whole or 1?

3. Present this story and ask students to model it with their circles. Again, use whole circles as the unit.

   Last night Margo ate $\frac{3}{4}$ of a large pizza. (Show that with circles). In the morning she ate some leftover pizza that equaled $\frac{2}{4}$ of a pizza.

4. Ask students to try and show the extra $\frac{2}{4}$. They realize that they don not have enough pieces. Have them work with a partner and use 2 sets of fractions circles to model the story.

Comments

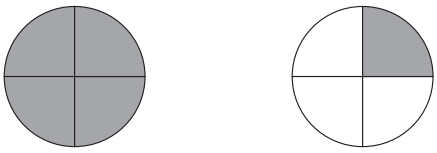
Modeling fractions greater than one using fraction circles, is easier than with chips, so we concentrate on developing the concept of changing improper fractions and vice versa with fraction circles.

Accept both names: 1 and $\frac{1}{4}$ or $\frac{5}{4}$.

Do not rush any rules about changing improper fractions to mixed fractions. Our goal is for students to change from one notation to another using circles and then just with mental images of circles. No paper/pencil rules.

Teaching Actions

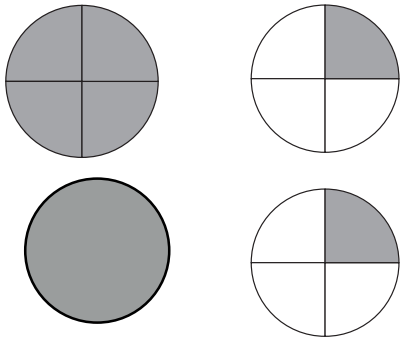
5. Continue with the story: How much pizza did Margo eat altogether?



Questions to lead discussion for naming amount of pizza:

- ∞ Did Margo eat more than 1 whole pizza? How do you know?
- ∞ Let's count how many fourths she ate: $\frac{1}{4}$, $\frac{2}{4}$, $\frac{3}{4}$, $\frac{4}{4}$, $\frac{5}{4}$.
- ∞ From the picture I see that $\frac{5}{4}$ equals $\frac{4}{4}$ and _____.
- ∞ What's another name for $\frac{4}{4}$?
- ∞ What's another way of describing the amount of pizza Margo ate?

6. Draw a picture of what you did and restate how $\frac{5}{4} = 1\frac{1}{4}$.



This shows $\frac{5}{4}$; point to $\frac{4}{4}$ and $\frac{1}{4}$ more.

This shows $\frac{5}{4}$; 1 whole and $\frac{1}{4}$ of another circle.

Comments

You may want to label 1 and $\frac{1}{4}$ a "mixed fraction;" $\frac{5}{4}$ an "improper fraction." Ask students why these names "make sense."

Students may try to name the picture of two circles with 5 parts shaded as $\frac{5}{8}$. If they do this, ask: What is the unit?

Since the unit is one pizza, each part is named by comparing it to one pizza or one circle.

Teaching Actions

6. Ask students to use their fraction circles to show these amounts. In each case, have the students name the amount in another way.

$\frac{8}{6}$

$\frac{3}{4}$

$\frac{3}{2}$

$1\frac{1}{3}$

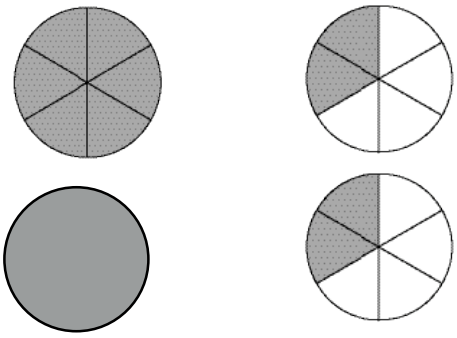
$1\frac{2}{4}$

$\frac{2}{3}$

7. As students explain their models, ask if the amount is greater or less than one. Try to get them to verbalize concrete actions that show when a fraction is greater than one.

8. Select students draw pictures for a few examples showing two ways to name the fraction.

Ex: $\frac{8}{6}$



10. To name fractions greater than 2, use pictures. Show transparency 1 and ask students to name each picture. [In each case, the unit is one circle or one rectangle.]

Small Group/Partner Work

11. Assign Student Pages A, B, C.

Comments

We want students to verbalize that, for example, $\frac{8}{6}$ is greater than 1 because they need more than 1 whole circle to model it. $\frac{4}{5}$ is less than 1 because they needed only 1 unit to model it.

You might want to make an overhead of Student Pages A and B to facilitate the review of answers.

Teaching Actions	Comments
Wrap Up 12. Select students to share their pictures for problems on Student Page C at the board. Students should explain how the picture was used to solve each problem. Ask students to name the fractional answer in two ways. 13. Ask students to imagine each fraction noted below and from their mental image name the amount in another way. $\frac{7}{5}$ $\frac{13}{6}$ $\frac{11}{2}$	

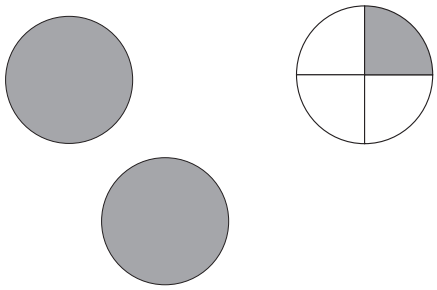
Translations

- ∞ Real world to manipulative to verbal
- ∞ Written symbols to manipulative to written symbols
- ∞ Written symbols to picture to verbal to written symbols
- ∞ Picture to written symbols
- ∞ Written symbols to pictures
- ∞ Written symbols to verbal

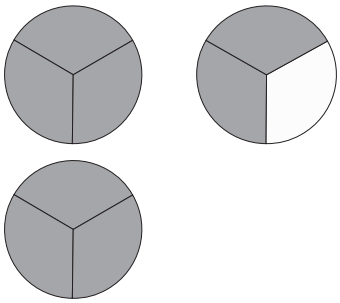
Transparency 1

Write Two Fraction Names

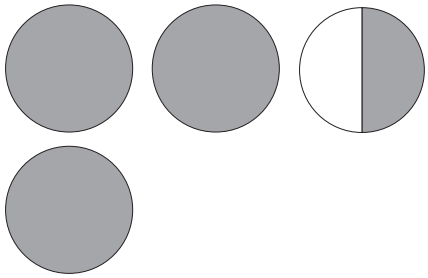
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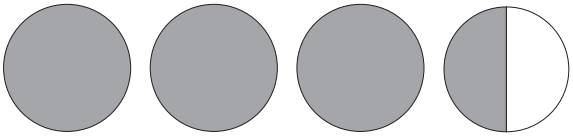
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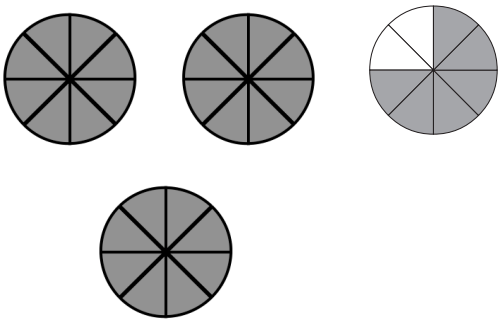
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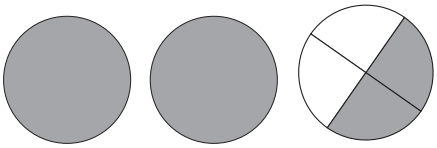
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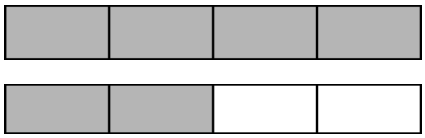
(5)



(6)



(7)



(8)



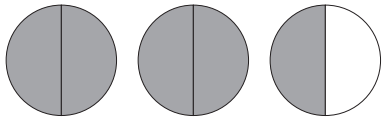
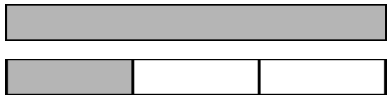
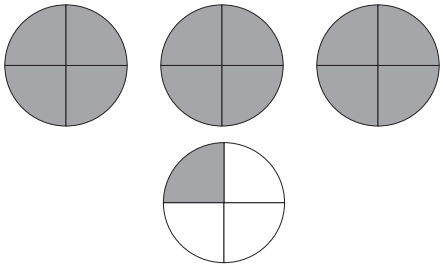
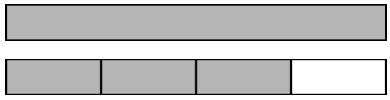
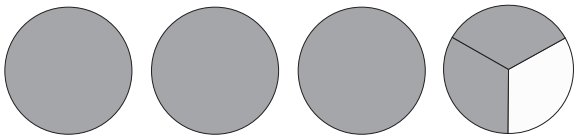
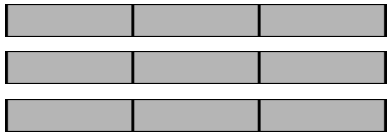
Imagine a tower made of 1- inch cubes.

You can't see my tower but I will tell you that 12 cubes would be $\frac{2}{3}$ the height of my tower.

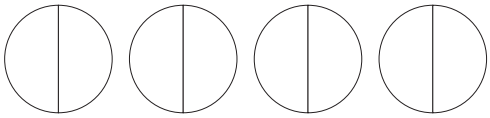
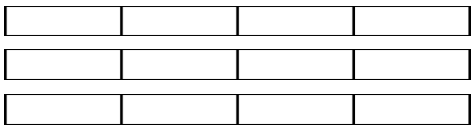
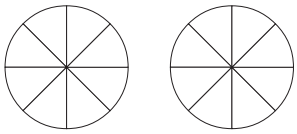
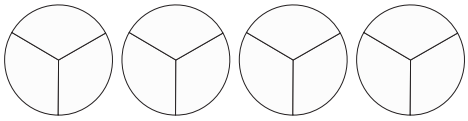
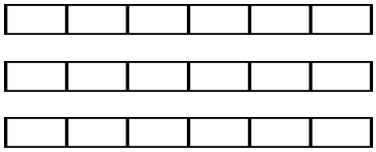
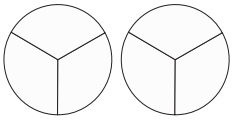
How many cubes in my tower?

Name_____ Lesson 17 / Student Page A

Write two fraction names for each picture.

Improper Fraction		Mixed Number
		
		
		
		
		
		

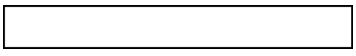
Shade in the pictures to show each fraction. Write another name for each amount

Mixed Number	Shade In	Improper Fraction
$3\frac{1}{2}$		
$2\frac{3}{4}$		
		$\frac{11}{8}$
		$\frac{10}{3}$
$2\frac{4}{6}$		
		$\frac{4}{3}$

Draw a picture for each story. Write the fraction name.

1. Brenda ate $\frac{2}{3}$ of a candy bar for lunch. She finished it after lunch and ate $\frac{1}{3}$ more of a second candy bar of the same type. How much candy did Brenda eat?

2. Marcia's dad was making pancakes. He added $\frac{2}{3}$ cup milk to the pancake mix. He decided to make a bigger batch so he poured another $\frac{2}{3}$ cup of milk in. How much milk did he use?
(Use  as a picture of a cup)

3. The dress designer needed some yellow ribbon for 3 dresses. He needs $\frac{2}{3}$ yard for one dress, $\frac{1}{3}$ for another, and $\frac{2}{3}$ for the third. Draw a picture to show how many yards of ribbon he bought.
( = 1 yard)

Rational Number Project

Initial Fraction Ideas **Lesson 18: Overview**	Materials
Students look at the numerical relationship between the numerators and denominators of fractions equal to $\frac{1}{2}$. They use this number pattern to determine if a given fraction is less than or equal to $\frac{1}{2}$.	∞ Fraction Circles for students and teacher ∞ Student Page A ∞ Student Pages A and B from Lesson 11

Teaching Actions

Warm Up

Draw two pictures for each fraction to show its two different names.

$\frac{7}{4}$ $\frac{3}{2}$ $\frac{8}{3}$

Large Group Introduction

1. Ask students to take out the fraction circles and find several equivalences for $\frac{1}{2}$ (use the black circle as unit).

2. Record them on chart.

   Fractions equal to $\frac{1}{2}$ $\frac{2}{4}, \frac{3}{6}, \frac{4}{8}, \frac{5}{10}, \frac{6}{12}$

3. Tell students that you can add to the list without using circles:

   $\frac{7}{14}, \frac{8}{16}, \frac{9}{18}, \frac{10}{20}, \frac{25}{50}, \frac{50}{100}, \frac{150}{300}$

4. Ask students to look at the numerator and denominator of each fraction equal to $\frac{1}{2}$ and ask them if they can see any pattern or relationship between numerator and denominator that's the same

Comments

Students with a quantitative sense of fractions use $\frac{1}{2}$ as a reference point to estimate fraction sums and differences.

Ex: $\frac{3}{6} + \frac{1}{3}$

"$\frac{3}{6}$ equals $\frac{1}{2}$, and $\frac{1}{3}$ is less than $\frac{1}{2}$, so the sum is greater than $\frac{1}{2}$ but less than 1."

Notice the role of fraction equivalence for $\frac{1}{2}$ in estimation as well as in the same numerator but different denominator strategy [Lessons 6 & 7]

At this point we won't look explicitly at examples like $\frac{2\frac{1}{2}}{5}$ but if students mention examples like this one acknowledge that it does equal $\frac{1}{2}$.

Teaching Actions

for each fraction.

5. Help students verbalize that in each case, the denominator is double (twice) the numerator.

6. Give students these fractions with parts missing and have them make them into fractions equal to $1\frac{1}{2}$

$$\frac{\quad}{24}, \frac{11}{\quad}, \frac{\quad}{30}, \frac{\quad}{28}, \frac{100}{\quad}$$

7. Ask students to show these fractions with their circular pieces.

$$\frac{1}{4} \quad \frac{2}{6} \quad \frac{3}{8} \quad \frac{4}{10} \quad \frac{5}{12}$$

Ask if they are greater or less than $\frac{1}{2}$. Have them tell you how far away from $\frac{1}{2}$ each amount is.

8. Without using the pieces, ask them to tell you numerators that would make each fraction greater than $\frac{1}{2}$.

9. Present these fractions to students. Ask them if they are $> \frac{1}{2}$, $< \frac{1}{2}$, or $= \frac{1}{2}$. Use fraction circles if needed. Have them verbalize their reasoning.

$$\frac{3}{10} \quad \frac{5}{12} \quad \frac{4}{6} \quad \frac{6}{10} \quad \frac{9}{20} \quad \frac{15}{18} \quad \frac{2}{4}$$

Small Group/Partner Work

10. Student Page A provides practice. You may want to use Student Pages A and B from Lesson 11 again. Now have students see if they can solve problems using number patterns for $\frac{1}{2}$.

Comments

Teaching Actions	Comments
Wrap Up 11. End the class with this problem. Ask students how they can use equivalence for ½ and other order ideas to estimate the following problem: $\frac{14}{30}+\frac{5}{10}$ Is $\frac{19}{40}$ a reasonable answer? Is the sum greater than 1 or less than 1?	

Translations

- ∞ Written symbols to verbal
- ∞ Real world to verbal

Draw two pictures for each fraction to show its two different names.

$\frac{7}{4}$ $\frac{2}{3}$ $\frac{8}{3}$

Comparing to 1-half

1. Margo and Jose shared a couple of large pizzas. Margo ate $\frac{5}{8}$ of a pizza. Jose ate $\frac{6}{16}$ of a pizza. Who ate more? Explain how you know.

2. Imagine that you shared your bag of mini doughnuts with your sister. You ate $\frac{3}{5}$ of the bag while your sister ate $\frac{4}{10}$ of the bag. Who ate more? Explain how you know.

3. Chou-Mei ran 2 and $\frac{7}{8}$ miles. Her sister ran 2 and $\frac{3}{10}$ miles. Who ran the shorter distance? Explain how you know.

4. Circle the larger fraction in each pair.

a)	$\frac{2}{3}$	$\frac{1}{5}$	b)	$\frac{9}{12}$	$\frac{6}{15}$	c)	$\frac{5}{9}$	$\frac{3}{7}$
d)	$\frac{1}{2}$	$\frac{3}{4}$	e)	$\frac{3}{5}$	$\frac{4}{9}$	f)	$\frac{11}{17}$	$\frac{3}{9}$
g)	$\frac{10}{22}$	$\frac{4}{5}$	h)	$\frac{3}{6}$	$\frac{2}{9}$	i)	$\frac{8}{13}$	$\frac{6}{16}$

Rational Number Project

Initial Fraction Ideas Lesson 19: Overview	Materials
Students are introduced to fraction addition through familiar contexts and estimating reasonable answers (by comparing sum to and 1).	∞ Fraction Circles for students and teacher ∞ Student Page A

Teaching Actions

Warm Up

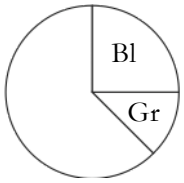
List 5 fractions greater than $\frac{1}{2}$. How do you know that they are greater than $\frac{1}{2}$?

Large Group Introduction

1. Present this story to the students: William ate $\frac{1}{4}$ of a pizza for dinner. The next morning he ate a piece that equaled $\frac{11}{8}$ of the pizza. How much of a pizza did he eat?

2. Explain that you don't want the exact answer, but just an estimate. Ask students to imagine $\frac{1}{4}$ of a pizza and $\frac{1}{8}$ of a pizza. Did William eat more or less than $\frac{1}{2}$ of a pizza? Have students to explain their responses by referring to their mental images for $\frac{1}{4}$ and $\frac{1}{8}$.

3. Show with fraction circles $\frac{1}{4}+\frac{1}{8}$:



Comments

The time spent developing fraction concepts, ordering and equivalence ideas will enable students to approach fraction addition and subtraction in a meaningful way. Initial experience with +, - operations will be through estimation.

Estimation skills depend on students' mental images for symbols as well as the context in which the operation is embedded.

Students' explanation of estimation may sound like this:

(a) He ate less than $\frac{1}{2}$. You need two-fourths to be $\frac{1}{2}$, and $\frac{1}{8}$ is less than $\frac{1}{4}$.

(b) $\frac{1}{4}$ of a pizza is like the blue piece. The gray is $\frac{1}{8}$ and it is smaller than the blue. Together they won't make $\frac{1}{2}$.

Teaching Actions	Comments
4. Explain to students that some people would say that $\frac{1}{8}+\frac{1}{4}$ is $\frac{2}{12}$. Ask: Does that make sense? If you ate $\frac{1}{4}$ and then $\frac{1}{8}$ of a pizza would that be the same as $\frac{2}{12}$? Show with circles $\frac{1}{4}$, $\frac{1}{8}$, and $\frac{2}{12}$ of the black circle.	
5. Repeat estimation process with the following story problems. In each case have students verbalize their reasoning. Point out when students use an ordering or equivalence idea previously learned. Maria received a chocolate chip cookie as big as a birthday cake for a present. She cut it into 6ths and shared the cookie with her friend LeAnna. Maria ate $\frac{3}{6}$ of the cookie,. Leanna ate $\frac{1}{3}$. Together, how much did they eat? Martin was making play dough. He added $\frac{3}{4}$ cup of flour to the bowl. Then he added another $\frac{3}{6}$ cup. How much flour did he use? (In this case also ask if the sum is greater or less than one).	Familiarity with context helps students to reason about appropriateness of answers.
6. Provide added practice by estimating these sums. In each case, estimate as $>\frac{1}{2}$ or $<\frac{1}{2}$, and >1 or <1. (a) $\frac{1}{8}+\frac{1}{4}$ (b) $\frac{3}{6}+\frac{1}{4}$ (c) $\frac{3}{4}+\frac{2}{4}$ (d) $\frac{4}{6}+\frac{1}{2}$	
Small Group/Partner Work	
7. Student Page A provides practice. Assign in groups so students can share strategies for estimation.	Students are asked to write out their explanations for estimating sums.
Wrap Up	
8. Have students share their estimation strategies. List the ordering and equivalence ideas mentioned in	

Teaching Actions	Comments
students' explanations. Discuss how important those order and estimation skills are when operating with fractions.	

Translations

- ∞ Real world to verbal
- ∞ Real world to verbal to manipulative
- ∞ Written symbols to verbal

List 5 fractions greater than $\frac{1}{2}$. How do you know that they are greater than $\frac{1}{2}$?

Fraction Addition and Estimation

1. Marty divided a candy bar into 12 equal parts. He ate 1-sixth of the candy bar before lunch. He ate 1-fourth of the candy bar after lunch. Did he eat more or less than 1-half of the candy bar? Did he eat the whole candy bar? **Explain your reasoning.**

2. Terri ate $\frac{5}{6}$ of a small pizza and $\frac{11}{12}$ of another small pizza. Did she eat more than one whole pizza? **Explain your reasoning.**

3. Alex used $\frac{1}{3}$ cup of flour in one recipe and $\frac{1}{4}$ cup of flour in another recipe. Together did he use more than $\frac{1}{2}$ cup of flour? **Explain your reasoning.**

4. Give a reasonable estimate for each sum. On the back of this sheet write out your reasoning for each problem.

$\frac{1}{3} + \frac{2}{6}$ $\frac{1}{8} + \frac{9}{10}$ $\frac{7}{8} + \frac{1}{6}$ $\frac{1}{5} + \frac{3}{12}$ $\frac{1}{3} + \frac{3}{4}$

Rational Number Project

Initial Fraction Ideas **Lesson 20: Overview**	**Materials**
Students use fraction circles to obtain exact answers to fraction addition.	∞ Fraction Circles for students and teacher ∞ Student Pages A, B, C

Teaching Actions

Warm Up

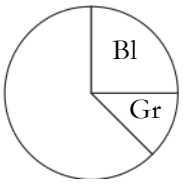
If you ran $\frac{3}{4}$ of a mile before lunch and ran $\frac{7}{8}$ of a mile after lunch, about how many miles did you run?

Large Group Introduction

1. Present this story from Lesson 19.

   William ate $\frac{1}{4}$ of a pizza for dinner. The next morning he ate a piece that equaled $\frac{1}{8}$ of the pizza. How much of a pizza did he eat?

2. Remind students that they already estimated the sum to be less than $\frac{1}{2}$. Explain that to find the exact sum, they are to use their fraction circles. Model the problem and ask students to do the same with their materials.

   Say: This blue represents $\frac{1}{4}$ of the pizza. This gray represents $\frac{1}{8}$ of the pizza.

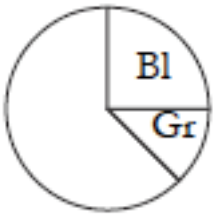
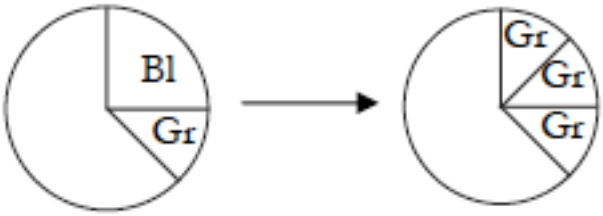
3. Explain that from this picture they can see that William ate less than $\frac{1}{2}$ of the pizza. Ask: Exactly how much pizza did William eat? Is there a way to

Comments

Addition of fractions with unlike denominators is introduced right from the start. Students are led to solve the problem of naming the amount of the circle covered using one fraction name.

At this point you are trying to develop an implicit understanding of the symbolic procedure.

Same denominator means, concretely,

Teaching Actions	Comments
express the amount of circle covered using one fraction name? Fore example, is $\frac{3}{4}$ covered? $\frac{2}{12}$?	using the same color pieces to show each fraction.
4. Ask students to try to find a way to cover $\frac{1}{4}$ and $\frac{1}{8}$ of the circle using only one color.	
5. Give clues if needed: (1) Think about what you know about equivalent fractions; (2) How many grays equal one blue?	
6. Building on students' ideas, lead to modeling the problem, verbalizing each step.	
 "I can tell how much of the whole circle is covered if I can cover the amount with pieces of the same color." "I know 2 grays cover the same amount as 1 blue, so I am going to exchange 1 blue for 2 grays."	At this point there is no need to record with symbols. Work at the verbal level.
	
7. Ask: what fraction of the circle is covered? How much pizza did William eat? Is 3/8 less than 1/2?	
8. Repeat with the chocolate chip cookie problem from Activity 20:	
Maria received a chocolate chip cookie as big as a birthday cake for a present. She cut it into 6ths and shared the cookie with her friend LeAnna. Maria ate $\frac{3}{6}$ of the cookie, LeAnna ate $\frac{2}{6}$. Together, how much did they eat?	Important for children to internalize that sometimes you need to find equivalent fractions and sometimes this step is not needed.
10. Ask students if they have to exchange circle pieces in this example to determine the amount of circle covered.	In all exercises so far only 1 of the

Teaching Actions	Comments
11. Repeat for these problems. Encourage children to verbalize each step with materials. Have them attend to the decision of whether or not an exchange is needed. (Don't forget to estimate!) (a) $\frac{1}{2}+\frac{1}{4}$ (b) $\frac{2}{4}+\frac{1}{8}$ (c) $\frac{2}{3}+\frac{1}{3}$ (d) $\frac{2}{3}+\frac{2}{3}$	fractions (or none) need to be changed. In *RNP: Fraction Operations and Decimals* module this work with addition of fractions is extended to where both fractions need to be renamed. [Student page C does provide problems like those for students who need a challenge]. You will want to extend this lesson over 2-3 days.
Small Group/Partner Work 12. Student Pages A and B provide added practice. Student Page C provides challenges.	
Wrap Up 13. Have each group create their own story problem for addition for of fractions to share with the class. Then model how to solve them with the fraction circles.	Mastery of the addition of fractions is not a major goal of this module. You are providing experience that will be extended in the *RNP: Fraction Operations and Decimals* module.

Translations

- ∞ Real world to manipulative to verbal
- ∞ Written symbols to manipulative to verbal

If you ran $\frac{3}{4}$ of a mile before lunch and ran $\frac{7}{8}$ of a mile after lunch, about how many miles did you run?

Fraction Addition and Estimation:
Finding the Exact Answer

1. Marty ate some candy. He ate 1-half of a whole Hershey bar before lunch. He ate 1-fourth of a whole Hershey bar after lunch. About how much of one candy bar did he eat? With your fraction circles, find out the exact amount of a Hershey bar that Marty ate. Draw pictures to show what you did with the circles. Estimate first!!!

Estimate: _____

2. Terri ate 1-half of a small pizza and 5-twelfths of another small pizza. About how much of a whole pizza did she eat? With your fraction circles, find out the exact amount. Draw pictures to show what you did with the circles. Estimate first!!!

Estimate: _____

3. Allie rode her bicycle 7-eights of a mile to school. Then she rode 1-fourth of a mile to her friend's house. About how far did she ride altogether? With your fraction circles, find out the exact amount. Draw pictures to show what you did with the circles. Estimate first!!! (Use back of the page for your drawing).

Estimate: _____

Fraction Addition Continued

4. Because of a rainstorm, the water level in a swimming pool rose $\frac{2}{3}$ of an inch. The following day it rained again. The pool rose another $\frac{11}{12}$ of an inch. About how high did the water level increase? With your fraction circles, find out the exact amount.

Estimate _____

5. Alex used $\frac{1}{4}$ cup of flour in one recipe and $\frac{3}{8}$ cup of flour in another recipe. Together about how much flour did he use? ? With your fraction circles, find out the exact amount.

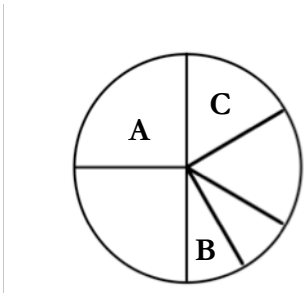
Estimate _____

6. With your fraction circles, find the exact answers.

$\frac{1}{3} + \frac{1}{6}$ $\frac{1}{8} + \frac{3}{4}$ $\frac{4}{10} + \frac{1}{5}$ $\frac{1}{6} + \frac{3}{12}$ $\frac{1}{2} + \frac{3}{4}$

Challenges

1.



Shade Parts A, B, and C

a) How much are parts A and B together?

b) How much of the circle are Parts A and C together?

c) How much of the circle are Parts B and C together?

2. Describe how you solved the problems.

3. Estimate a sum for each problem. Then find the exact answer using your fraction circles. Explain how these problems are different from the others you solved.

$\frac{1}{4} + \frac{1}{3}$ $\frac{1}{2} + \frac{2}{5}$ $\frac{2}{4} + \frac{5}{6}$ $\frac{1}{2} + \frac{5}{6}$

Rational Number Project

Initial Fraction Ideas Lesson 21: Overview	Materials
Students explore fraction subtraction within take away and difference contexts. Students estimate and solve story problems with fraction circles.	∞ Fraction Circles for students and teacher ∞ Student Page A, B and C

Teaching Actions

Warm Up

Estimate – Is the sum greater or less than 1?
Use fraction circles to add these fractions.

$\frac{3}{4}+\frac{1}{6}$ $\frac{1}{5}+\frac{2}{3}$ $\frac{5}{6}+\frac{5}{12}$

Large Group Introduction

1. Consider the development of fraction subtraction as a problem solving activity. Let students explore, in groups, using the circle pieces and creating their own solution strategies.

2. Introduce this take away problem. Ask students to act out the problem in their groups. Explain that you will be soliciting, at random, someone to explain their group's work to the whole class.

   Alice noticed that there was 3/4 of a pizza left after the party. She ate a slice of pizza that was the size of 1/8 of a whole pizza. How much pizza was left after Alice ate a slice?

3. Before group work, ask for estimates. Will there be more or less than $\frac{1}{2}$ pizza left? Try picturing the $\frac{3}{4}$ pizza in your mind. Does this help in your estimate? Explain your thinking. (Record estimates on the board).

Comments

This lesson will take two class periods.

Developing subtraction of fractions from contexts is more difficult than for fraction addition. There is more than one context for subtraction: take away, finding the difference, how many more.

You should expect students to solve each story problem type at the concrete level differently. Students will match their actions with the fraction circles with the action from the story problem.

In this module you should be only interested in concrete solutions to subtraction problems and estimation of reasonable answers.

Teaching Actions

4. Have students try to act out the problem. As you monitor their work, consider these hints:

   - ∞ Can you show $\frac{3}{4}$ of a pizza with the circles?
   - ∞ Is $\frac{1}{8}$ > or < than $\frac{1}{4}$?
   - ∞ What is the action in the story?
   - ∞ Can you take away $\frac{1}{8}$ if the model is shown with 3 blues?
   - ∞ Can you use your ideas of equivalence to help you?

5. Randomly select a student to explain how his/her group attempted to act out the problem. Deemphasize an exact answer, but help students verbalize their group's reasoning and discussion.

6. Repeat the same story problem with these new numbers:

   $\frac{1}{2}, \frac{1}{8}$

   $1, \frac{3}{4}$

   $\frac{2}{3}, \frac{1}{6}$

6. Close the introduction with a story problem that represents a difference model. Ask students to act out this story problem:

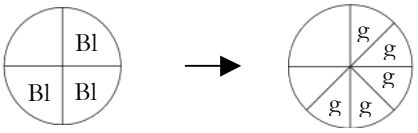
   Joe & Renata each receive the same allowance. Joe spent $\frac{2}{3}$ of his allowance on records. Renata spent $\frac{1}{6}$ of her allowance repairing her bicycle. How much more did Joe spend than Renata?

   Ask: Is it easier to compare amounts when they are shown with the same colored pieces?

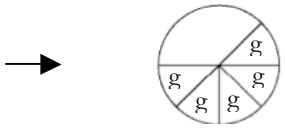
   Estimate answer. Will the difference be greater or less than $\frac{1}{2}$? Why?

Comments

In the take-away model, students may build the first fraction and take away the second.

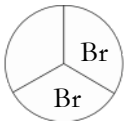


Take away 

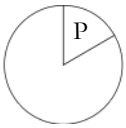


In the difference model, students may construct 2 separate models for the 2 fractions in the story and compare models.

For example:



Joe's amount



Renata's amount

Teaching Actions

7. As you monitor their group work consider these hints:
   - ∞ Is the action in this story different than in the pizza problem?
   - ∞ Can you show Joe's amount? Renata's amount?
   - ∞ Can you use your ideas of equivalence?

8. Have students share solution strategies. Talk about differences in actions in the two problem types.

9. Repeat the same story with these numbers:

$\frac{1}{2}$	$\frac{1}{4}$
$\frac{2}{3}$	$\frac{2}{3}$
$\frac{6}{8}$	$\frac{1}{2}$

Group/Partner Work

10. Student Pages A and B provide practice with story problems. Spend time in class having students explain their strategies for solving these problems. Student Page C provides some challenges.

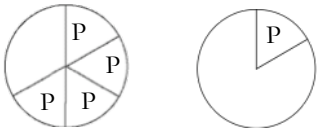
Wrap Up

11. Select problems from Student Pages A and B and have students share how they used fraction circles to solve the problems.

12. The last four problems on Student Page C involve ordering ideas and subtraction. Pick a couple to do together in large group.

Comments

Change to pink:



Joe spent $\frac{3}{6}$ more.

Some may still use the take-away model.

Again, you will want to use this lesson plan over 2 days. You may want to introduce the take-away model one day & the difference model the next day.

The goal is for students to be able to act out the problems with fraction circles, talking about what they are doing.

Translations:

- ∞ Real life to manipulatives to verbal
- ∞ Written symbols to manipulatives to verbal
- ∞ Real life to manipulative to pictures

Estimate – Is the sum greater or less than 1?

$$\frac{3}{4}+\frac{1}{6} \qquad \frac{1}{5}+\frac{2}{3} \qquad \frac{5}{6}+\frac{5}{12}$$

Use fraction circles to add these fractions.

Fraction Operations: Finding the Exact Answer

1. Joe lives $\frac{4}{10}$ a mile from school. Mary lives $\frac{1}{5}$ of a mile away. How much farther from school does Joe live than Mary? Draw pictures to show what you did with the circles.

Estimate: _____

2. Because of a rainstorm the water level in a swimming pool rose by $\frac{9}{12}$″. The following day it dropped by $\frac{4}{6}$″. What was the total change in water level? Draw pictures to show what you did with the circles.

Estimate: _____

3. Velicia spent $\frac{1}{3}$ of her allowance on a CD and $\frac{4}{6}$ of her allowance on a movie. What fraction of her allowance did she have left? Draw pictures to show what you did with the circles.

Estimate: _____

Fraction Operations Continued

1. A clerk sold three pieces of ribbon. The red piece was $\frac{1}{3}$ of a yard long. The blue piece was $\frac{1}{6}$ of a yard long. The green piece was $\frac{10}{12}$ of a yard long.

   a) How much longer was the green ribbon than the red ribbon?

   b) How much longer was the green ribbon than the blue ribbon?

   c) Are the red ribbon and blue together greater than, less than, or equal in length to the green ribbon?

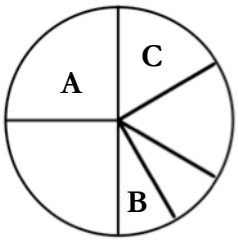
   d) If the red and blue together are greater than the green, how much greater are they? If shorter, how much shorter are they?

5. With your fraction circles, find the exact answers

$$\frac{4}{5} - \frac{3}{10} \qquad \frac{1}{4} + \frac{5}{8} \qquad \frac{7}{8} - \frac{1}{3} \qquad \frac{1}{2} + \frac{3}{8} \qquad \frac{3}{5} - \frac{2}{5}$$

Challenges

1.



Shade Parts A, B, and C

a) How much larger is part A than part B?

b) How much larger is part A than part C?

c) How much larger is part C than part B?

d) How much are parts A and C together?

e) How much greater are parts A and C together than part B?

f) Are parts C and B together greater than A? If so, how much greater? If not, how much smaller?

2. Describe how you solved the problems.

3. Circle the larger fraction in each pair. Then find out how much larger that fraction is.

$\frac{1}{4}$ $\frac{1}{3}$ $\frac{1}{2}$ $\frac{2}{5}$ $\frac{2}{4}$ $\frac{5}{6}$ $\frac{1}{2}$ $\frac{5}{6}$

Rational Number Project

Initial Fraction Ideas Lesson 22: Overview	Materials
Students explore fraction subtraction within difference and how many more contexts. Students estimate and solve story problems with fraction circles.	∞ Fraction Circles for students and teacher

Teaching Actions

Warm Up

Which fraction is larger? How much larger?

$$\frac{3}{4} \quad \frac{2}{3}$$

Large Group Introduction

1. Continue with the difference story problems. Students act out the problem in a group with fraction circles.

   Joe and Renata each receive the same allowance. Joe spent $\frac{1}{2}$ on records. Renata spent $\frac{1}{8}$ on repairing her bicycle. How much more did Joe spend?

2. Ask for estimates. Is it reasonable to expect the answer to be less than $\frac{1}{2}$? Greater or less than $\frac{1}{4}$?

3. As you monitor their group work consider these hints:

   - ∞ How can you show Joe and Renata's amounts with the fraction circles?
   - ∞ Is $\frac{1}{8}$ greater or less than $\frac{1}{2}$?
   - ∞ How many 8ths equal $\frac{1}{2}$?

4. Have student share solutions again, emphasizing the group's thinking rather than the exact answer.

Comments

You may want to spend more than one day on this lesson.

Teaching Actions

5. Present these two problems. Ask students to act out both problems and explain how they are different.

   ∞ *José ate $\frac{3}{4}$ of a pizza. Mara ate $\frac{5}{8}$. Who ate more and how much more?*

   ∞ *A movie costs Josie $\frac{1}{2}$ of her allowance. If she only had $\frac{3}{4}$ of her allowance left before the move, how much of her allowance did she have left after the movie?*

6. Present this "how many more" problem and ask students to act it out.

   Hamdi agreed to mow about $\frac{1}{2}$ of the lawn before dinner. At 3:00 she finished about $\frac{3}{8}$ of the lawn. How much of the job does she have left?

7. As you monitor group work, consider these questions:

   a) Can you show how much she completed by 3:00?
   b) Even if you can reason the answer in your head, can you show how to do it with circles?
   c) If she has completed $\frac{3}{8}$, how much more to $\frac{1}{2}$?

8. Share responses and actions on circles. Discuss how this problem is different from previous ones.

9. Talk about how all 3 problem-types (take away, difference, missing addend) are a form of subtraction. Pick any 3 examples and substitute whole numbers less than 19 in for fractions. Ask students to solve with whole numbers. Note that in each case students use their subtraction facts.

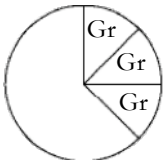
Small Group/Partner Work

10. In groups ask students to create a story problem for addition and one for each subtraction category. Ask them to solve the problems using their fraction

Comments

The pizza problem models subtraction as "finding a difference". The movie problem is a "take away" problem.

Expect students to act out this story differently from the others. Students may start with $\frac{3}{8}$ (amount completed.



They will try to fill in the space left to make $\frac{1}{2}$ ($\frac{1}{8}$ more).

Even though actions in story problems are different, they are all subtraction problems. Symbolically, all will use "-" (minus sign). It's important for students to see that these are subtraction contexts.

Teaching Actions	Comments
circles, recording steps symbolically if you think they are ready to do so. **Wrap Up** 11. Spend the time in large group for sharing these stories and the strategies for solving them. 12. Provide symbolic problems for students to act out with the fraction circles. They can use any of the three ways shown. Many might choose the take-away model. Examples: $\frac{2}{3} - \frac{1}{3}$ $\frac{4}{5} - \frac{1}{10}$ $\frac{1}{2} - \frac{3}{8}$ $\frac{7}{12} - \frac{1}{2}$ $\frac{6}{9} - \frac{1}{3}$	You decide whether students are ready to record their work symbolically. Their symbol records may not look like "standard algorithm," but reflect each step they act out with the fraction symbols. For example, for the lawn problem, the students might record as follows: $\frac{3}{8}$ $\frac{1}{2} = \frac{4}{8}$ $\frac{4}{8} - \frac{3}{8} = \frac{1}{8}$

Translations

- ∞ Real life to manipulative to verbal
- ∞ Written symbols to manipulative to written symbols

Which fraction is larger? How much larger?

$$\frac{3}{4} \qquad \frac{2}{3}$$

Rational Number Project

Initial Fraction Ideas Lesson 23: Overview	Materials
This lesson helps to highlight for students how much they know about fractions: meaning for symbols as well as how to add and subtract them in a meaningful way.	∞ Fraction Circles for students ∞ Student Page A

Teaching Actions

Warm Up

Order these fractions:

$\frac{1}{8}$ $\frac{1}{4}$ $\frac{2}{3}$ $\frac{5}{12}$ $\frac{11}{12}$ $\frac{8}{9}$

Large Group Introduction

1. Show students this problem and ask them to write a paragraph explaining how they could estimate a reasonable answer.

   Marge ran $\frac{3}{4}$ *mile and stopped to catch her breath. She then ran another* $\frac{1}{8}$ *mile. Did Marge run more or less than one mile? Did Marge run* $\frac{4}{12}$ *mile?*

2. Have students share their paragraph with others in their group and then select a sample for large group sharing.

3. Have students act out the problem with fraction circles.

4. Repeat for this subtraction story.

   Allie ran $\frac{2}{3}$ *of a mile. Mark ran* $\frac{5}{12}$ *of a mile. Who ran the farthest? How much further?*

   (Don't be surprised if students can mentally find the exact answer).

Comments

The estimation problems here allow you to assess to what extent students' mental images of fractions help them reason about adding and subtracting fractions.

First problem looks at students' understanding of fractions > $\frac{1}{2}$ as well as comparing fractions with the same numerator ($\frac{1}{4}$ vs. $\frac{1}{8}$).

Possible student response:

∞ It has to be greater than $\frac{1}{2}$ but less than 1. To make one whole you'd need to add $\frac{1}{4}$ more. But $\frac{1}{8} < \frac{1}{4}$ so it's not 1 mile.

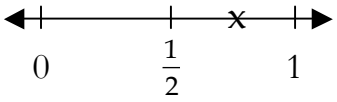
∞ $\frac{4}{12}$ doesn't make sense because $\frac{4}{12} < \frac{1}{2}$.

Teaching Actions

Small Group/Partner Work

5. In groups have students estimate a reasonable answer to each problem on Student Page A. They can show their estimates by marking the interval on the number line with an "x". Encourage students to talk about their strategies for estimation.

$$\frac{2}{3} + \frac{1}{20}$$



   - ∞ 2-thirds is greater than $\frac{1}{2}$. 1-twentieth is a very small fraction (much smaller than $\frac{1}{3}$). Therefore the sum has to be greater than $\frac{1}{2}$ but less than one.

Wrap Up

6. End class by sharing their thinking strategies. Congratulate children for learning so much about fractions. Point out to them examples of their quantitative thinking that shows just how much they know about these new numbers.

Comments

Students are not to find the exact location on the number line. The number line is being used only as a way to record a reasonable range for the problem.

You may want to use a multiple choice format:

a) between 0 and $\frac{1}{2}$

b) between $\frac{1}{2}$ and 1

The problems may be challenging, but use them to assess the strength of students' mental representations for fraction symbols.

Translations

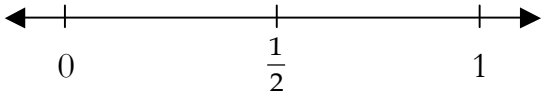
- ∞ Symbols to verbal
- ∞ Real life to written symbols

Order these fractions:

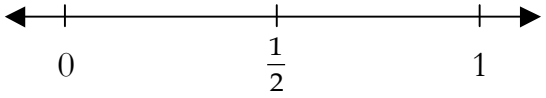
$\frac{1}{8}$ $\frac{1}{4}$ $\frac{2}{3}$ $\frac{5}{12}$ $\frac{11}{12}$ $\frac{8}{9}$

Estimating Fraction Sums and Differences

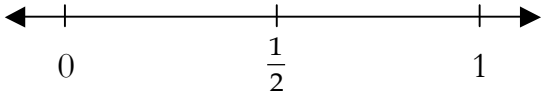
1) $\frac{2}{3} + \frac{1}{20}$



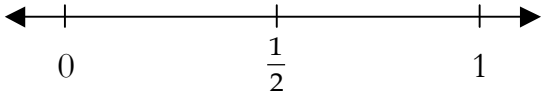
2) $\frac{2}{3} - \frac{1}{20}$



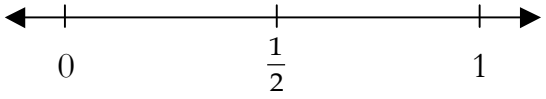
3) $\frac{9}{10} - \frac{8}{9}$



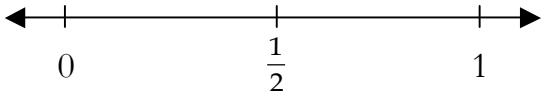
4) $\frac{5}{6} + \frac{1}{15}$



5) $\frac{1}{3} + \frac{1}{100}$

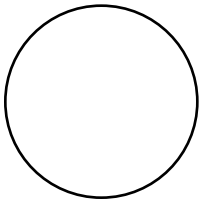


6) $\frac{3}{4} - \frac{4}{5}$

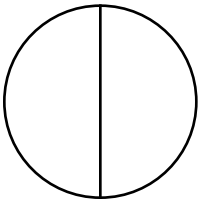


Appendix

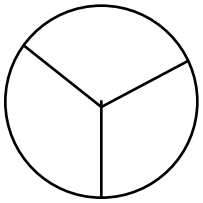
Pictures of the Fraction Circles



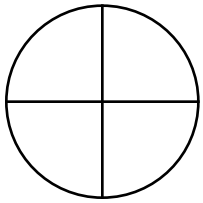
Black



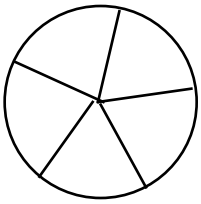
Yellow
2 pieces



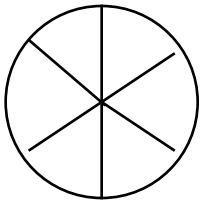
Brown
3 pieces



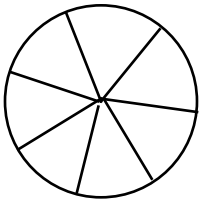
Blue
4 pieces



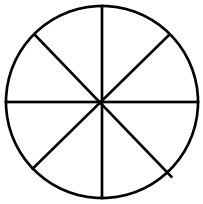
Orange
5 pieces



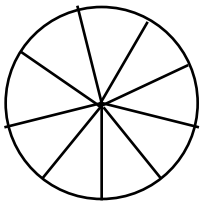
Pink
6 pieces



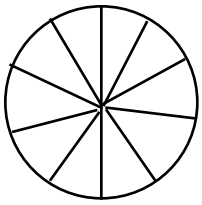
Light Blue
7 pieces



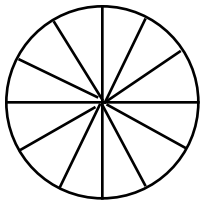
Gray
8 pieces



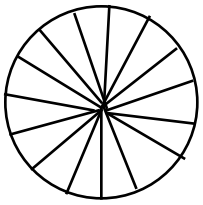
White
9 pieces



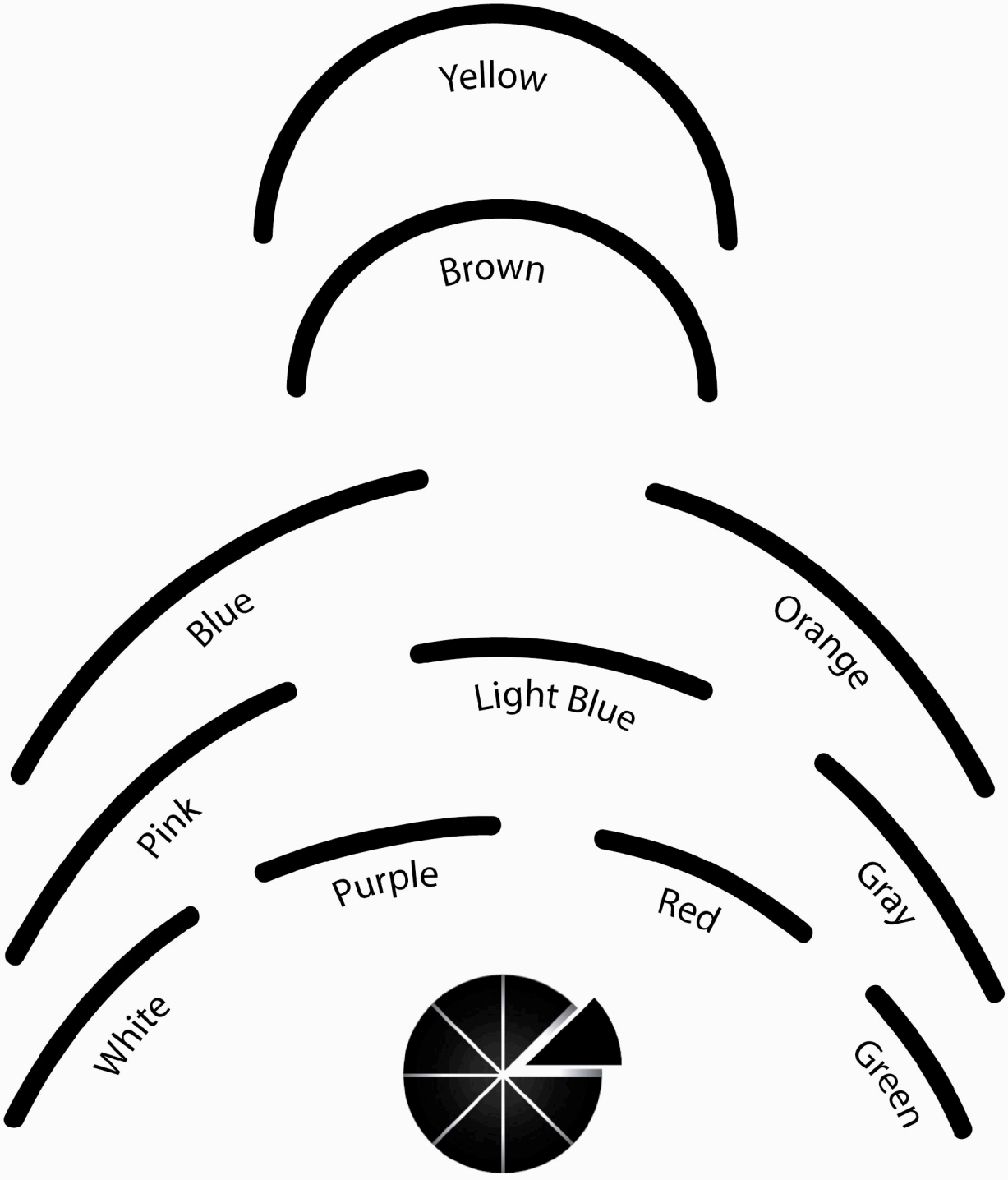
Purple
10 pieces



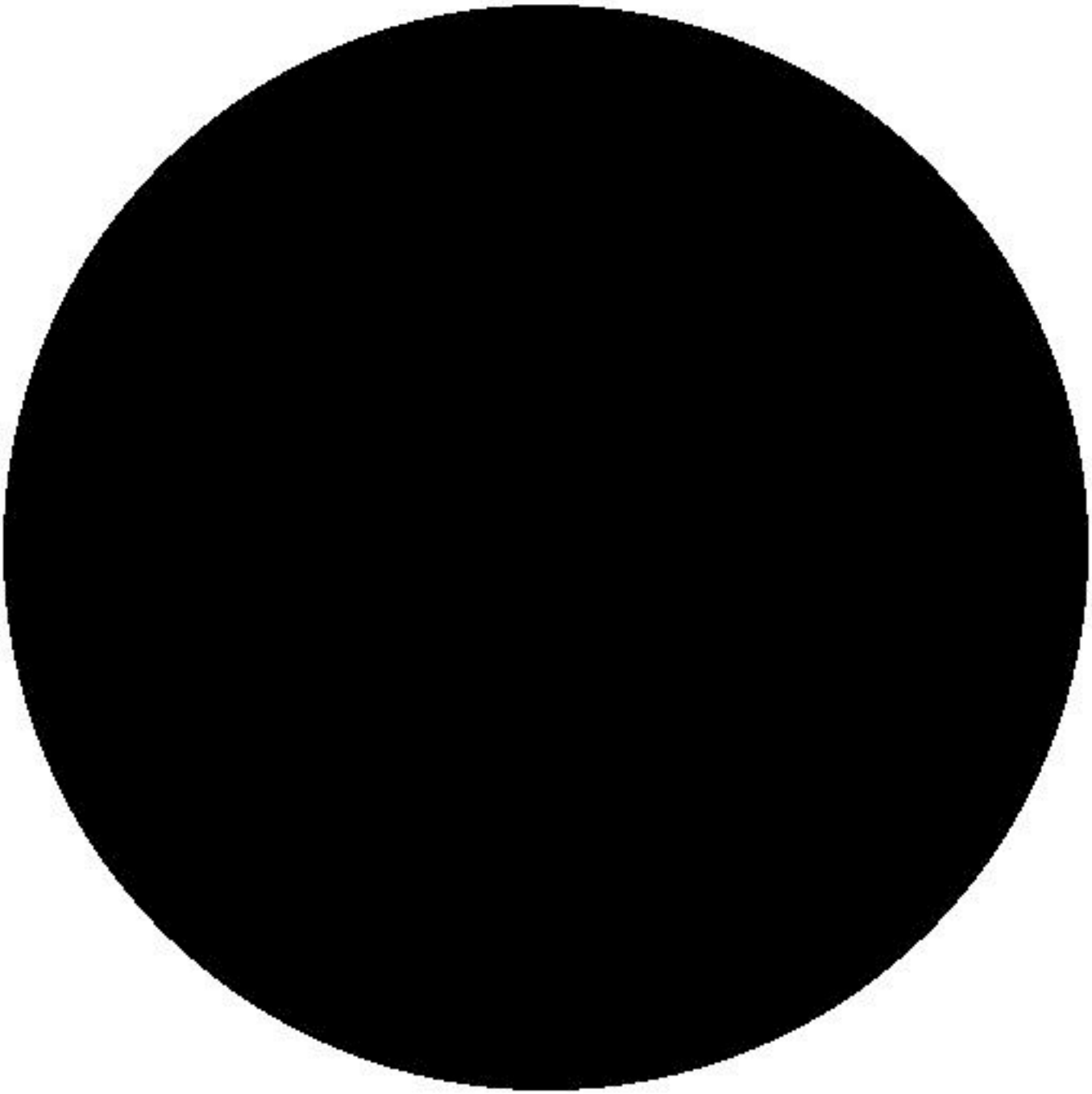
Red
12 pieces



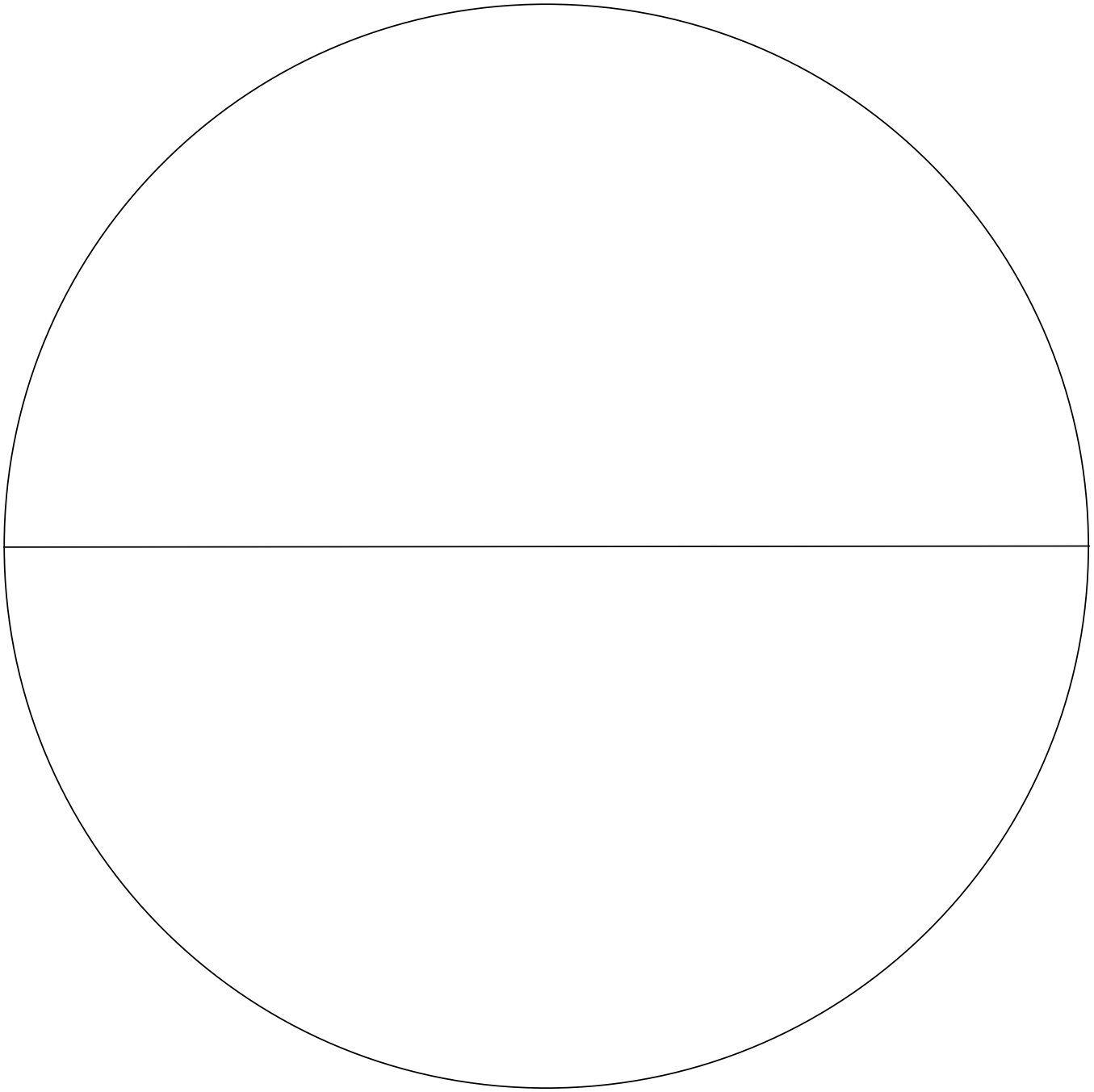
Green
15 pieces


Yellow
Brown
Blue
Orange
Light Blue
Pink
Gray
Purple
Red
White
Green

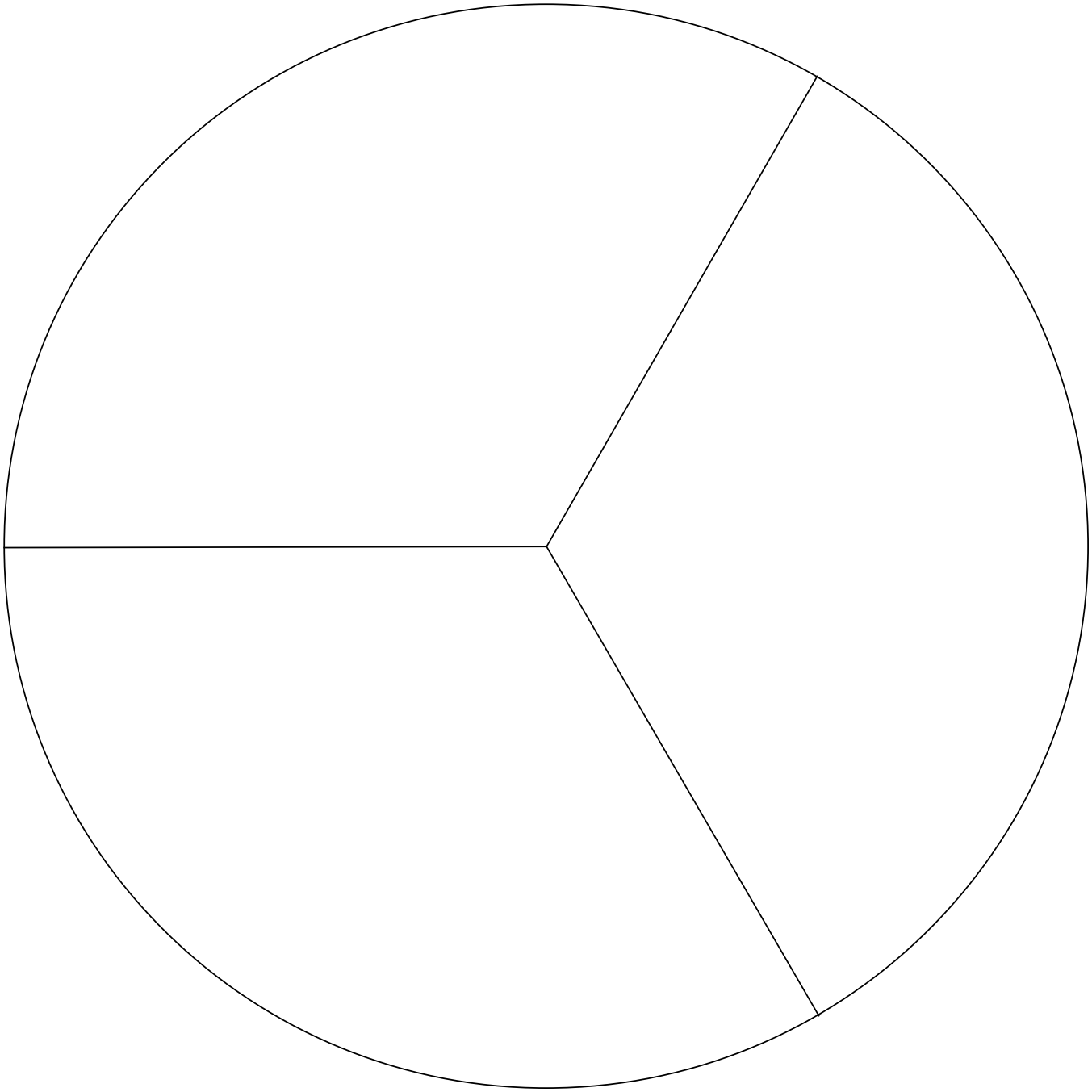
BLACK



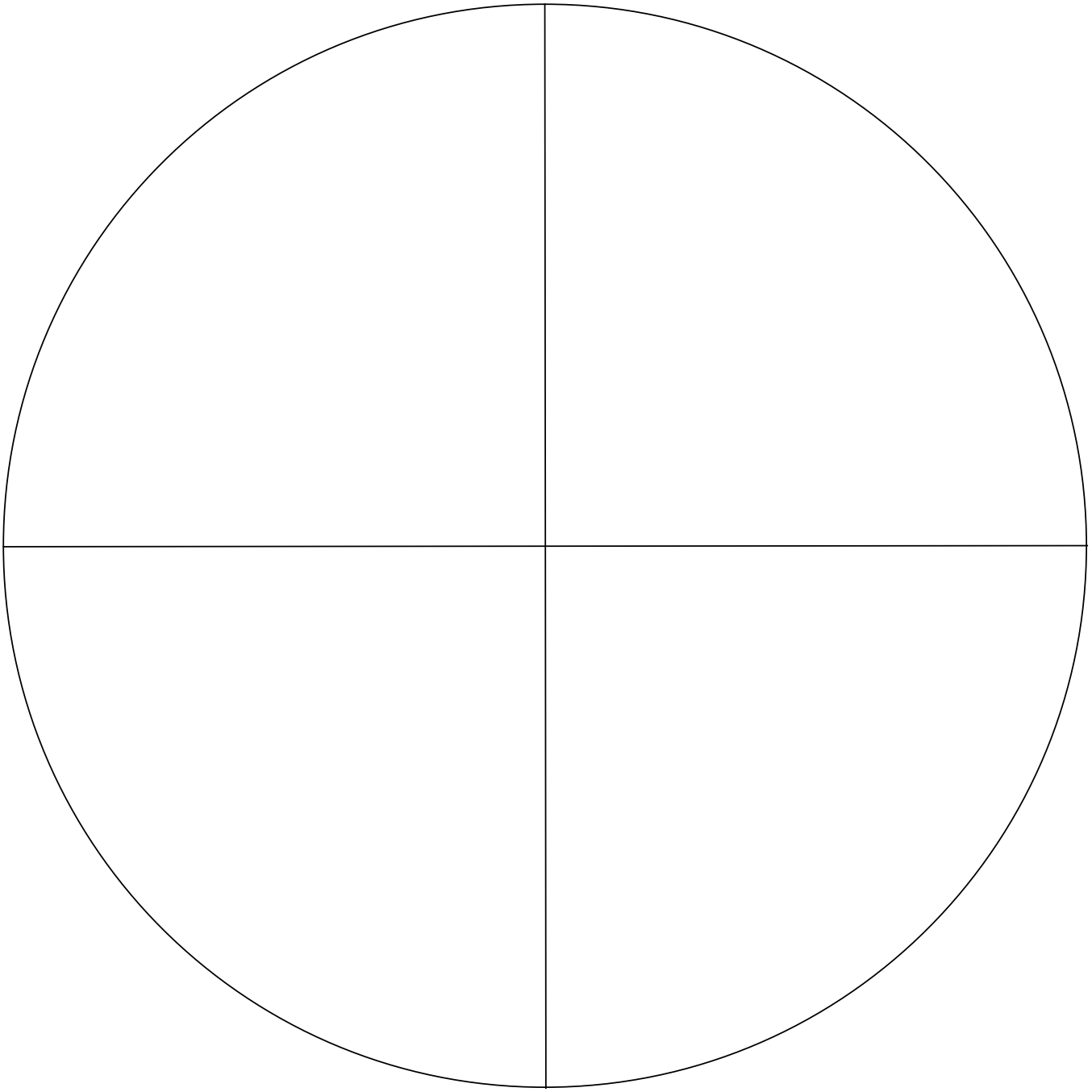
YELLOW



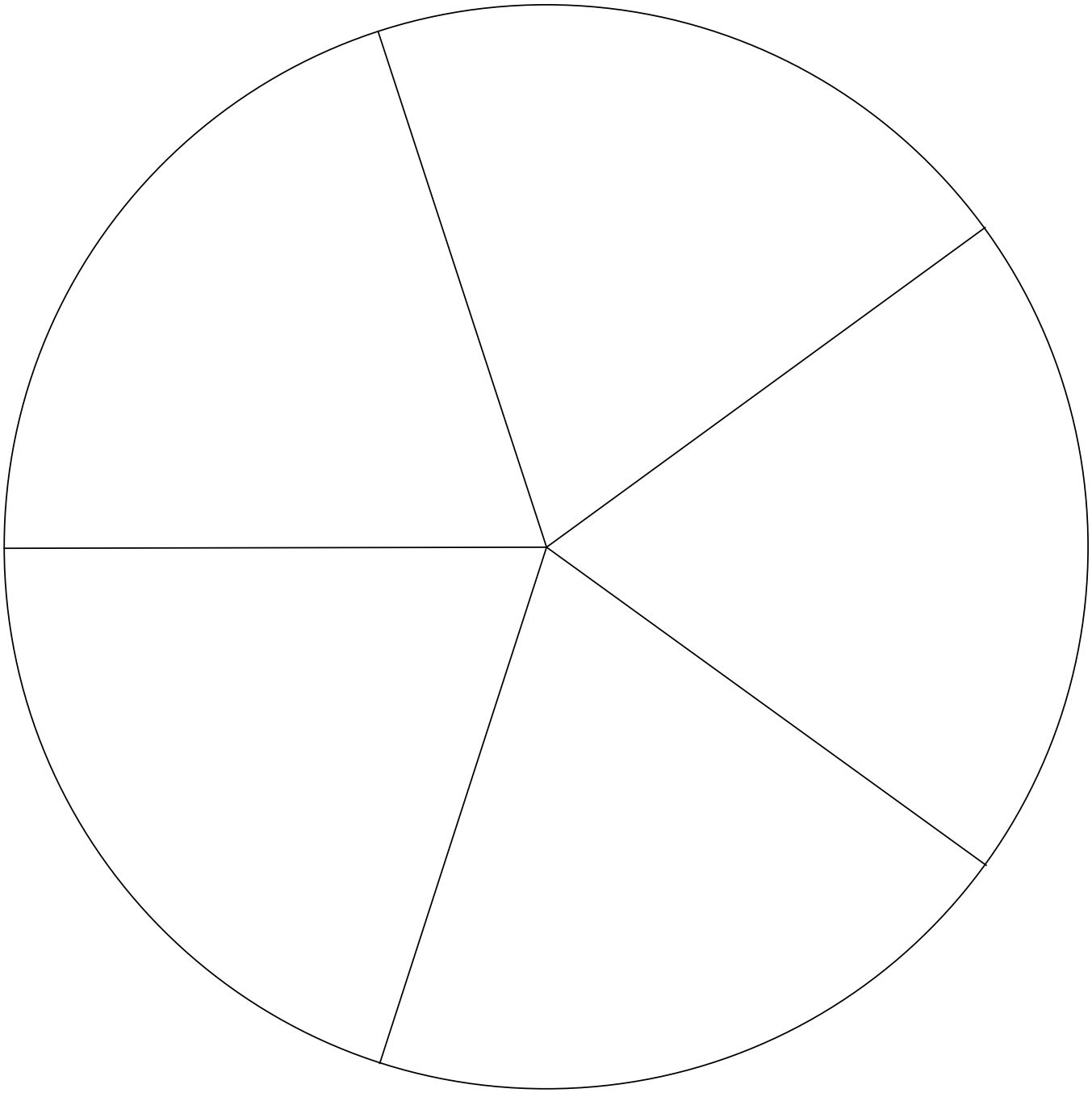
BROWN



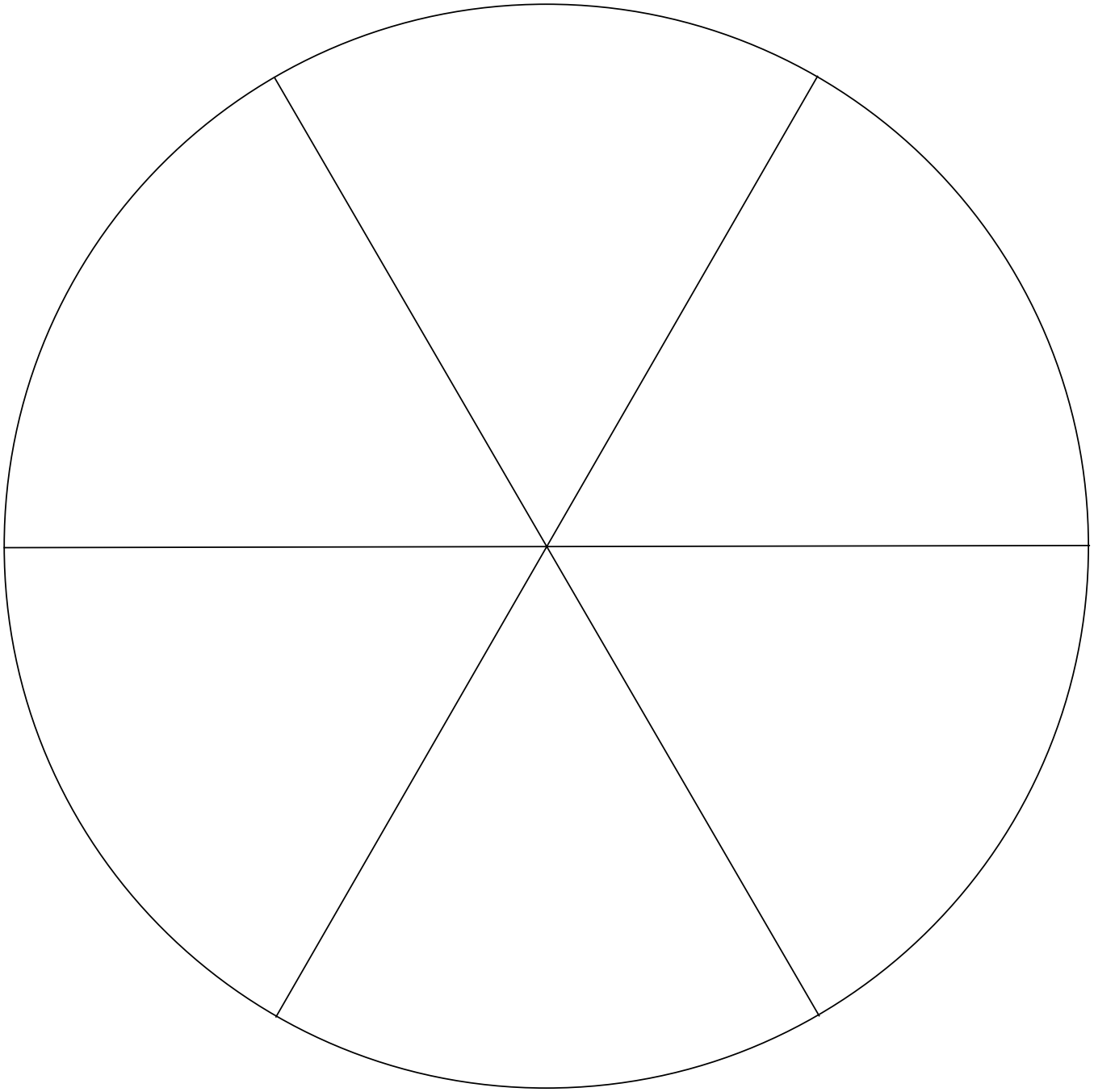
BLUE



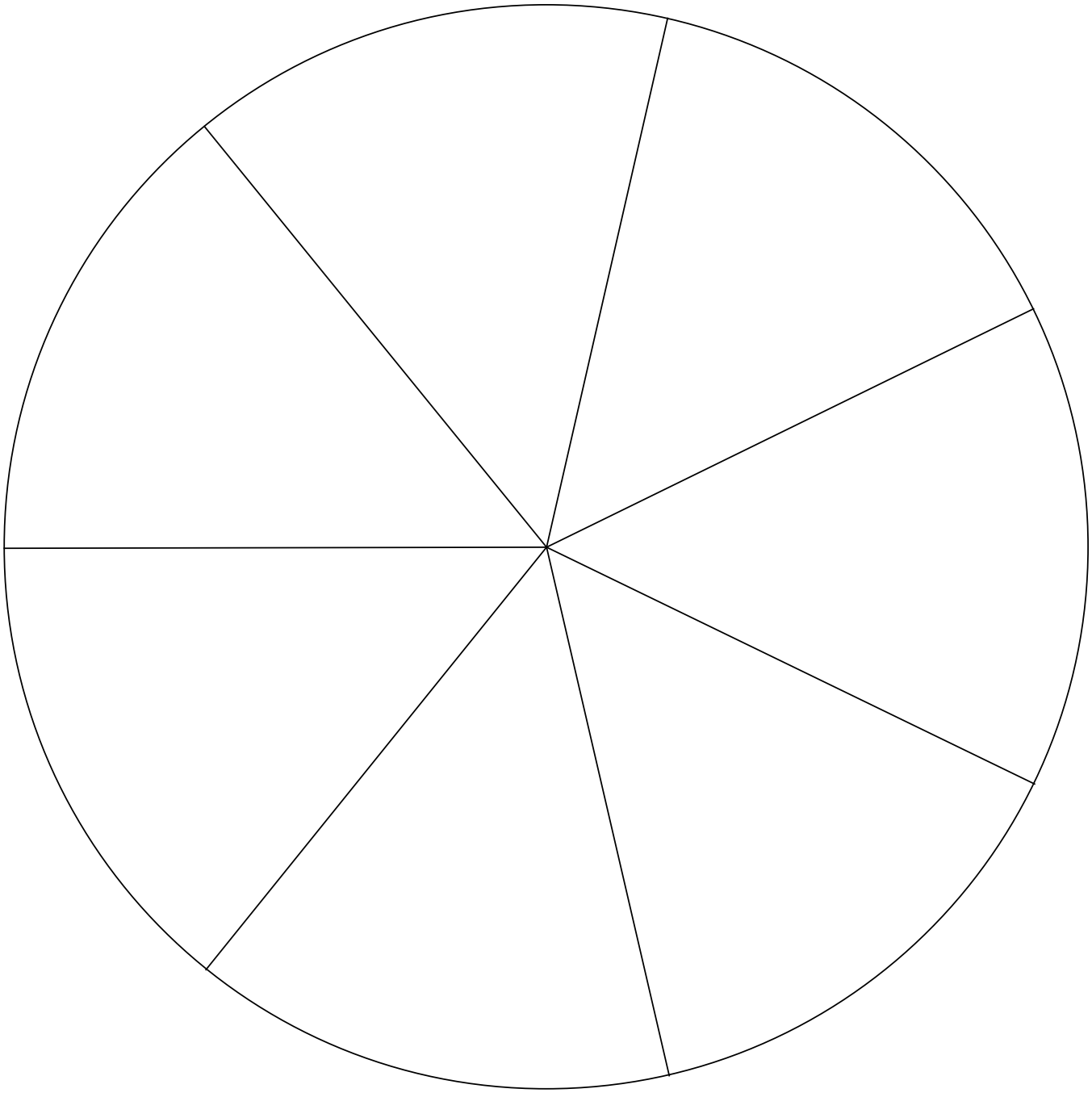
ORANGE



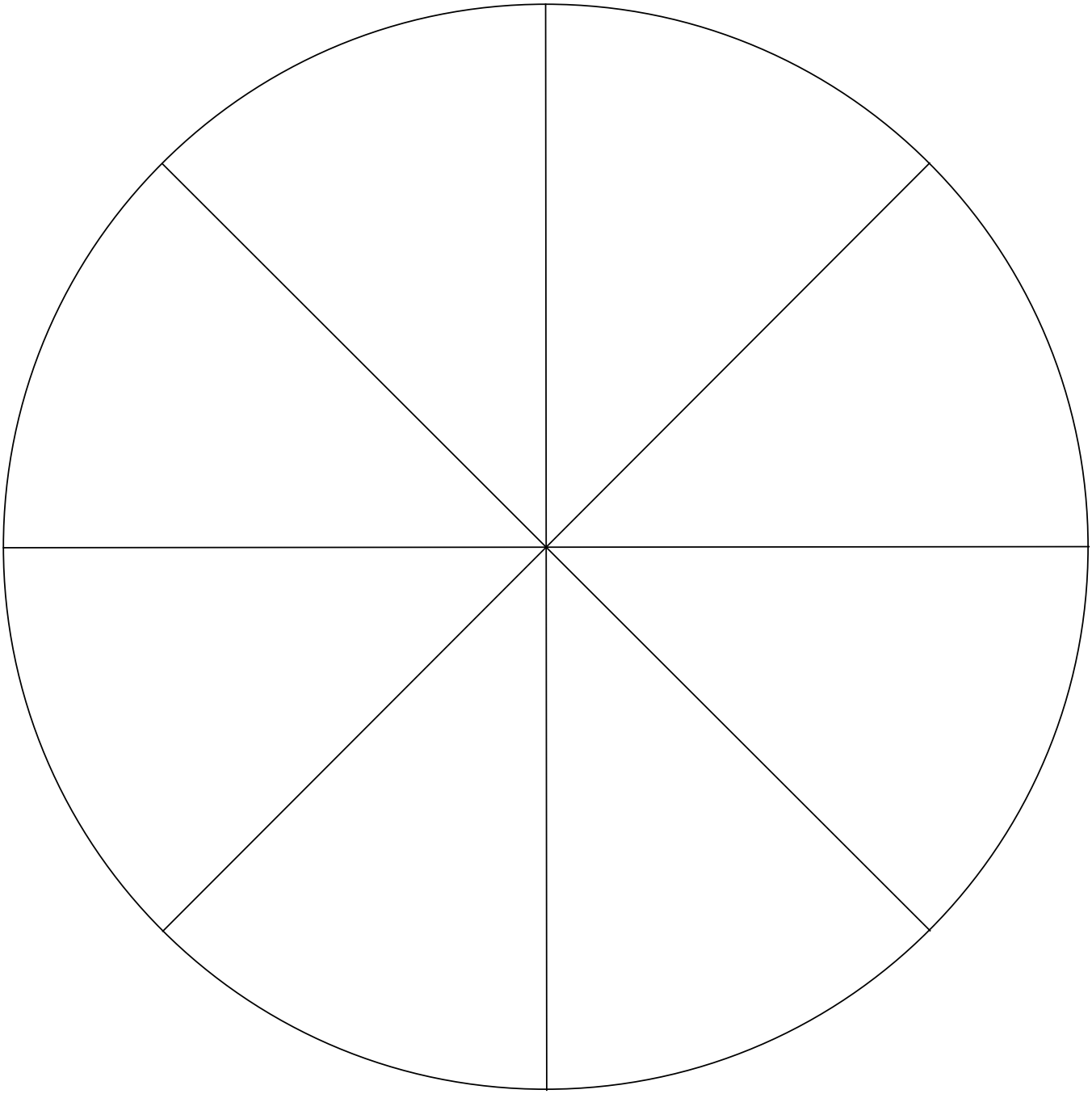
PINK



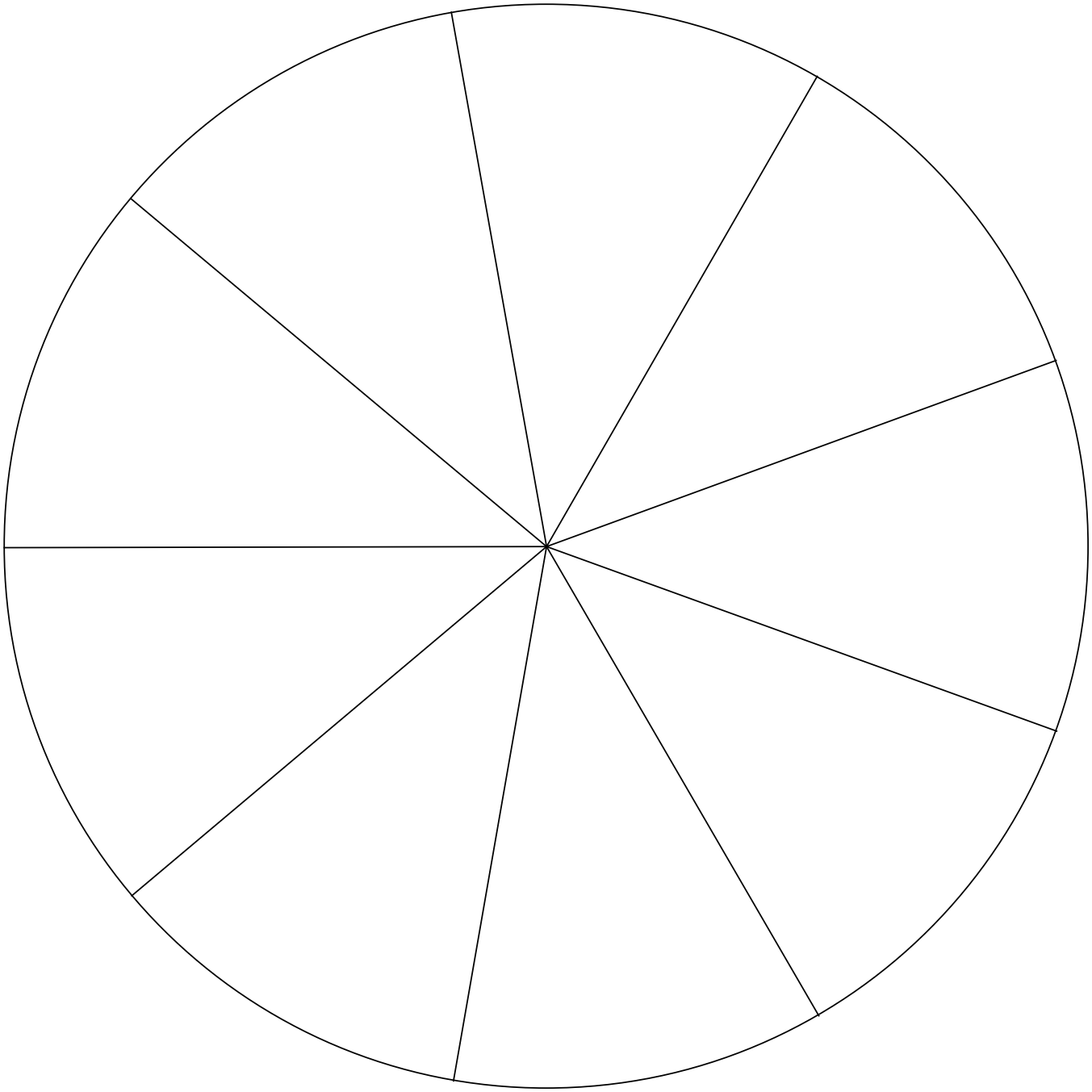
LIGHT BLUE



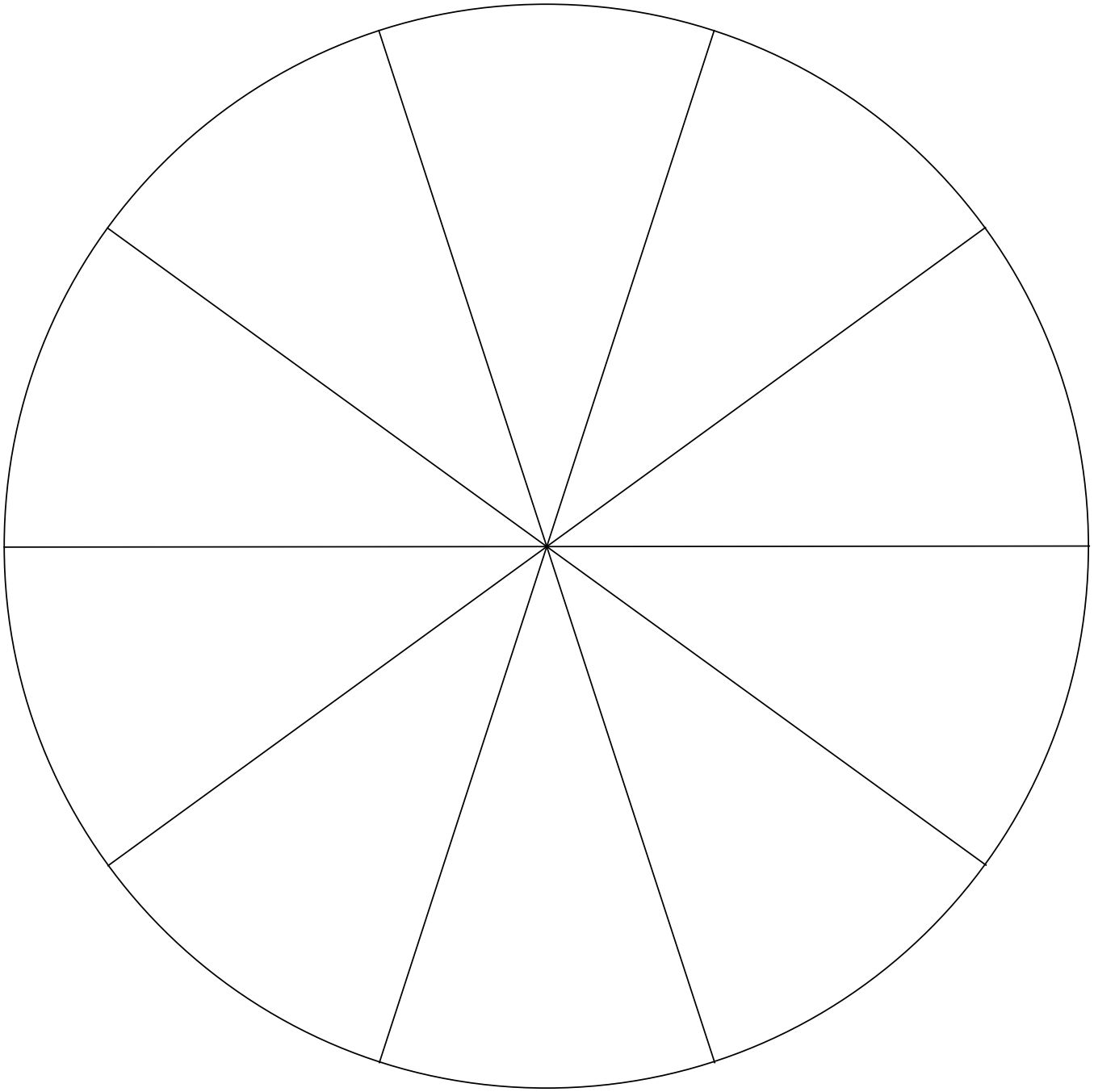
GRAY



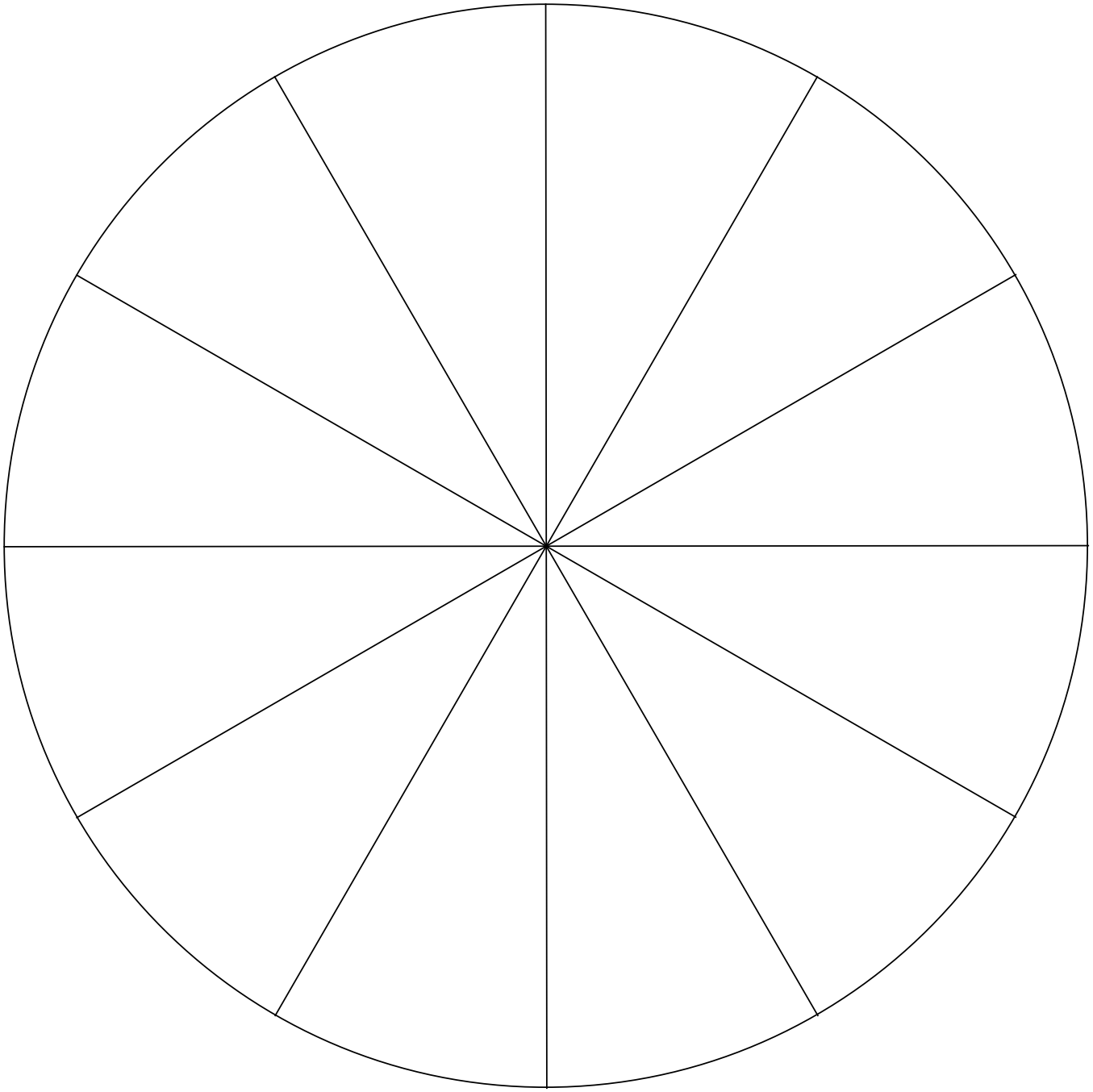
WHITE



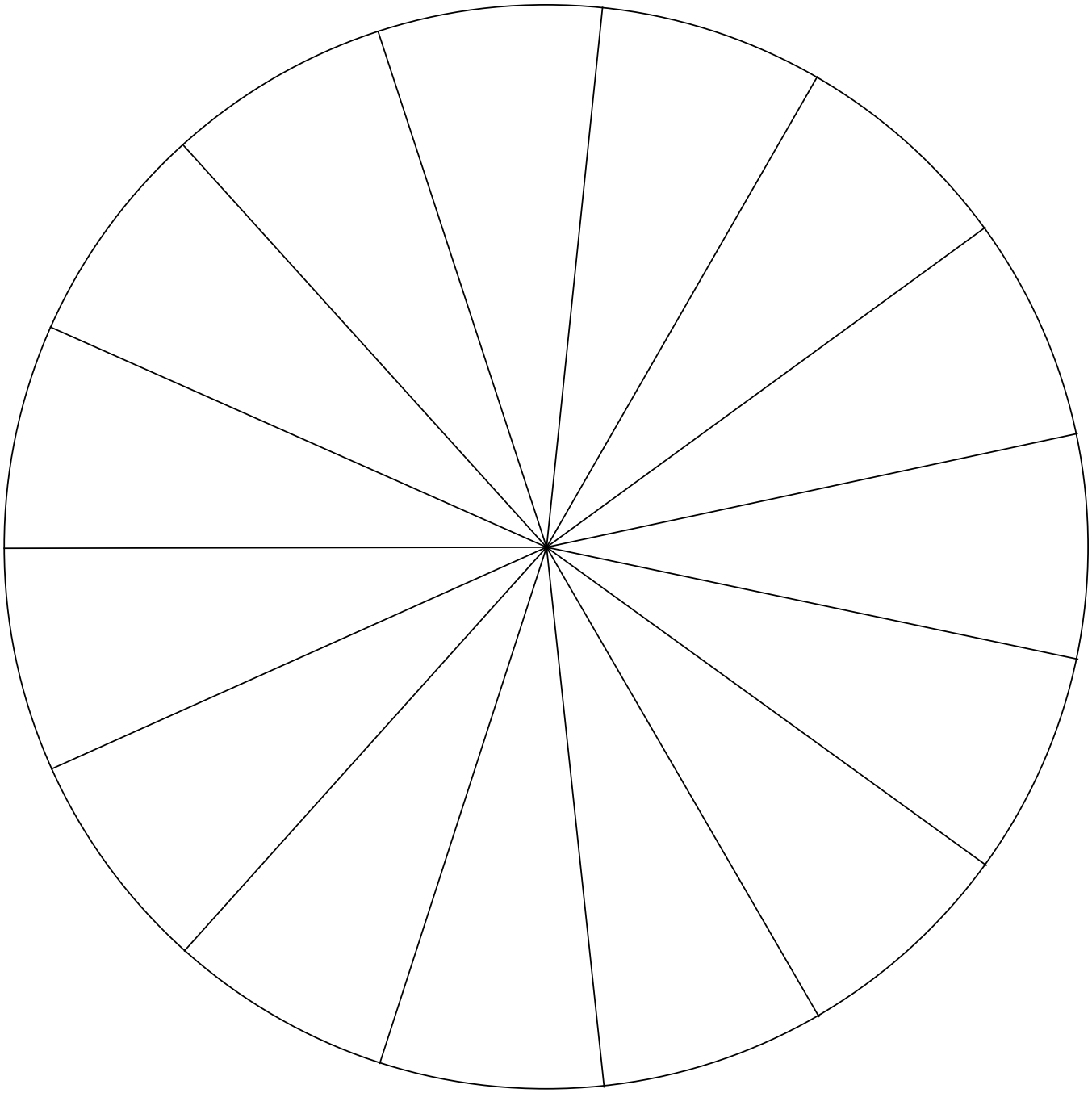
PURPLE



RED

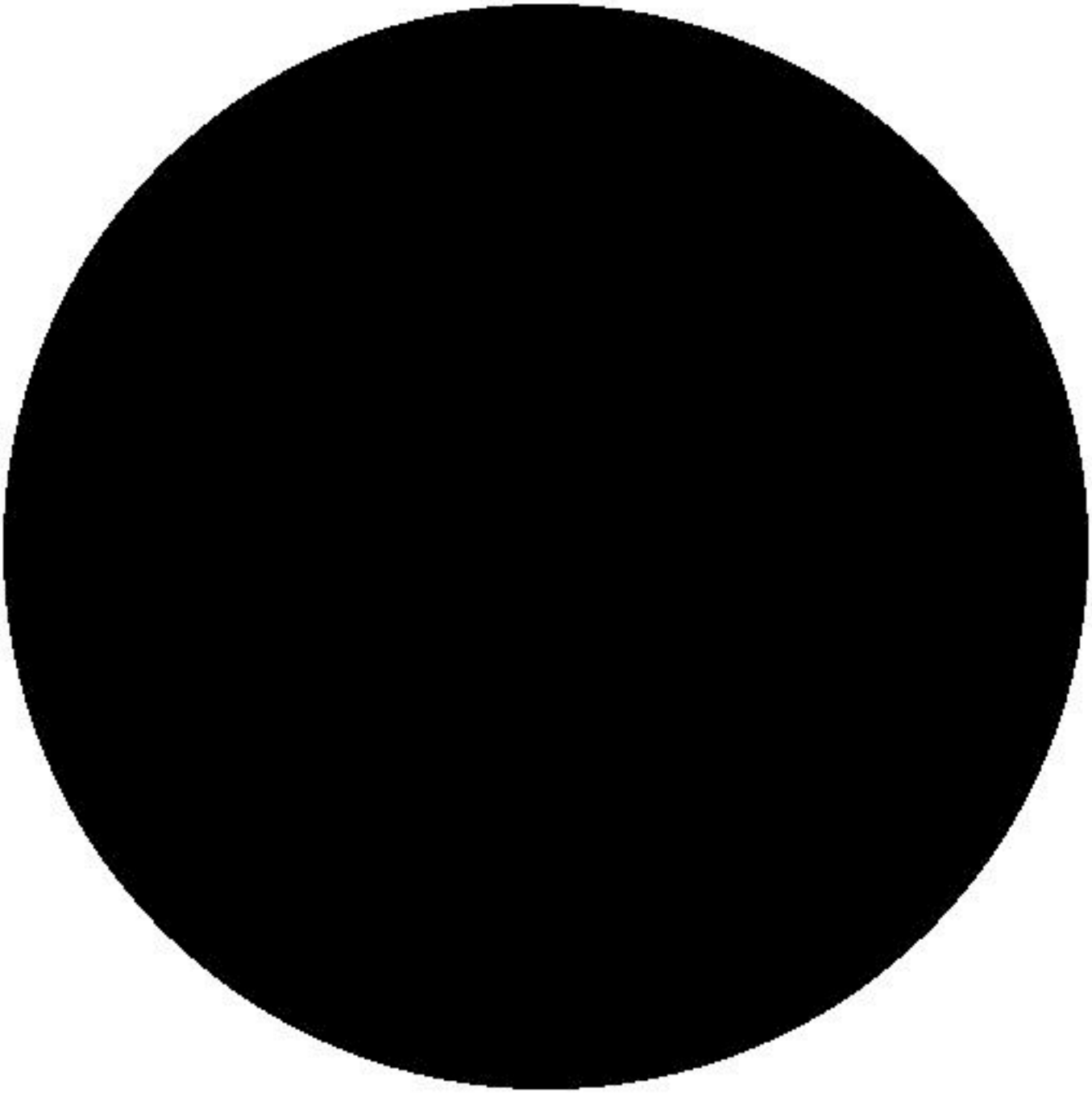


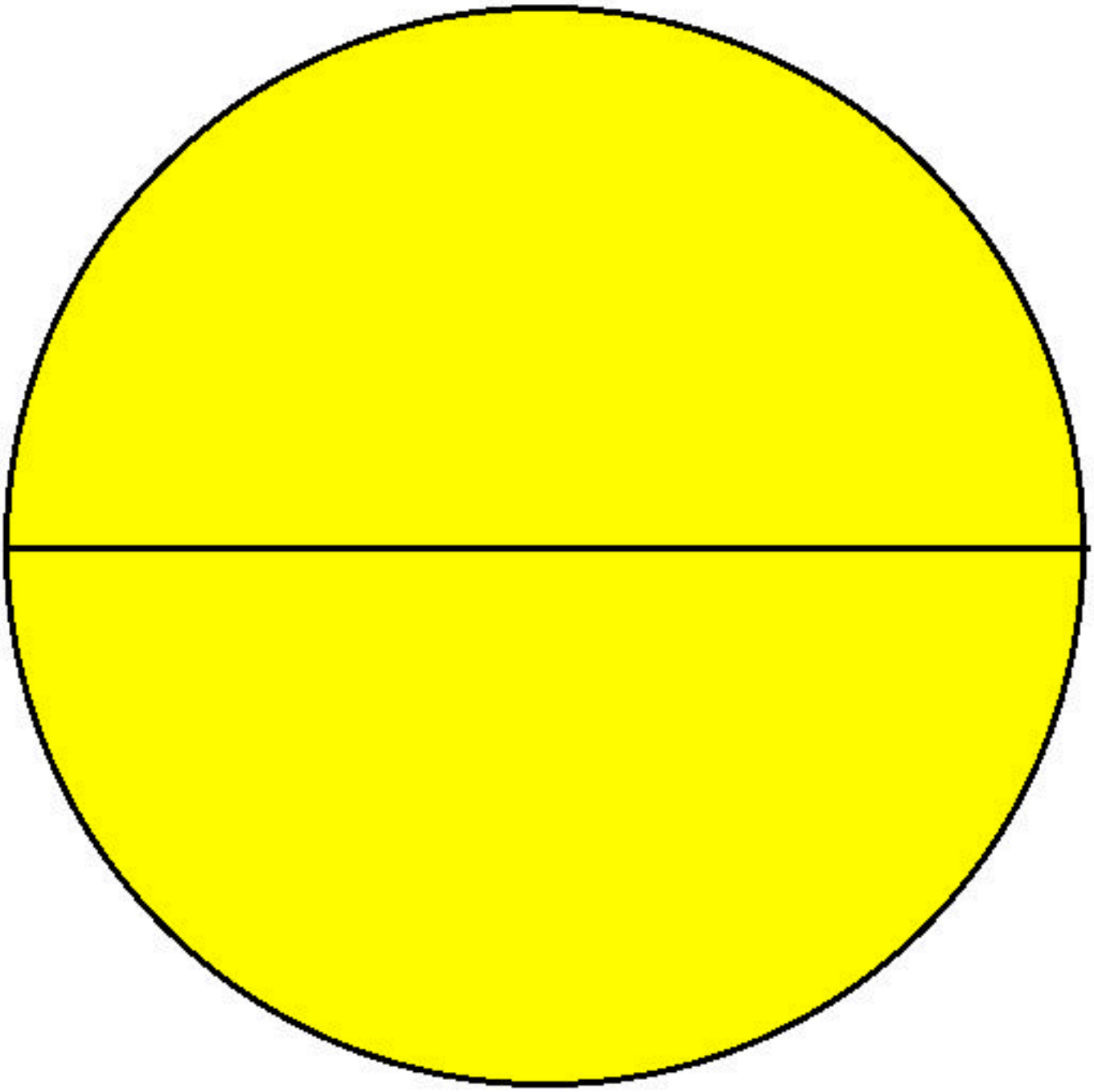
GREEN

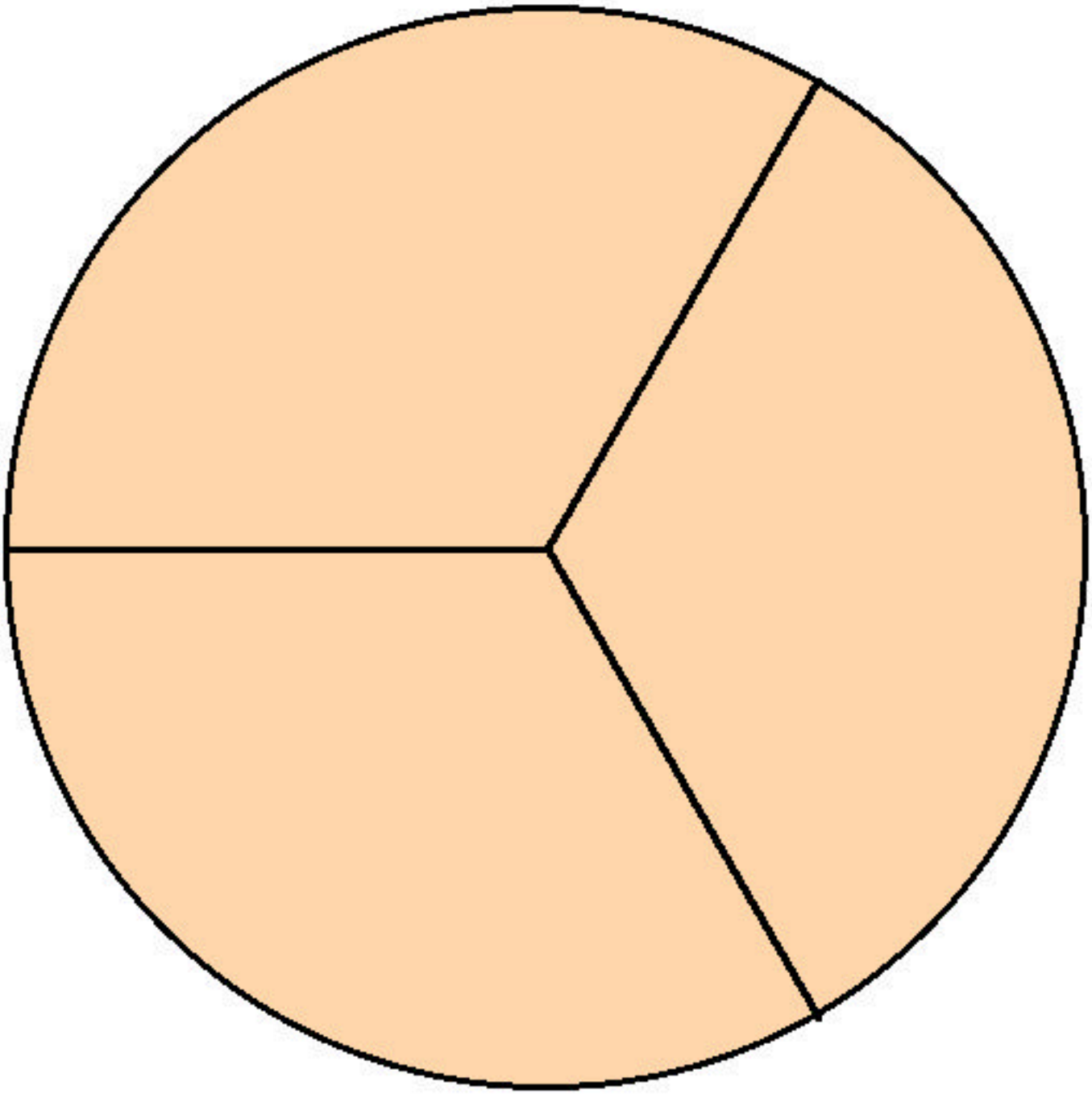


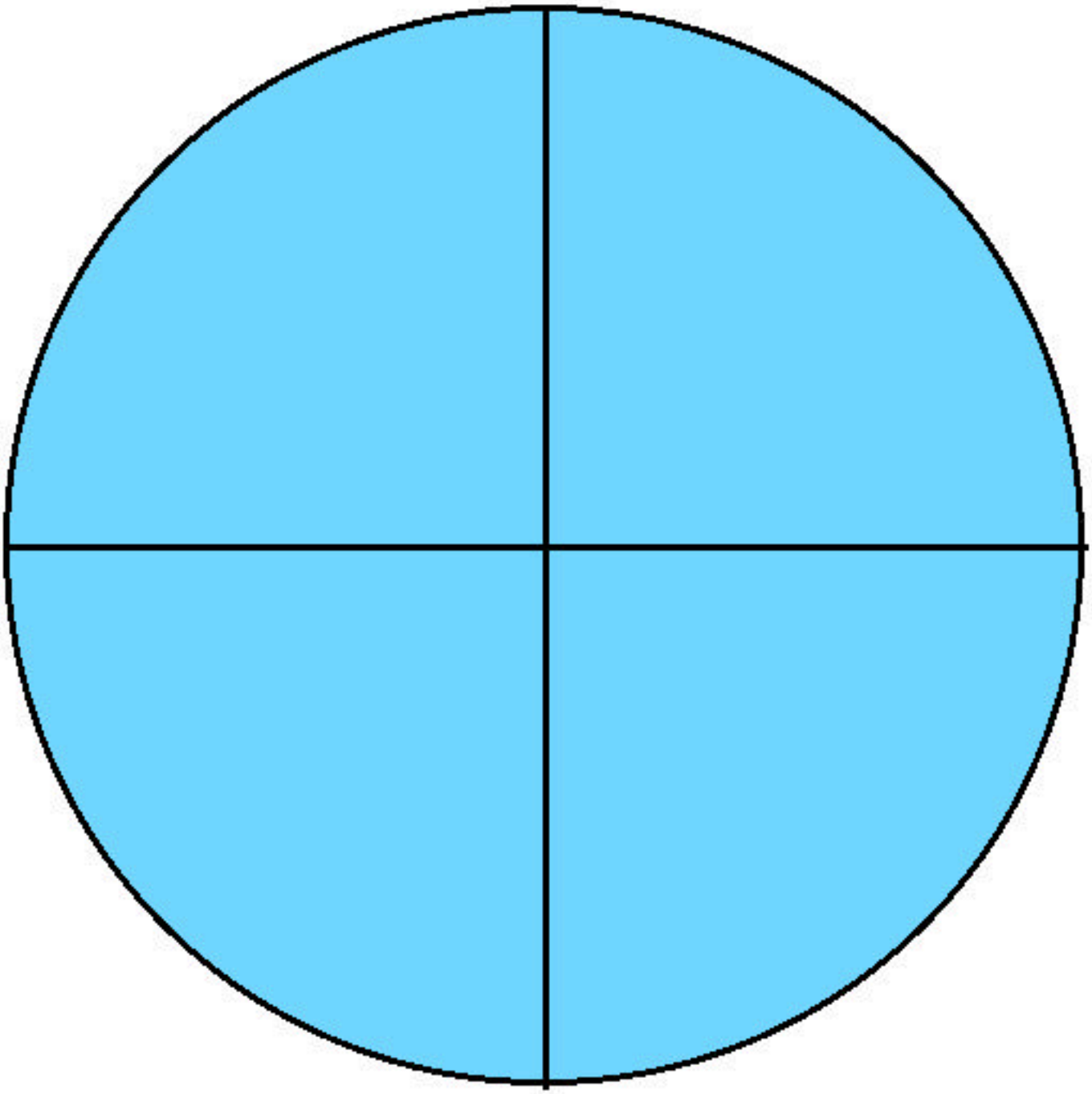
Fraction Circles
Colored Masters

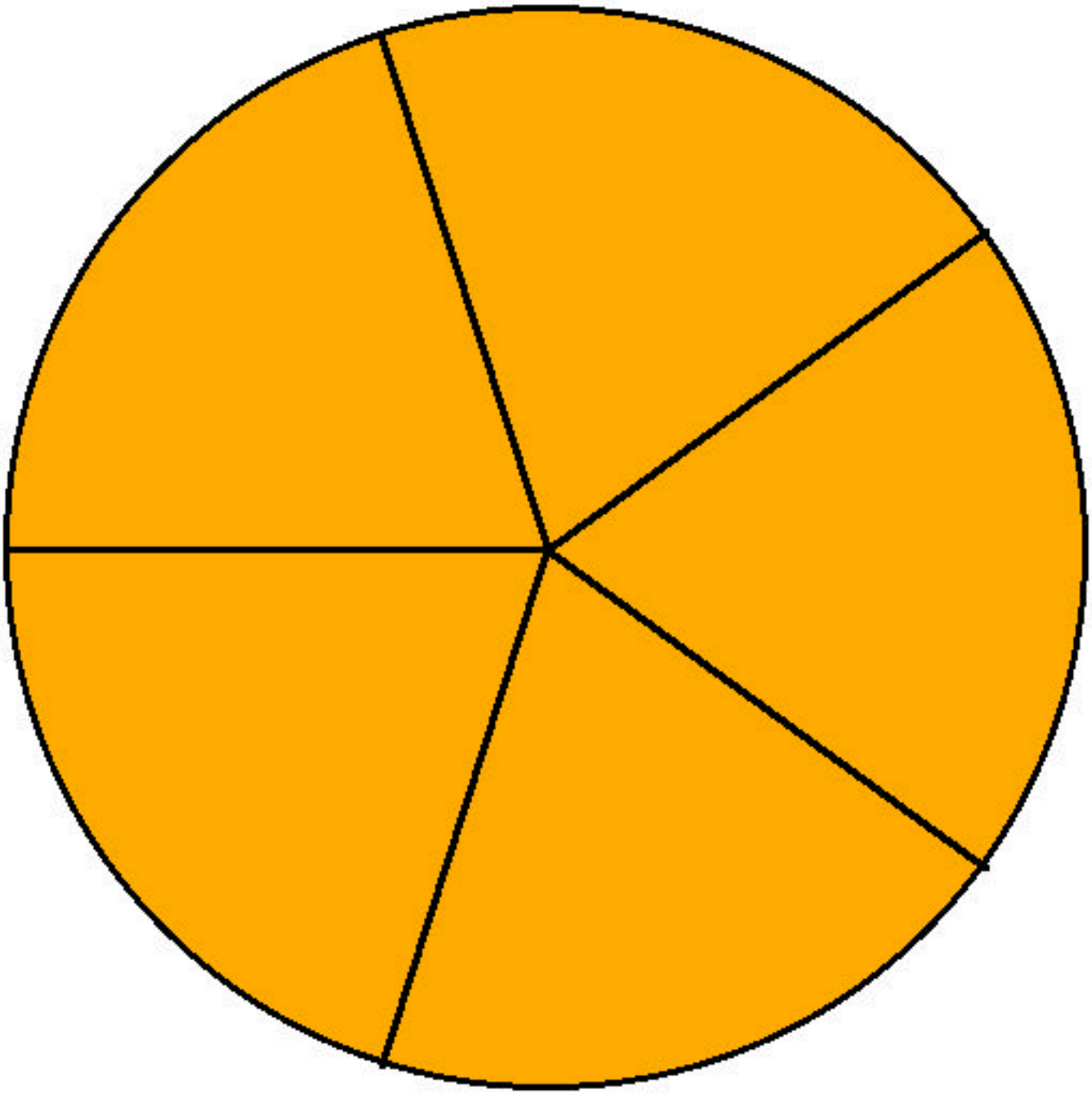
The following pages can be used to print
colored fraction circles on transparencies.

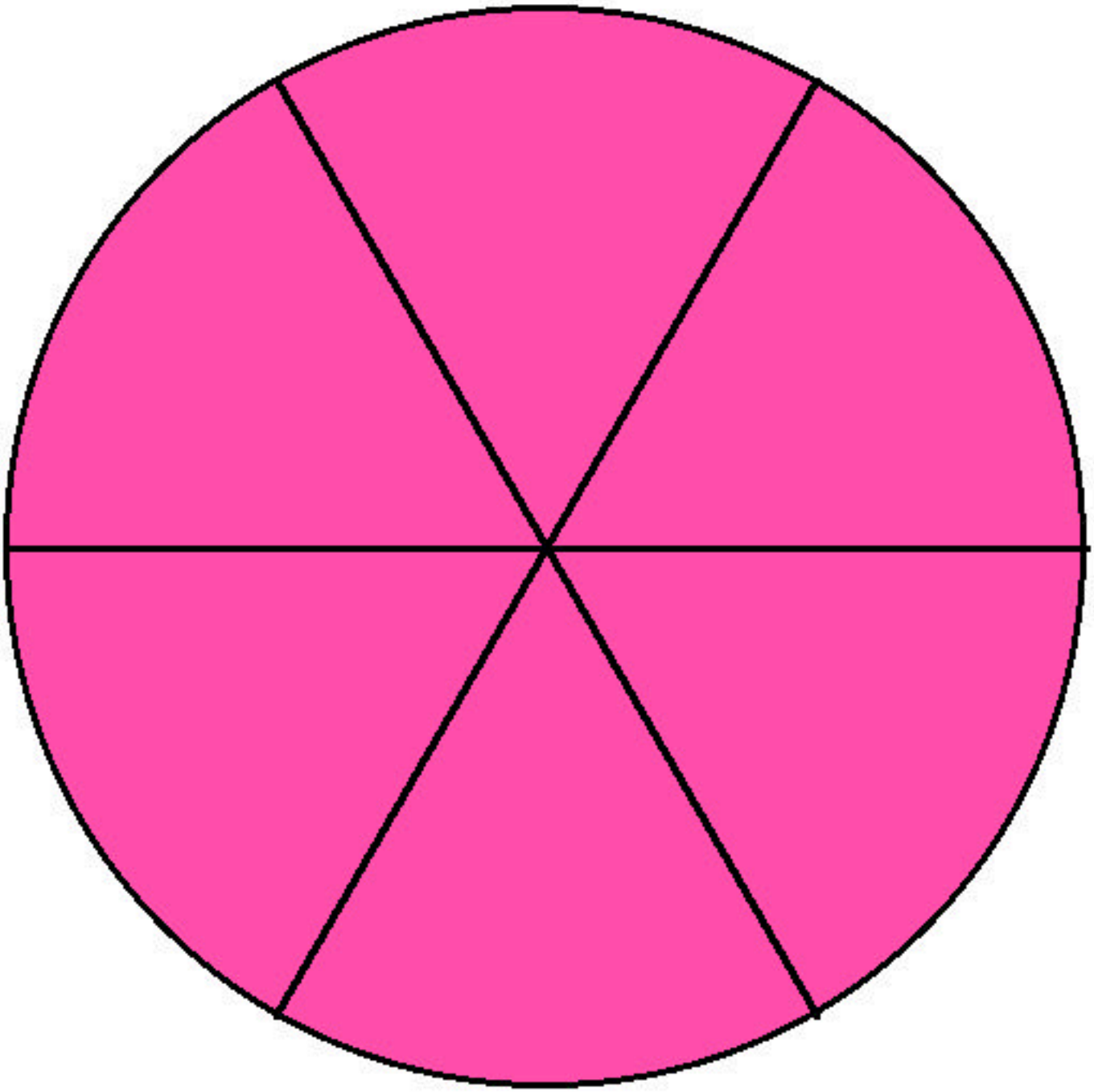


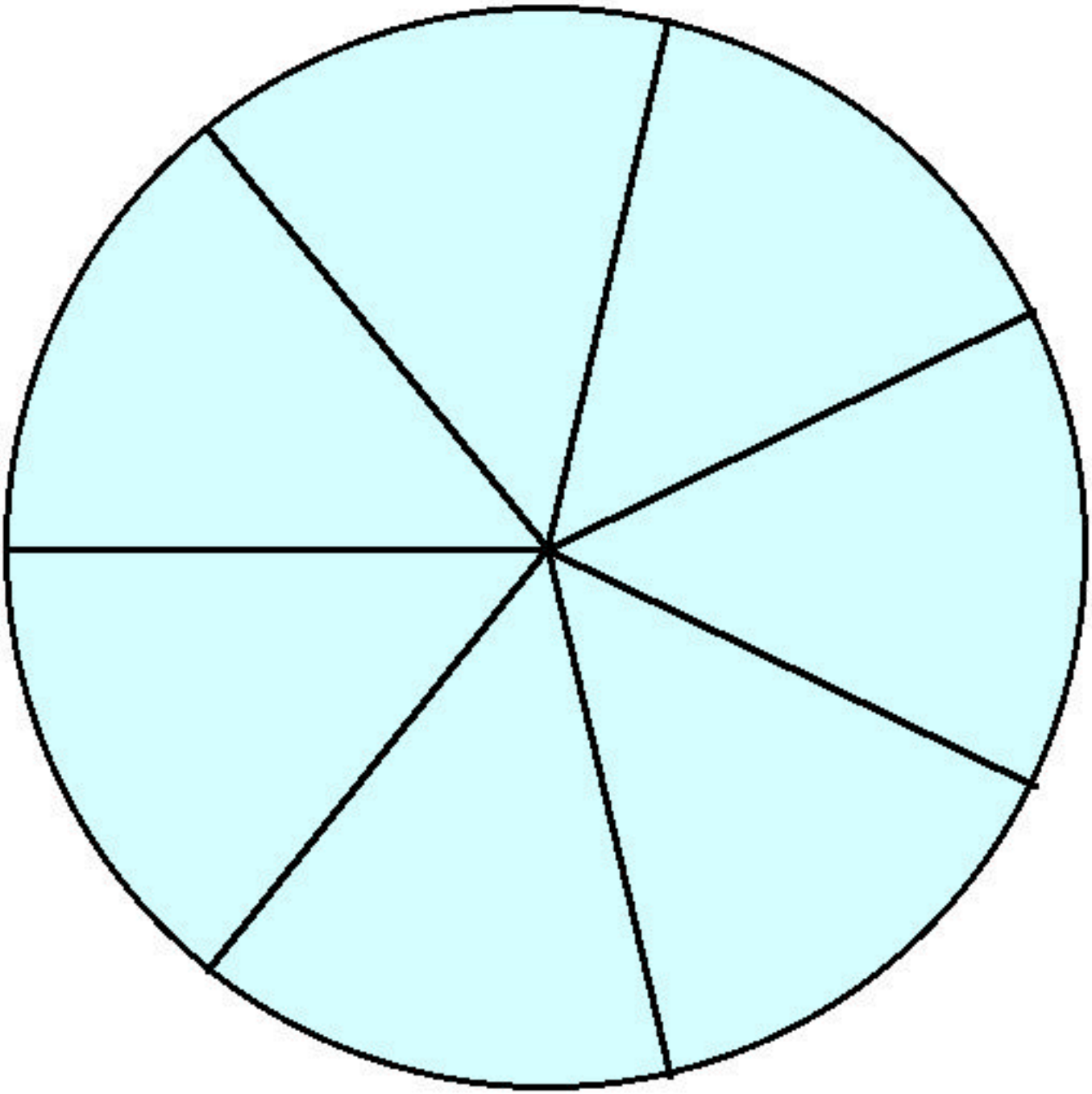


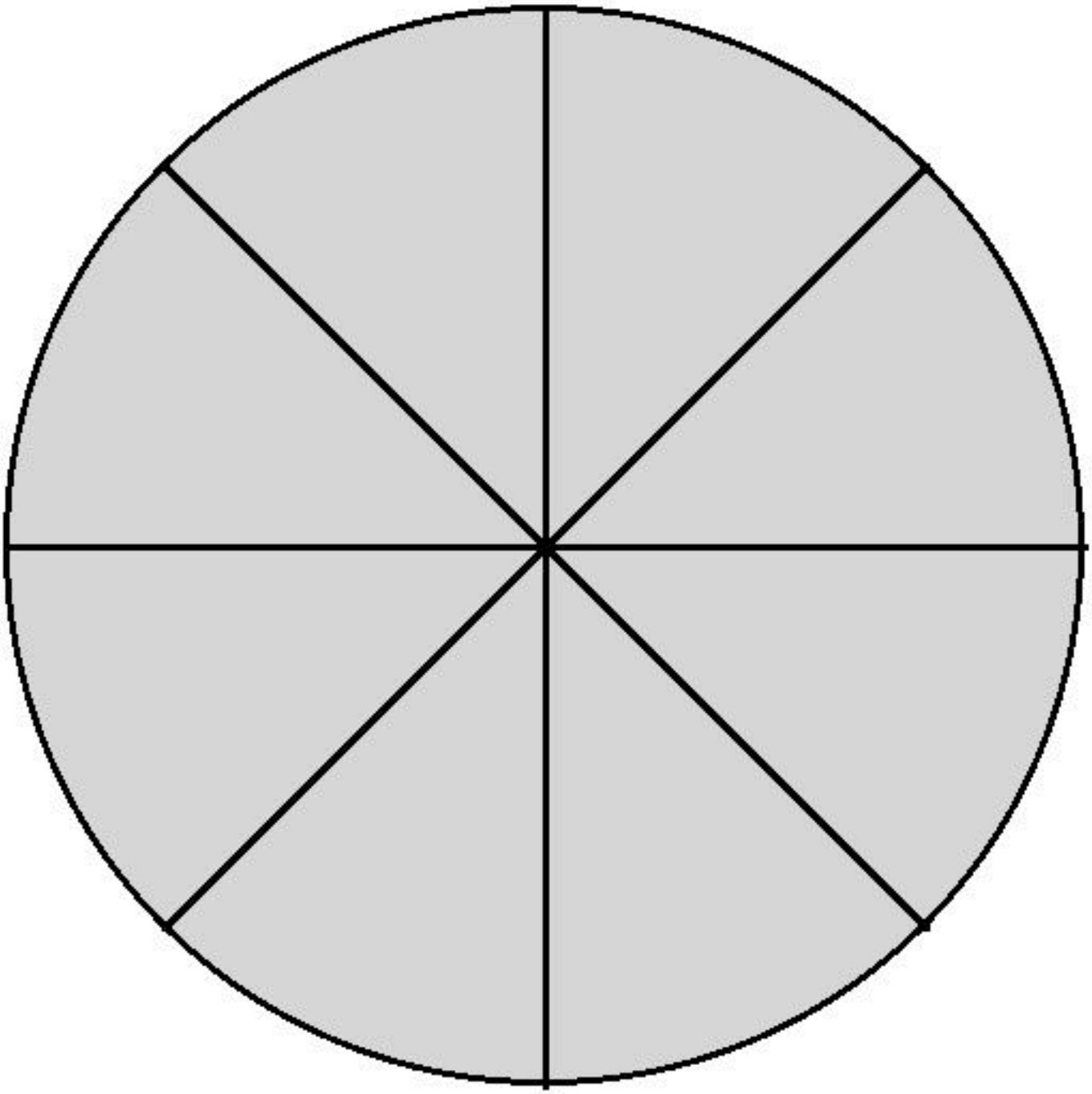


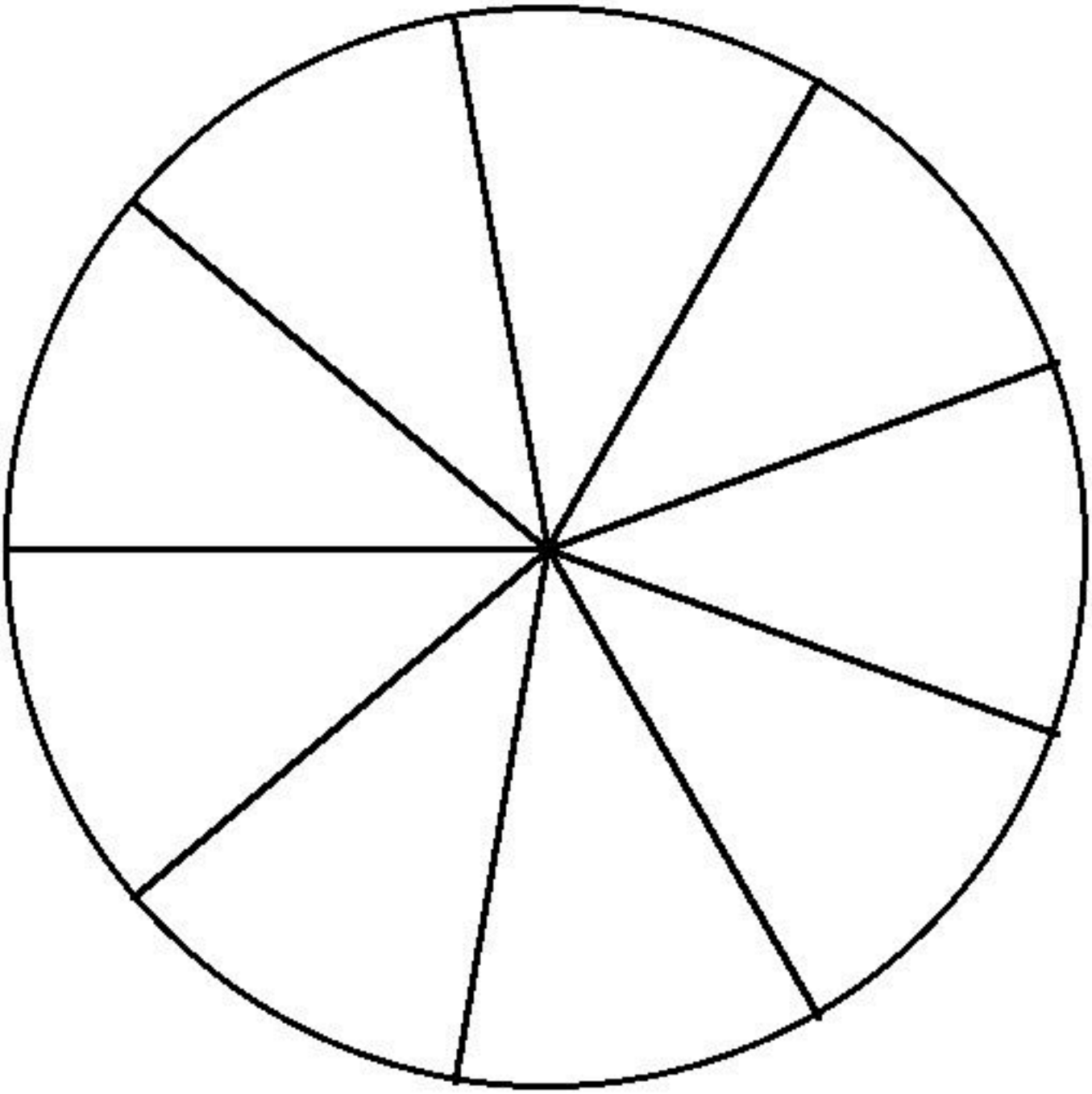


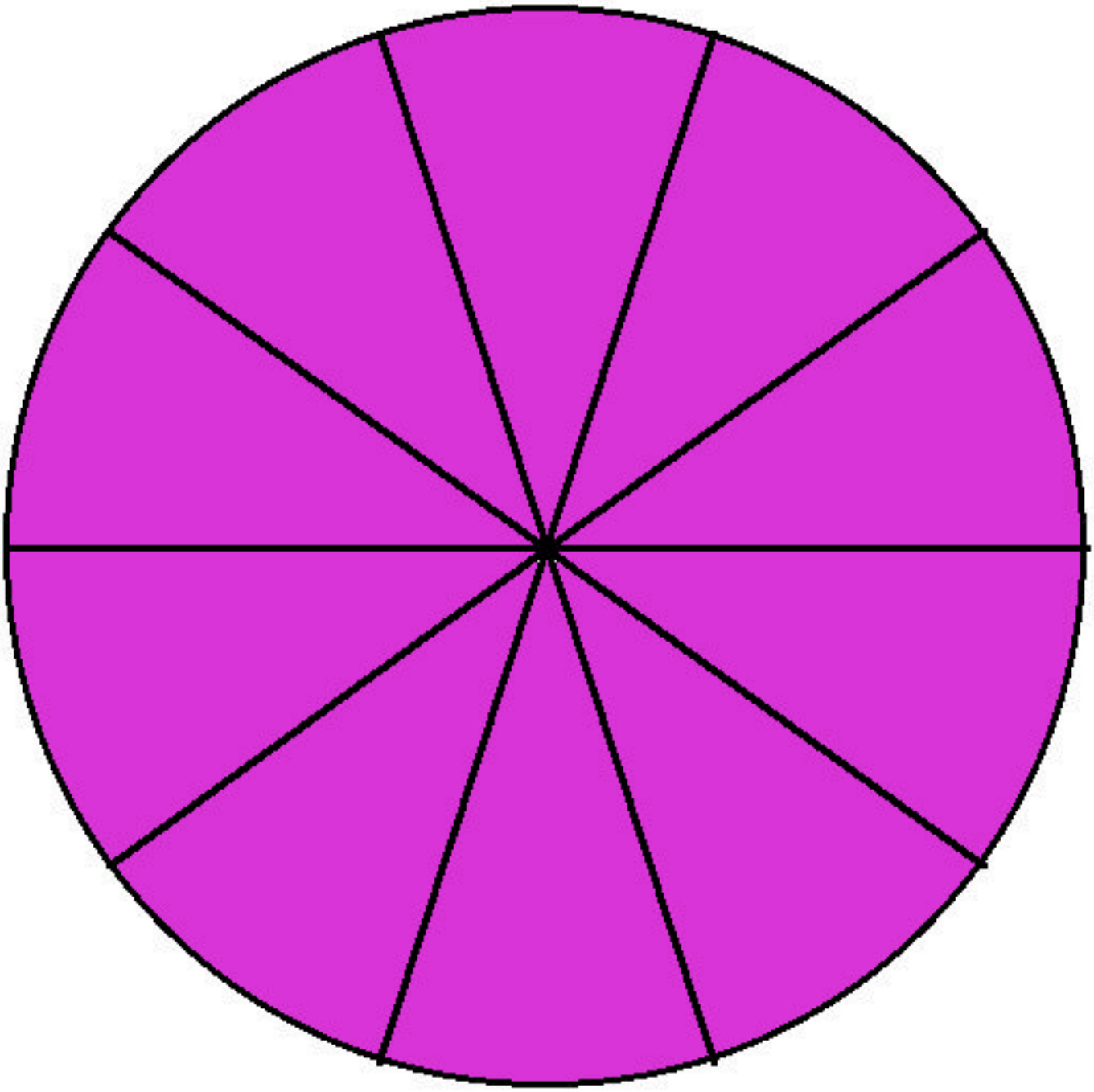


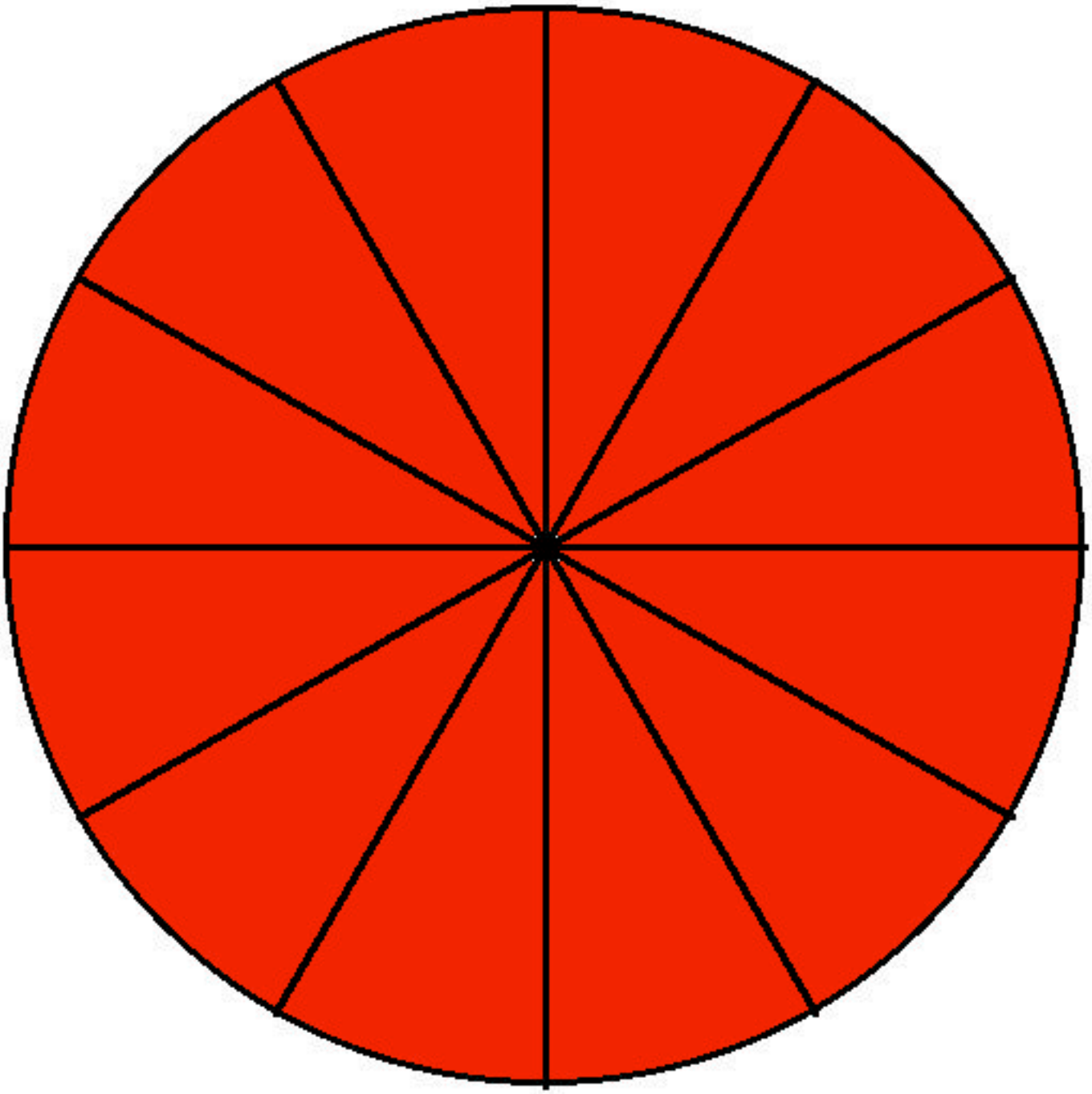


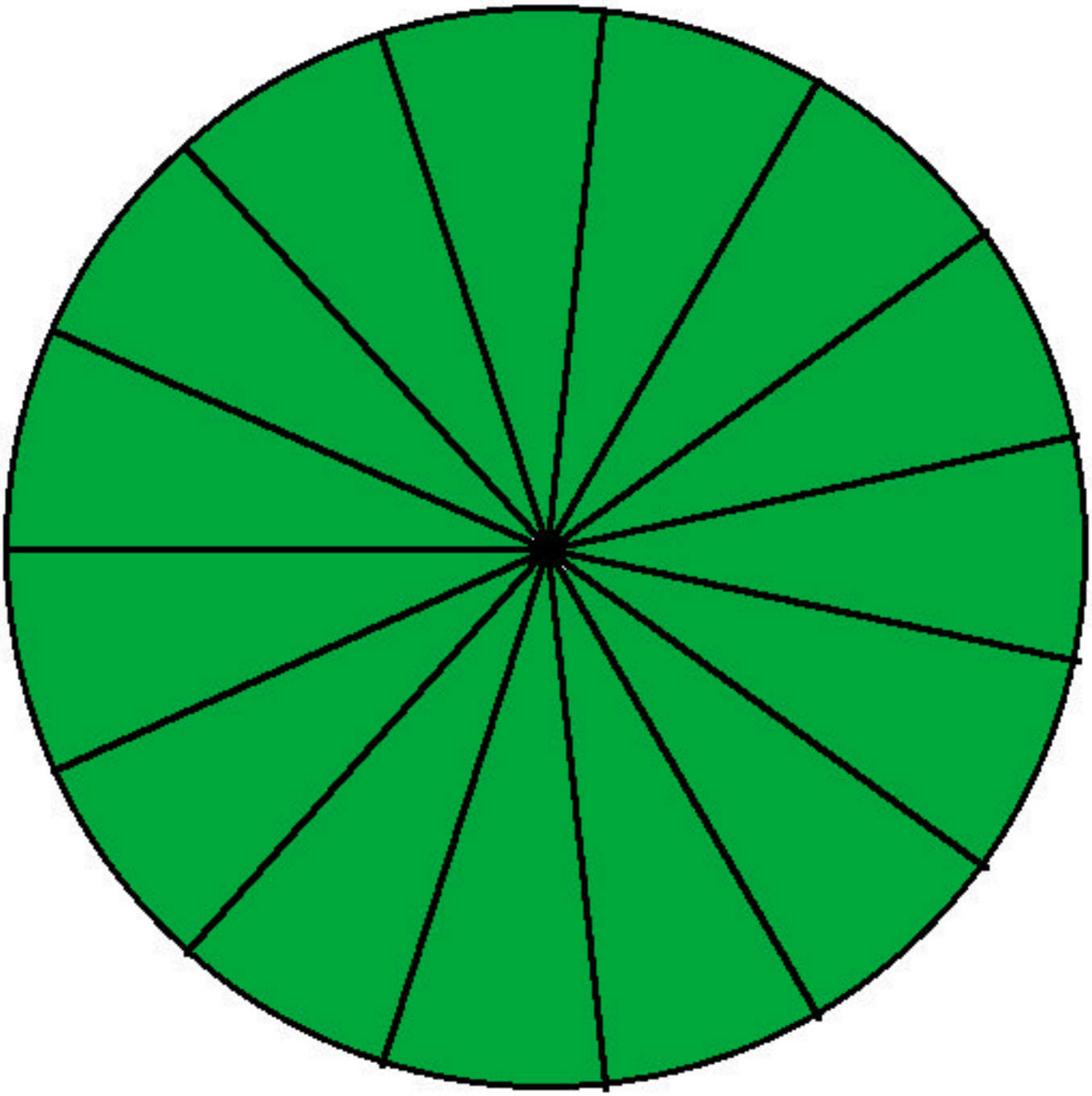












Name _____ Parent/Child Activity Sheet

Explorations With the Fraction Circles

1. How many different colors are in your folder? _____
2. Which color has the most pieces? _____
3. Which color has the fewest pieces? _____
4. How many light blue pieces are there? _____

   How many dark blue pieces are there? _____
5. How many dark blue pieces does it take to cover a yellow piece? _____
6. Can you find two different pieces to cover a yellow piece?

   Which colors did you use? _____
7. Cover the yellow piece using smaller pieces of one color.

   What color did you use? _____

   How many did you use? _____
8. Is a pink piece larger, equal to or smaller than two red pieces? _____
9. Cover a brown piece by using several smaller pieces.

   Which colors did you use? _____

   How many did you use? _____

Name _____ Quiz 1 Naming Fractions

1. The yellow piece is the unit. What fraction name can you give each of these pieces:

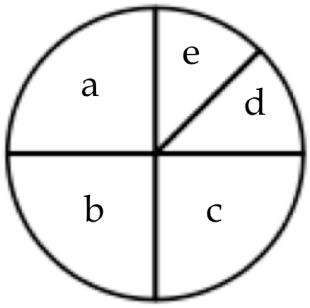
   1 blue _____

   2 grays _____

   4 reds _____

2. Draw a circle divided into 6 equal parts. Shade 5 of those parts. What fraction of the circle is shaded?

Name _____ Quiz 1 Naming Fractions

3. 

What fraction circle color would a, b, or c be?_____

What fraction of the circle is a? _____

What fraction of the circle is a, b and c combined?_____

What fraction of the circle is e?_____

4. Draw a rectangle divided into 8 equal parts. Shade in 5 parts. Write the fraction amount shaded in word.

5. You want to share your pan of brownies equally among yourself and your 8 friends. Draw a picture of your pan of brownies showing how you would divide it up to share. What fraction of the pan will each get?

Name _____ Quiz 2 Ordering fractions

1. Imagine the brown piece and the pink piece. Which is bigger?

2. Which is larger, $\frac{1}{6}$ or $\frac{1}{3}$? Explain your thinking.

3. Spinner A was divided into 4 equal parts with 3 parts green. Spinner B was divided into 10 equal parts with 3 parts green. Which spinner had the larger amount green? Explain your thinking.

Name _____ Quiz 2 Ordering fractions

4. Circle the larger fraction. Explain your thinking for each example.

$\frac{2}{4}$ $\frac{2}{3}$

$\frac{1}{10}$ $\frac{1}{20}$

$\frac{4}{5}$ $\frac{2}{5}$

$\frac{3}{9}$ $\frac{6}{9}$

Name _____ Quiz 3 Fraction Equivalence

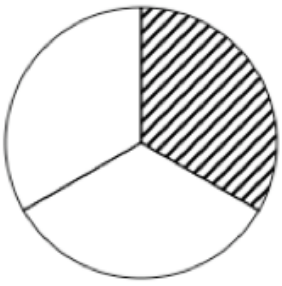
1. How many pinks equal 1 brown? _____

2. $\frac{1}{3} = \frac{\quad}{6}$

3. Give two fraction names for the shaded amount in each picture. You may draw on the pictures to help you find equivalent fractions.







Name _____ Quiz 3 Fraction Equivalence

4. Use your fraction circles to find equivalent fractions

$$\frac{1}{4} = \frac{\quad}{8}$$

$$\frac{3}{4} = \frac{9}{\quad}$$

$$\frac{1}{2} = \frac{\quad}{6}$$

Name _____ Quiz 4 Chip (tile) model

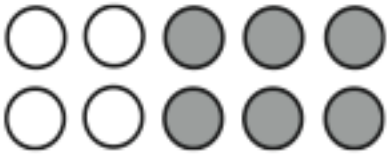
1. Draw a picture of chips (or tiles) to show each fraction below. Use 8 chips as the unit for each example.

$\frac{3}{4}$	
$\frac{1}{2}$	
$\frac{3}{8}$	

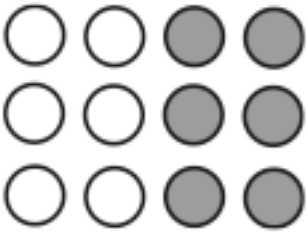
2. Show $\frac{2}{3}$ with chips (or tiles) using two different units.

Name _____ Quiz 4 Chip (tile) model

3. How many fifths are shaded?



4. Give three fraction names for the amount shaded.



5. Draw a picture of $\frac{1}{3}$ using 12 chips (or tiles). What is another fraction amount for the picture drawn?

Name _____ Quiz 5 ½ and fractions > 1

1. If the whole circle is the unit, name the amount shaded in two different ways.



2. Draw a picture to show $\frac{6}{4}$. Name that amount using a mixed fraction.

3. List three fractions equal to $\frac{1}{2}$.

Name _____ Quiz 5 ½ and fractions > 1

4. Circle the larger fraction in each pair:

$\frac{1}{2}$ $\frac{2}{3}$

$\frac{2}{4}$ $\frac{5}{12}$

$\frac{3}{4}$ $\frac{2}{12}$

$\frac{12}{15}$ $\frac{5}{10}$

$\frac{2}{8}$ $\frac{10}{12}$

5. Challenge: Which is bigger: $\frac{3}{4}$ or $\frac{2}{3}$? Explain your thinking.

Name _____ Quiz 6 Adding & subtracting

1. Use fraction circles to solve this problem. Draw a picture of what you did with the circles.

$$\frac{3}{4}+\frac{1}{6}$$

2. Use your fraction circles to solve this problem. Draw a picture of what you did with the circles.

Alex needed $\frac{3}{4}$ *cup of sugar to bake cookies. When he measured out the sugar, he had only* $\frac{1}{2}$ *cup of sugar. How much more sugar did he need?*

Name _____ Quiz 6 Adding & subtracting

3. Ty noticed that there was $\frac{6}{8}$ of a pizza left over. He ate an amount equal to $\frac{1}{4}$ of the pizza. How much of a whole pizza was left? Use your fraction circles to solve and draw a picture of what you did.

4. Is the answer to this problem greater than one or less than one? Explain your thinking.

$$\frac{4}{5}+\frac{1}{6}$$

5. Is this a reasonable answer? Explain your thinking.

$$\frac{8}{9}-\frac{4}{6}=\frac{4}{3}$$

Name _____ Date _____

End of Module Fraction Assessment

The purpose of this test is to find out what you know about fractions. You will be shown 3 problems on the overhead. Estimate the answer by recording in each box the whole number the answer is closet to.

1.

2.

3.

Name_____

Initial Fraction Ideas End of the Module Test continued

Continue to work on the rest of the test on your own. Show all the work you do to solve each problem.

4. This is the unit: 

   You have two pieces this size: 

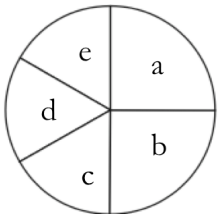
   What fraction of the unit would these two pieces be? _____

5. Bert's father cuts a cake into 8 equal sized pieces. He is going to take $\frac{3}{4}$ of the cake to the party. How many pieces of cake will he take with him?

   (a) Draw a picture below to solve the problem.

   (b) The number of pieces of cake taken to the party is _____.

6. Draw picture of $\frac{2}{3}$ using chips (or tiles) in the space below. Use a unit greater than 3 chips (or tiles).

7. 

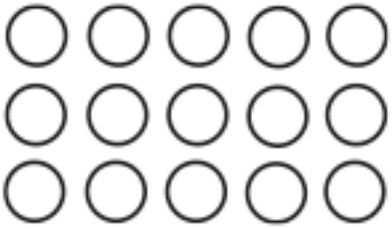
What fraction of the circle above is part c? _____

8. How many fifths are shaded?

$\frac{\quad}{5}$

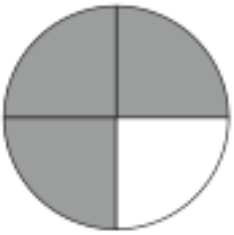


9. Circle $\frac{2}{3}$ of the set below:

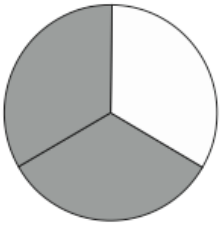


For problems 10 – 12 give two fraction names for the shaded amount. You may draw on the pictures to help you.

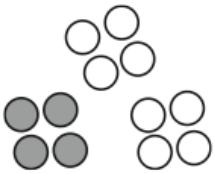
10.



11.



12.



For problems 13 – 17 circle the larger fraction. If equal, circle both. Explain your reasoning for each one.

13.	$\frac{3}{4}$ $\frac{2}{3}$	
14.	$\frac{1}{2}$ $\frac{5}{8}$	
15.	$\frac{3}{12}$ $\frac{7}{12}$	
16.	$\frac{4}{9}$ $\frac{4}{11}$	
17.	$\frac{8}{14}$ $\frac{4}{9}$	

You may use your fraction circles on the last five problems. Draw pictures to show what you did with the circles.

18. Liana ate $\frac{3}{8}$ of a small pizza. The next day she ate $\frac{1}{4}$ of a small pizza. How much of a pizza did she eat altogether?

19. Ann and Josie receive the same allowance. Josie spent 4-ninths of hers on CDs. Ann spent 1-third of her allowance repairing her bicycle. Josie spent how much more of her allowance than Ann?

Show Pictures Here

20. $\begin{array}{r} \frac{5}{6} \\ +\frac{2}{6} \\ \hline \end{array}$

21. $\begin{array}{r} \frac{1}{2} \\ -\frac{3}{8} \\ \hline \end{array}$

22. $\begin{array}{r} \frac{1}{3} \\ +\frac{2}{6} \\ \hline \end{array}$

Transparency
[These are the first three items on the test]

(1) $\frac{7}{8} + \frac{12}{13}$

(2) $\frac{3}{8} + \frac{5}{13}$

(3) $\frac{8}{9} - \frac{7}{8}$

Name _____ Date _____

Interview #1:

[Used after Lesson 10 or 11]

I am going to ask you some questions about fractions. I am very interested in how you come up with the answers, so it is important for you to tell me what you are thinking about. The interview will not be graded, so you do not have to worry about wrong answers. Are you ready?

Concept Questions

1.

(A) Use your fraction circles to show a model for the fraction 3/5.

(B) Now I want you to explain how you know that this models 3/5.

(C.) Now can you show me another way to model $\frac{3}{5}$ using the same fraction circles? Explain how the two models are alike and different.

Interview #1 continued

2. Display 15 tiles without counting or telling the child how many there are. [This is an extension. Tiles have not been introduced as a model yet].

(A) Say: You can arrange the tiles any way you want. I want you to show me the fraction $\frac{3}{5}$ with these tiles.

(B) Explain what you were thinking in order to solve this problem.

(C) Show me a model for $\frac{3}{5}$ using a different number of tiles. How are the two tile-models alike? How are they different?

(D) How is this model for 3/5 like your fraction circle models? How are they different?

Order Questions

[Same numerator]

3. Say: I'm going to show you word names for two fractions. I'll ask you to tell me whether they are equal or one is less.

Ready? 1-fifth and 1-sixth. Are they equal or is one less? Which is less? Explain your reasoning. [Do you imagine or picture something in your mind to help you tell which is less?]

Interview #1 continued

[Same denominator]

4. Say: I'm going to show you word names for two new fractions. I'll ask you to tell me whether they are equal or if one is less.

   Ready? 3-ninths and 4-ninths. Are they equal or is one less? Which is less? Explain your reasoning. [Do you imagine or picture something in your mind to help you tell which is less?]

[Transitive]

5. Read this story to the student:

   Jon and Lara each ordered a small pizza at Domino's. Jon's pizza was cut into 8 equal-sized parts. Lara's was cut into 6 equal-sized parts. Jon ate 5 pieces; Lara ate 2 pieces. Did they eat the same amount, or did one eat less?

   Explain your reasoning. [Do you imagine or picture something in your mind to help you tell which is less?]

[Residual]

6. Read this story to the student:

   Mark and William both had bags of M&M peanut candies. The bags held the same number of candies. Mark ate 2-thirds of his bag. William ate 3-fourths of his bag. Did they eat the same amount or did one eat less?

   Explain your reasoning. [Do you imagine or picture something in your mind to help you tell which is less?]

Concept of Unit Questions

7. Show and read the statement:

Tim used 6 cubes to build $\frac{1}{5}$ *of an orange tower. How tall will the whole tower be when it's finished?*

(A) Provide unifix cubes and ask students to show you how to use these cubes to solve the problem. Ask students to talk aloud as they solve the problem.

(B) **[If correct repeat changing the data: 8 cubes;** $\frac{2}{3}$**]**

Name _____ Date _____

Interview #2
Use after Lesson 15 or 16

I am going to ask you some questions about fractions. I am very interested in how you come up with the answers, so it is important for you to tell me what you are thinking about. The interview will not be graded, so you do not have to worry about wrong answers. Are you ready?

Concept Questions

1. Display 15 tiles without counting or telling the child how many there are.

(A) Say: You can arrange the tiles any way you want. I want you to show me the fraction $\frac{2}{3}$ with these tiles.

(B) Explain what you were thinking in order to solve this problem.

(C) Show me a model for $\frac{2}{3}$ using a different number of tiles. How are the two tile-models alike? How are they different?

2. Read this story to the student:

> *Mary and Jose both have some money to spend. Mary spends $\frac{1}{4}$ of hers and Jose spends $\frac{1}{4}$ of his. Is it possible that Mary and Jose spent the same amount of money?* Tell me what you are thinking.

Interview #2 continued

Order Questions

[Same numerator]
3. Say: Here are two fractions.

   Show: $\frac{3}{5}$ $\frac{3}{4}$

   Say: Are they equal or is one less? Which one is less? Tell me how you know. Did you picture anything in your mind as you thought about these fractions?

[Same numerator]
4. Say: Here are two fractions.

   Show: $\frac{1}{17}$ $\frac{1}{19}$

   Say: Are they equal or is one less? Which one is less? Tell me how you know. Did you picture anything in your mind as you thought about these fractions?

[Same denominator]
5. Say: Here are two fractions.

   Show: $\frac{14}{26}$ $\frac{18}{26}$

   Say: Are they equal or is one less? Which one is less? Tell me how you know. Did you picture anything in your mind as you thought about these fractions?

Interview #2 continued

[Residual]
6. Read this story to the student:

Alice and Janis both receive the same allowance. Alice spent $\frac{4}{5}$ *of hers on a movie. Janis spent* $\frac{9}{10}$ *of hers on a new CD. Did they spend the same amount or did one spend less?* [If less ask: who spent less?]

Say: Tell me how you know. Did you picture anything in your mind as you thought about these fractions?

[Transitive]
7. Read this story to the student:

Mark and Jenny walk home from school. Mark walks $\frac{3}{8}$ *of a mile. Jenny walks* $\frac{6}{9}$ *of a mile. Do they walk the same amount or does one walk less?*

Say: Tell me how you know. Did you picture anything in your mind as you thought about these fractions?

Interview #2 continued

Concept of Unit Questions

8. Show the red fraction piece.

   Say: This is $\frac{1}{6}$ of my unit. With your fraction circles, show me the unit. Talk aloud as you sole the problem explaining each step. Record answer.

   [If correct change data: pink is $\frac{2}{3}$; find the unit].

9. Read this story to the student:

   Ten children went to a party in a group. This group of 10 children was $\frac{2}{5}$ of all the children who were invited. How many children were invited?

   Provide tiles and ask the student to show you how to use these tiles to solve the problem. Ask students to talk aloud as they solve the problem. Record answer.

Name _____ Date _____

Interview #3
Use before Lesson 19

I am going to ask you some questions about fractions. I am very interested in how you come up with the answers, so it is important for you to tell me what you are thinking about. The interview will not be graded, so you do not have to worry about wrong answers. In fact, you have not worked on the type of problems I will ask about. I am interested in how you try to solve them before you learn from your teacher how to do them.

1. Read this story to the student:

   Sally ate $\frac{2}{3}$ of a pizza for dinner. The next morning she ate another $\frac{1}{6}$ of a pizza. How much of a pizza did she eat altogether?

   (A) Say: Without working out the exact answer, give me an estimate that is reasonable. (If needed, provide clues: Is the answer >1/2 or <1/2? Is the answer >1 or <1?

   (B) Say: Tell me what you were thinking to reach this estimate.

   (C) Say: Using fraction circles, act out how you would find the exact answer. Talk aloud as you solve the problem.

   (D) If the student was successful, ask student to record each step with the fraction circles with symbols.

Interview #3 continued

2. Read this story to the student:

A movie costs Jane $\frac{1}{6}$ *of her allowance. If she only had* $\frac{11}{12}$ *of her allowance before the movie, what fraction of her allowance did she have after the movie?*

(A) Say: Without working out the exact answer, give me an estimate that is reasonable. (If needed, provide clues: Is the answer $>\frac{1}{2}$ or $<\frac{1}{2}$? Is the answer >1 or <1?).

(B) Say: Tell me what you were thinking to reach this estimate.

(C) Say: Using fraction circles, act out how you would find the exact answer. Talk aloud as you solve the problem.

(D) If the student was successful, ask student to record each step with the fraction circles with symbols.

Interview #3 continued

3. Read this story to the student:

Josie and Al each receive the same allowance. Al spent $\frac{6}{10}$ *of his allowance on a book. Josie spent* $\frac{4}{5}$ *of her allowance to fix her skateboard. How much more did Josie spend?*

(A) Say: Without working out the exact answer, give me an estimate that is reasonable. (If needed, provide clues: Is the answer $> \frac{1}{2}$ or $< \frac{1}{2}$? Is the answer >1 or <1?).

(B) Say: Tell me what you were thinking to reach this estimate.

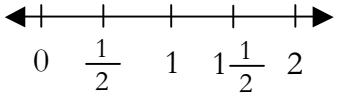
(C) Say: Using fraction circles, act out how you would find the exact answer. Talk aloud as you solve the problem.

(D) If the student was successful, ask student to record each step with the fraction circles with symbols.

Interview #3 continued

4. (A) Say: Tell me about where the answer to this addition problem would be on this number line.

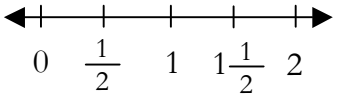
$\frac{3}{4} + \frac{1}{3}$



(B) Say: Tell me how you know.

5. (A) Say: Tell me about where the answer to this addition problem would be on this number line.

$\frac{7}{8} - \frac{1}{2}$



(B) Say: Tell me how you know.

6. Say: Jon calculated the problem as follows: $\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$

Ask: Do you agree? Tell me all you can.

Name _____ Date _____

Interview #4: End of Unit

I am going to ask you some questions about fractions. I am very interested in how you come up with the answers, so it is important for you to tell me what you are thinking about. The interview will not be graded, so you do not have to worry about wrong answers. Are you ready?

Concept Questions

1. Display a bag of chips without counting or telling the child how many there are.

(A) Say: I want you to show me $\frac{2}{3}$ using the chips as your model. Use as many chips as you want.

(B) Explain what you were thinking in order to solve this problem.

(C) Can you do this in another way? How are your two alike and different?

Interview #4 continued

2. Read this to the student:

Martin ate 1 $\frac{2}{3}$ *pizza.*

(A) Name the amount Martin ate in another way.

(B) Draw a picture to verify your answer. Explain what the picture shows.

Order Questions

[Same numerator]
3. Say: I am going to write 3 fractions

Write: $\frac{1}{5}$ $\frac{1}{3}$ $\frac{1}{4}$

(A) Say: I want you to write them in order from smallest to largest.

(B) Say: Tell me how you know.

[Same numerator]
4. Say: I'm going to write two fractions.

Write: $\frac{4}{35}$ $\frac{4}{29}$

(A) Say: Are they equal or is one less? Which one is less?

(B) Say: Tell me how you know.

Interview #4 continued

[Transitive]
5. Say: I'm going to write two fractions.

   Write: $\frac{8}{12}$ $\frac{3}{7}$

   (A) Say: Are they equal or is one less? Which one is less?

   (B) Say: Tell me how you know.

[Same Denominator]
6. Say: I'm going to write two fractions.

   Write: $\frac{27}{64}$ $\frac{19}{64}$

   (A) Say: Are they equal or is one less? Which one is less?

   (B) Say: Tell me how you know.

Interview #4 continued

[Residual]
7. Say: I'm going to write two fractions.

   Write: $\frac{4}{5}$ $\frac{9}{10}$

   (C) Say: Are they equal or is one less? Which one is less?

   (D) Say: Tell me how you know.

[Equivalence Questions]
8. Say: I'm going to write two fractions.

   Write: $\frac{3}{4}$ $\frac{9}{12}$

   (C) Say: Are these fractions equal or is one less? Which one is less?

   (D) Say: Tell me how you know.

Interview #4 continued

[Equivalence Questions]
9. Say: I'm going to write two fractions.

   Write: $\frac{6}{9}$ $\frac{4}{6}$

   (E) Say: Are they equal or is one less? Which one is less?

   (F) Say: Tell me how you know.

Concept of Unit Questions

10. Show and read the statement:

   Tina was building towers with cubes. Tina finished building $1\frac{1}{3}$ *towers. She used 12 cubes for these towers. How many cubes is one tower?*

   Ask student to solve the problem. Provide Unifix cubes. Ask students to talk aloud as they solve the problem.

11. (A) Say: This brown piece is $\frac{4}{6}$ of some unit. What is the unit? Use the fraction circles to show me. Talk aloud as you do this.

   (B) If correct ask this: This yellow piece is 1 and $\frac{1}{2}$ of some unit. What is the unit? Use the fraction circles to show me. Talk aloud as you do this.

Interview #4 continued

Operations

12. Read this story to the student:

Marty was making two types of cookies. He used $\frac{1}{4}$ *cup of flour for one recipe and* $\frac{2}{3}$ *cup of flour for the other. How much flour did he use altogether?*

(A) Say: Without working out the exact answer, give me an estimate that is reasonable. (If needed, provide clues: Is the answer $>\frac{1}{2}$ or $<\frac{1}{2}$? Is the answer >1 or <1?).

(B) Say: Tell me what you were thinking to reach this estimate.

(C) Say: Using fraction circles, act out how you would find the exact answer. Talk aloud as you solve the problem.

(D) If the student was successful, ask student to record each step with the fraction circles with symbols.

Interview #4 continued

13. Read this story to the student:

Martin and Jane each ordered small pizzas at Dominos. Jane ate $\frac{5}{8}$ *of her small pizza. Martin ate* $\frac{3}{4}$ *of his small pizza. Who ate more? How much more?*

(A) Say: Without working out the exact answer, give me an estimate that is reasonable. (If needed, provide clues: is the answer > $\frac{1}{2}$ or < $\frac{1}{2}$? Is the answer >1 or <1?)?

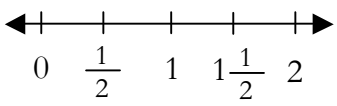
(B) Say: Tell me what you were thinking to reach this estimate?

(C) Say: Using fraction circles, (or paper folding) act out how you would find the exact answer. Talk aloud as you solve the problem.

(D) If the student was successful, ask the student to record each step with the fraction circles with symbols.

14. (A) Say: Tell me about where the answer to this problem would be on this number line.

$\frac{2}{3} + \frac{1}{6}$



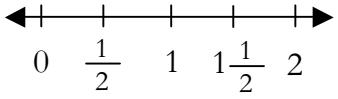
(B) Say: Tell me how you know.

Interview #4 continued

(C) Say: Using paper and pencil, how can you figure out the exact answer? Explain what you are doing.

15. (A) Say: Tell me about where the answer to this problem would be on this number line.

$\frac{8}{9} - \frac{1}{3}$



(B) Say: Tell me how you know.

(C) Say: Using paper and pencil, how can you figure out the exact answer? Explain what you are doing.